Vector & Three-Dimensional Geometry

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Vectors

VECTORS AND SCALARS :

The physical quantities (we deal with) are generally of two types:

Scalar Quantity:

A quantity which has magnitude but no sense of direction is called scalar quantity (or scalar), e.g., mass, volume, density, speed etc.

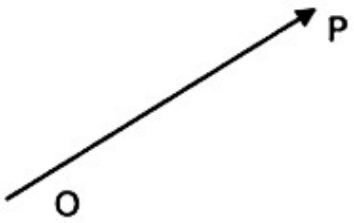
Vector Quantity:

A quantity which has magnitude as well as a sense of direction in space and obey the laws of vector algebra is called a vector quantity, e.g., velocity, force, displacement etc.

Notation and Representation of Vectors :

Vectors are represented by $\vec{a}, \vec{b}, \vec{c}$ and their magnitude (modulus) are represented by a, b, c or $|\vec{a}|, |\vec{b}|, |\vec{c}|$. The vectors are represented by directed line segments.

For example, line segment $\overline{OP}$ represents a vector with magnitude OP (length of line segment), arrow denotes its direction. O is initial point and P is terminal point, also called as head and tail of vector respectively.



KINDS OF VECTORS :

1. Zero or null vector:

A vector whose magnitude is zero is called zero or null vector and it is denoted by **0** or $\vec{0}$. The initial and terminal points of the directed line segment representing zero vector are coincident and its direction is arbitrary.

2. Unit vector:

A vector of unit magnitude is called a unit vector. A unit vector in the direction of $\vec{a}$ is denoted by $\hat{a}$. Thus

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = \frac{\text{Vector a}}{\text{Magnitude of a}}$$

Point to Remember!!!

(i) $|\hat{a}| = 1$

(ii) Unit vectors parallel to x - axis, y - axis and z - axis are denoted by $\hat{i}, \hat{j}$ and $\hat{k}$ respectively.

(iii) Two unit vectors may not be equal unless they have the same direction.

3. Equal Vectors:

Two vectors $\vec{a}$ and $\vec{b}$ are said to be equal, if

(a) $|\vec{a}| = |\vec{b}|$

(b) they have the same sense of direction

4. Co-initial vectors:

Vectors having same initial point.

5. Free vectors:

All such vectors are those which when transformed into space from one point to another point without affecting their magnitude and direction, can be considered as equal. i.e. the physical effects produced by them remains unaltered.
e.g. displacement, velocity.

6. Localised vectors:

e.g. force, different physical effect if line of application is changed.

Know the facts

In mathematics we mainly deal with free vectors.

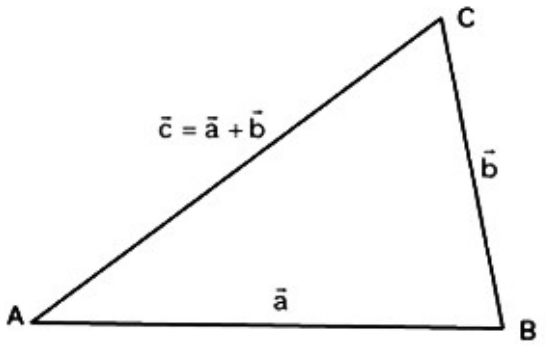
ADDITION OF VECTORS :

Triangle law of addition :

If two vectors are represented by two consecutive sides of a triangle, then their sum is represented by the third side of the triangle but in opposite direction. This is known as triangle law of addition of vectors.

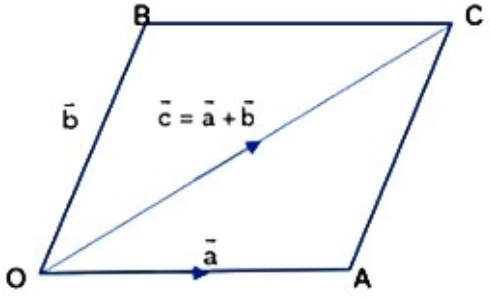
Thus, if $\overline{AB} = \vec{a}, \overline{BC} = \vec{b}$ and $\overline{AC} = \vec{c}$ then $\overline{AB} + \overline{BC} = \overline{AC}$ i.e. $\vec{a} + \vec{b} = \vec{c}$

Converse of triangle law is also true.



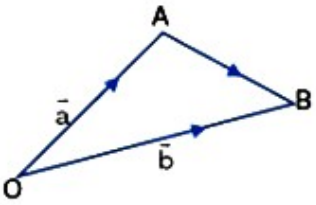
Parallelogram Law of Addition:

If two vectors are represented by two adjacent sides of a parallelogram, then their sum is represented by the diagonal of the parallelogram whose initial point is the same as the initial point of the given vectors. This is known as parallelogram law of addition of vectors.



Thus if $\overline{OA} = \vec{a}, \overline{OB} = \vec{b}$ and $\overline{OC} = \vec{c}$ then $\overline{OA} + \overline{OB} = \overline{OC}$

i.e. $\vec{a} + \vec{b} = \vec{c}$



POSITION VECTORS:

Let O be fixed point in space, then vector $\overline{OP}$ (P is any point in space) is called position vector of point P w.r.t. O. If A and B are any two point in space then

$\overline{AB}$ = p.v of B − p.v of A = $\overline{OB} - \overline{OA}$

i.e. $\overline{AB} = \vec{b} - \vec{a}$

MULTIPLICATION OF A VECTOR BY SCALAR:

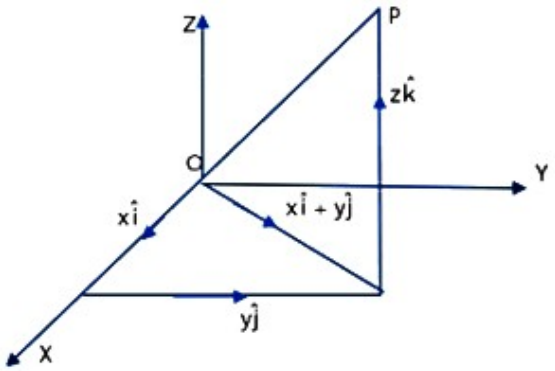
If $\vec{a}$ is a vector and m is a scalar, then $m\vec{a}$ is a vector parallel to $\vec{a}$ whose modulus is $|m|$ times that of $\vec{a}$. This multiplication is called **SCALAR MULTIPLICATION**. If $\vec{a}$ and $\vec{b}$ are vectors and m, n are scalars, then :

$m(\vec{a}) = (\vec{a})m = m\vec{a}$

$m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$

$(m+n)\vec{a} = m\vec{a} + n\vec{a}$

$m(\vec{a} + \vec{b}) = m\vec{a} + m\vec{b}$



Point to Remember!!!

Position vector of a point $P(x, y, z)$ in terms of its cartesian coordinate is $\overline{OP} = x\hat{i} + y\hat{j} + z\hat{k}$

DISTANCE BETWEEN TWO POINTS :

Let A and B be two given points whose coordinate are respectively (x_1, y_1, z_1) and (x_2, y_2, z_2)

If $\vec{a}$ and $\vec{b}$ are p.v. of A and B relative to point O, then

$\vec{a} = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$

$\vec{b} = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$

Now,

$\overline{AB} = \overline{OB} - \overline{OA} = \vec{b} - \vec{a} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

Distance between the points $\vec{A}$ and $\vec{B}$ = magnitude of

$$\overline{AB} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

COLLINEAR VECTORS OR PARALLEL VECTORS :

Vectors which are parallel to the same line are called collinear vectors or parallel vectors. Such vectors have either same direction or opposite direction. If they have the same direction they are said to be like vectors, and if they have opposite directions, they are called unlike vectors.

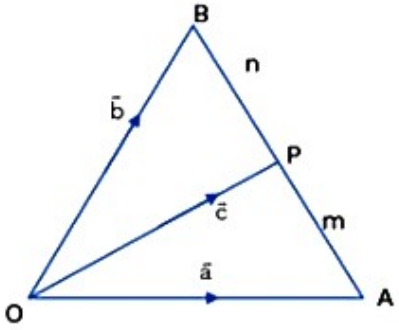


In the diagram $\vec{a}$ and $\vec{c}$ are like vectors whereas $\vec{a}$ and $\vec{b}$ are unlike vectors.

i.e. $\vec{a} = k_1\vec{c}\ (k_1 > 0)$, $\vec{a} = k_2\vec{b}\ (k_2 < 0)$

COPLANAR VECTORS :

If the directed line segments of some given vectors are parallel to the same plane then they are called coplanar vectors. It should be noted that two vectors are always coplanar but three or more vectors may or may not be coplanar.



SECTION FORMULAE :

If $\vec{a}$ and $\vec{b}$ are the position vectors of two points A and B, then the position vector $\vec{c}$ of a point P dividing AB in the ratio m : n is given by

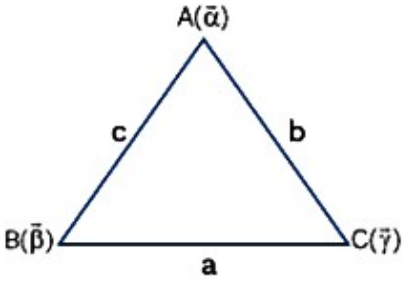
$$\vec{c} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

Point to Remember!!!

Particular Case:

1. Position vector of the mid point of AB is $\frac{\vec{a}+\vec{b}}{2}$.
2. If the point P divides AB in the ratio m : n externally, then p.v. of P is given by $\vec{c} = \frac{m\vec{b} - n\vec{a}}{m-n}$

Note:



1. P.V. of the centroid of a triangle ABC $= \frac{\vec{\alpha}+\vec{\beta}+\vec{\gamma}}{3}$

   (Concurrency of medians)

2. P.V. of incentre of the $\bar{A} = \frac{a\vec{\alpha}+b\vec{\beta}+c\vec{\gamma}}{a+b+c}$

   (Concurrency of internal angle bisectors)

   Excentres of the A are $\frac{-a\vec{\alpha}+b\vec{\beta}+c\vec{\gamma}}{-a+b+c}; \frac{a\vec{\alpha}-b\vec{\beta}+c\vec{\gamma}}{a-b+c}$

   and $\frac{a\vec{\alpha}+b\vec{\beta}-c\vec{\gamma}}{a+b-c}$.

3. P.V. of circumcentre of the

   $\bar{A} = \frac{\vec{\alpha}\sin 2A+\vec{\beta}\sin 2B+\vec{\gamma}\sin 2C}{\sum \sin 2A}$

   (Concurrency of perpendicular bisectors of sides)

4. P.V. of orthocenter of the

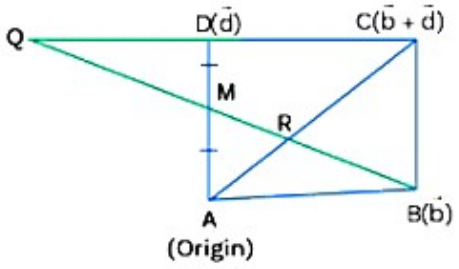
   $\bar{A} = \frac{\vec{a}\tan A+\vec{b}\tan B+\vec{c}\tan C}{\sum \tan A}$

   (Concurrency of altitudes)

Q. **Through the middle point M of the side AD of ||gm ABCD, a straight line BM is drawn intersecting AC at R and CD produced at Q. use vectors to prove that QR = 2RB.**

Sol. $\frac{DM}{MA} = \frac{QD}{AB} \Rightarrow \overline{QD} = \overline{AB}$

$\vec{d}-\vec{Q} = \vec{b} \Rightarrow \vec{Q} = \vec{d}-\vec{b}$



Let R divides AC in ratio $\lambda : 1$ and QB in $\mu : 1$ ratio by section formula

$\vec{R} = \frac{\lambda(\vec{b}+\vec{d})}{\lambda+1} = \frac{\mu\vec{b}+\vec{Q}}{\mu+1} = \frac{\mu\vec{b}+(\vec{d}-\vec{b})}{\mu+1} = \frac{(\mu-1)\vec{b}+\vec{d}}{\mu+1}$

$\lambda(\mu+1)\vec{b}+\lambda(\mu+1)\vec{d} = (\mu-1)(\lambda+1)\vec{b}+(\lambda+1)\vec{d}$

$\lambda(\mu+1) = (\mu-1)(\lambda+1)$ & $\lambda(\mu+1) = \lambda+1$

$\lambda\mu+\lambda = \mu\lambda-\lambda+\mu-1$ & $\lambda\mu+\lambda = \lambda+1$

$2\lambda = \mu-1 \qquad \lambda = \frac{1}{\mu}$

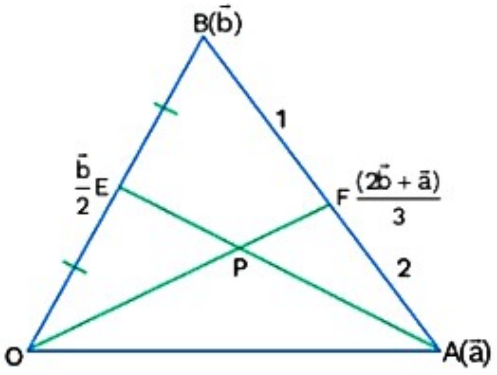
$\frac{2}{\mu} = \mu-1$

$2 = \mu^2-\mu \Rightarrow \mu^2-\mu-2 = 0 \Rightarrow \mu = -1, 2$

$\therefore \quad \mu = 2 \Rightarrow QR = 2RB$

Q. **In ΔAOB, E is the mid point of OB and F divides BA in the ratio 1 : 2 use vectors to prove that $\frac{OP}{PF} = \frac{3}{2}$.**

Sol. Let P divides of in ratio $\lambda : 1$ & EA in ratio



$$\mu : 1 - \vec{p} = \frac{\lambda\left(\frac{2\vec{b}+\vec{a}}{3}\right)}{\lambda+1} = \frac{\mu(\vec{a})+\frac{\vec{b}}{2}}{\mu+1}$$

$$\frac{\lambda}{3}(\mu+1)(2\vec{b}+\vec{a}) = (\lambda+1)\left(\mu\vec{a}+\frac{\vec{b}}{2}\right)$$

$$\frac{2\lambda(\mu+1)}{3}\vec{b} + \frac{\lambda(\mu+1)}{3}\vec{a} = (\lambda+1)\mu\vec{a} + \frac{(\lambda+1)}{2}\vec{b}$$

$$\frac{2\lambda(\mu+1)}{3} = \frac{\lambda+1}{2} \;\&\; \frac{\lambda(\mu+1)}{3} = (\lambda+1)\mu$$

$4\lambda(\mu+1) = 3\lambda + 3$
$4\lambda\mu + 4\lambda = 3\lambda + 3$
$4\lambda\mu = 3 - \lambda$
$\mu = \frac{3-\lambda}{4\lambda}$

$\lambda\mu + \lambda = 3\mu(\lambda+1)$
$\lambda\mu + \lambda = 3\mu\lambda + 3\mu$
$\lambda = 2\mu\lambda + 3\mu$
$\lambda = 2\lambda\frac{(3-\lambda)}{4\lambda} + 3\frac{(3-\lambda)}{4\lambda}$

$$\lambda = \frac{6\lambda - 2\lambda^2 + 9 - 3\lambda}{4\lambda}$$

$4\lambda^2 + 2\lambda^2 = 3\lambda + 9$
$6\lambda^2 = 3\lambda + 9$
$2\lambda^2 = \lambda + 3$
$\lambda = \frac{3}{2}, -1$

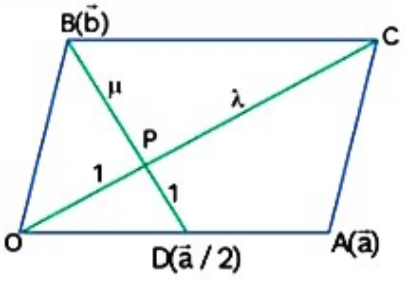
$\therefore \lambda = \frac{3}{2} \Rightarrow \frac{OP}{PF} = \frac{3}{2}$

Q. **Let OACB be parallelogram with O at the origin & OC a diagonal. Let D be the mid point of OA. Using vector method prove that BD & CO intersect in the same ratio. Determine this ratio.**

Sol. OACB is parallelogram with O as origin.

Let $\overline{OA} = \vec{a}, \overline{OB} = \vec{b}, \overline{OC} = \vec{a} + \vec{b}$ and $\overline{OD} = \frac{\vec{a}}{2}$

$\overline{CO}$ and $\overline{BD}$ meets at P .



$\therefore \overline{OP} = \frac{\lambda \cdot 0 + 1(\vec{a} + \vec{b})}{\lambda + 1}$ [along $\overline{OC}$]

$\overline{OP} = \frac{\vec{a} + \vec{b}}{\lambda + 1}$ ...(i)

Again, $\overline{OP} = \frac{\mu\left(\frac{a}{2}\right) + b}{\mu + 1}$ [along $\overline{BD} \Rightarrow \overline{OP} = \frac{\mu\vec{a} + 2\vec{b}}{2(\mu + 1)}$ ...(ii)]

From Eqs. (i) and (ii), $\frac{\vec{a} + \vec{b}}{\lambda + 1} = \frac{\mu\vec{a} + 2\vec{a}}{2(\mu + 1)}$

$\frac{1}{\lambda + 1} = \frac{\mu}{2(\mu + 1)}$ and $\frac{1}{\lambda + 1} = \frac{1}{\mu + 1}$

On solving, we get $\mu = \lambda = 2$ Thus, required ratio is 2 : 1

Q. **If the vectors $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}, \vec{b} = -\hat{i} + 3\hat{j} + 4\hat{k}$ & $\vec{c} = 4\hat{i} - 2\hat{j} - 6\hat{k}$ constitute the sides of a ΔABC, then the length of the median bisecting the vector $\vec{c}$ is**

(A) $\sqrt{2}$ **(B)** $\sqrt{14}$ **(C)** $\sqrt{74}$ **(D)** $\sqrt{6}$

Sol. **(D)**

$a = \sqrt{14}$ $b = \sqrt{26}$ $c = \sqrt{56}$

Length of Median $= \frac{1}{2}\sqrt{2b^2 + 2a^2 - c^2}$

$= = \frac{1}{2}\sqrt{2(26) + 2(14) - 56}$

$= \frac{1}{2}\sqrt{52 + 28 - 56} = \frac{1}{2}\sqrt{24}$

$= \frac{1}{2} \times 2\sqrt{6} = \sqrt{6}$

Q. **Consider the points A, B and C with position vectors $(-2\hat{i} + 3\hat{j} + 5\hat{k})$, $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $7\hat{i} - \hat{k}$ respectively.**

Statement-1 : The vector sum $\overline{AB} + \overline{BC} + \overline{CA} = \vec{0}$

Statement-2 : A, B and C form the vertices of a triangle.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

Sol. **(C)**

$\overline{AB} = (\hat{i} + 2\hat{j} + 3\hat{k}) - (-2\hat{i} + 3\hat{j} + 5\hat{k})$

$= 3\hat{i} - \hat{j} - 2\hat{k}$

$\overline{BC} = (7\hat{i} - \hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k}) = 6\hat{i} - 4\hat{k} - 2\hat{j}$

$\overline{CA} = (-2\hat{i} + 3\hat{j} + 5\hat{k}) - (7\hat{i} - \hat{k})$

$= -9\hat{i} + 3\hat{j} + 6\hat{k}$

$\therefore \overline{AB} + \overline{BC} + \overline{CA} = (3\hat{i} - \hat{j} - 2\hat{k}) + (6\hat{i} - 4\hat{k} - 2\hat{j}) + (-9\hat{i} + 3\hat{j} + 6\hat{k}) = \vec{0}$

So, A, B, C are collinear. So they will not form vertices of a triangle

Q. **ABCD is a parallelogram whose diagonals meet at P. If O is a fixed point, then $\overline{OA} + \overline{OB} + \overline{OC} + \overline{OD}$ equals**

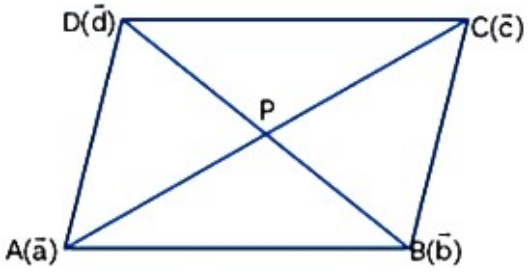
(A) $\overline{OP}$ **(B)** $2\overline{OP}$ **(C)** $3\overline{OP}$ **(D)** $4\overline{OP}$

Sol. **(D)**

Since, P bisects both the diagonal AC and BD, so

$\therefore \overline{OA} + \overline{OC} = 2\overline{OP}$ and

$\overline{OB} + \overline{OD} = 2\overline{OP} \Rightarrow \overline{OA} + \overline{OB} + \overline{OC} + \overline{OD} = 4\overline{OP}$



Q. **If $\vec{a}, \vec{b}$ are represented by the sides AB and BC of a regular hexagon ABCDEF, then vector represented by FA will be**

(A) $\vec{a} + \vec{b}$ **(B)** $\vec{b} - \vec{a}$ **(C)** $\vec{a} - \vec{b}$ **(D)** $2\vec{b} - \vec{a}$

Sol. **(C)**

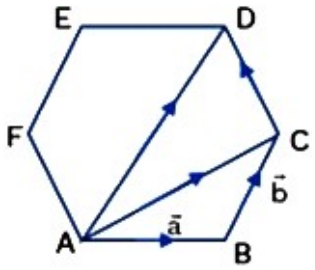
$\because \overline{AC} = \vec{a} + \vec{b}$

$\overline{AD} = 2\overline{BC} = 2\vec{b}$ ($\because$ By property of hexagon AD = 2BC)

$\therefore \overline{DC} = \overline{DA} + \overline{AC}$

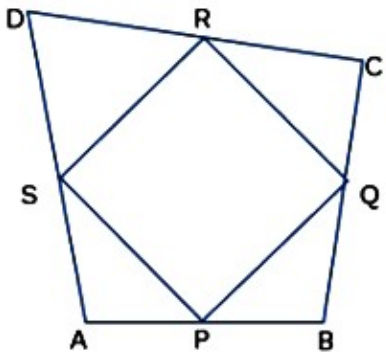
$= -2\vec{b} + (\vec{a} + \vec{b}) = \vec{a} - \vec{b}$

But $\overline{FA} = \overline{DC} \Rightarrow \overline{FA} = \vec{a} - \vec{b}$



Q. **If the mid-points of the consecutive sides of a quadrilateral are joined, prove that the resulting quadrilateral is a parallelogram.**

Sol. Let ABCD be the given quadrilateral and P, Q, R, S be the mid-points of sides AB, BC, CD and AD respectively. Let O be origin of reference and let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ be the position vectors of A, B, C and D respectively.



Then, $\overline{OP} = \frac{1}{2}(\vec{a}+\vec{b}), \overline{OQ} = \frac{1}{2}(\vec{b}+\vec{c})$

$\overline{OR} = \frac{1}{2}(\vec{c}+\vec{d}), \overline{OS} = \frac{1}{2}(\vec{d}+\vec{a})$

Now, $\overline{PQ} = \overline{OQ} - \overline{OP} = \frac{1}{2}(\vec{b}+\vec{c}-\vec{a}-\vec{b}) = \frac{1}{2}(\vec{c}-\vec{a})$

and $\overline{SR} = \overline{OR} - \overline{OS} = \frac{1}{2}(\vec{c}+\vec{d}-\vec{d}-\vec{a}) = \frac{1}{2}(\vec{c}-\vec{a})$

$\therefore \overline{PQ} = \overline{SR}$

Thus, PQ = SR and PQ ∥ SR i.e., two opposite sides of PQRS are equal and parallel. Hence PQRS is a parallelogram.

Q. **A rigid body rotates about an axis through the origin with an angular velocity $10\sqrt{3}$ m/s. If $\vec{\omega}$ points in the direction with position vector $(\hat{i}+\hat{j}+\hat{k})$, then find $\vec{\omega}$.**

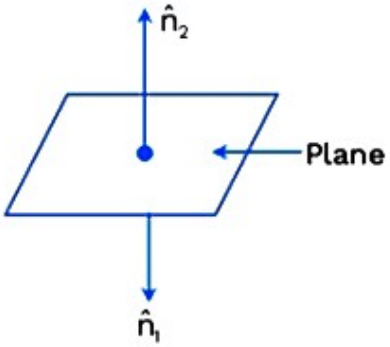
Sol. $\hat{n} = \frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}} \Rightarrow \vec{\omega} = |\vec{\omega}|\,\hat{n}$

$= (10\sqrt{3})\left(\frac{\hat{i}+\hat{j}+\hat{k}}{\sqrt{3}}\right)$

$= 10(\hat{i}+\hat{j}+\hat{k})$

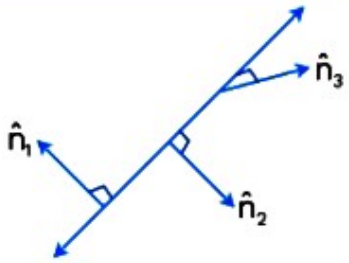
Q. **Find the number of distinct unit vectors in space perpendicular to a given plane.**

Sol. There will be two distinct unit vectors in space perpendicular to a given plane



Q. **How many unit vectors to a given line in space are possible ?**

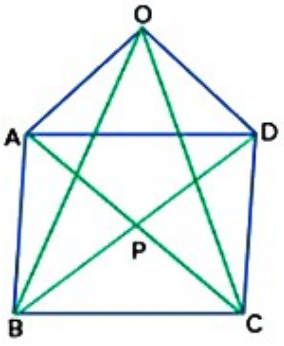
Sol. There will be infinite directions perpendicular to a given line.
So infinite unit vectors are possible



Q. **ABCD is parallelogram whose diagonals meet at P. If O is a fixed point and $\overline{OA} + \overline{OB} + \overline{OC} + \overline{OD} = \mu\overline{OP}$ find μ.**

Sol. Since, P bisects both the diagonals AC and BD, so
$\therefore\ \overline{OA} + \overline{OC} = 2\overline{OP}$ and $\overline{OB} + \overline{OD} = 2\overline{OP}$
(P is the mid point of AC & BD)

$\overline{OA} + \overline{OC} + \overline{OB} + \overline{OD} = 4\overline{OP}$



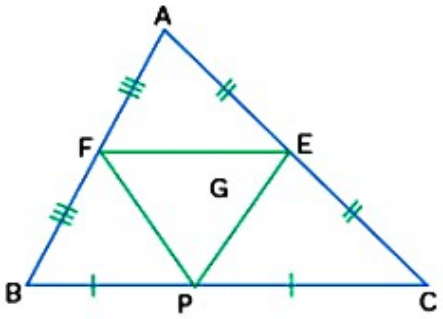
Q. **If mid points of BC, CA, AB are D, E, F then find the position vector of centroid of triangle DEF. Given that when position vector of A, B, C are $\hat{i} + \hat{j}, \hat{j} + \hat{k}$ and $\hat{k} + \hat{i}$ respectively.**

Sol. centroid of ΔABC & ΔDEF are same

$\therefore$ position vector of centroid of

$\Delta DEF = \frac{1}{3}[(\hat{i}+\hat{j})+(\hat{j}+\hat{k})+(\hat{k}+\hat{i})]$

$= \frac{2}{3}(\hat{i}+\hat{j}+\hat{k})$



Q. **Three co-initial vectors of magnitade a, 2a, 3a meet at a point and their directions are along the diagonals of 3 adjacent faces of a cube. Determine their resultant.**

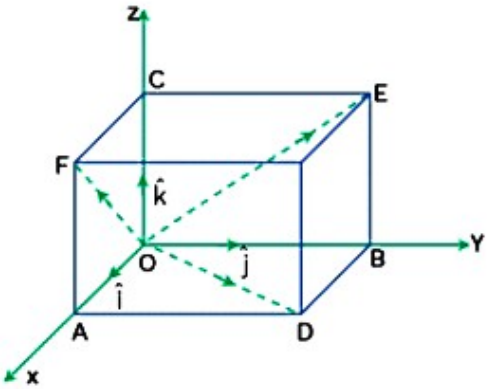
Sol. Consider a unit cube whose one vertex at the origin and three coterminous edges OA, OB and OC along the coordinate exes OX, OY and OZ respectively. Then, $\overline{OA} = \hat{i}, \overline{OB} = \hat{j}$ and $\overline{OC} = \hat{k}$.

Let OD, OE and OF be the diagonals of three adjacent faces of the cube passing through O along which act the vectors of magnitude a, 2a and 3a respectively. We have

$\overline{OD} = \overline{OB} + \overline{BD} = \overline{OB} + \overline{OA} = \hat{j} + \hat{i}$

$\therefore \quad |\overline{OD}| = \sqrt{1+1} = \sqrt{2}$

Thus, the unit vector along $\overline{OD}$ is $\frac{1}{\sqrt{2}}(\hat{i}+\hat{j})$



Similarly, unit vectors along $\overline{OE}$ and $\overline{OF}$ are $\frac{1}{\sqrt{2}}(\hat{j}+\hat{k})$ and $\frac{(\hat{i}+\hat{k})}{\sqrt{2}}$ respectively.

A vector of magnitude 'a' along OD is given by

$\vec{r}_1 = a \times \frac{1}{\sqrt{2}}(\hat{i}+\hat{j}) = \frac{a}{\sqrt{2}}(\hat{i}+\hat{j})$

Similarly, a vector of magnitude 2a and 3a along OE and OF are given by

$\vec{r}_2 = 2a \times \frac{1}{\sqrt{2}}(\hat{j}+\hat{k}) \quad = \frac{2a}{\sqrt{2}}(\hat{j}+\hat{k})$

$\vec{r}_3 = \frac{3a}{\sqrt{2}}(\hat{i}+\hat{k})$ respectively.

Let $\vec{r}$ be the resultant of $\vec{r_1}, \vec{r_2}, \vec{r_3}$.

Then,

$\vec{r} = \vec{r_1} + \vec{r_2} + \vec{r_3}$

$\Rightarrow \quad \vec{r} = \frac{a}{\sqrt{2}}(\hat{i} + \hat{j}) + \frac{2a}{\sqrt{2}}(\hat{j} + \hat{k}) + \frac{3a}{\sqrt{2}}(\hat{k} + \hat{i}) = \frac{a}{\sqrt{2}}(4\hat{i} + 3\hat{j} + 5\hat{k})$,

$\therefore \quad |\vec{r}| = \frac{a}{\sqrt{2}}\sqrt{16 + 9 + 25} = 5a$

Q. **Centre of a Room(cube) is joined to 8 corners of cube Room(cube). 3 vectors out of thus formed 8 vectors are selected at random. Find probability that selected vectors are coplanar.**

Sol. Total ways to select 3 vectors out of 8 = ${}^8C_3$. Favourable number of ways

$= ({}^4C_3) \times 2$

$p = \frac{{}^4C_3 \times 2}{{}^8C_3} = \frac{4 \times 2}{8 \times 7} = \frac{1}{7}$

Q. **ABCDE is regular pentagon.**

$\overline{AB} + \overline{AE} + \overline{BC} + \overline{CD} + \overline{DE} = \lambda\overline{AE}$

Then find the value of λ ?

Sol. $\overline{AB} + \overline{AE} + \overline{BC} + \overline{CD} + \overline{DE} = \lambda\overline{AE}$

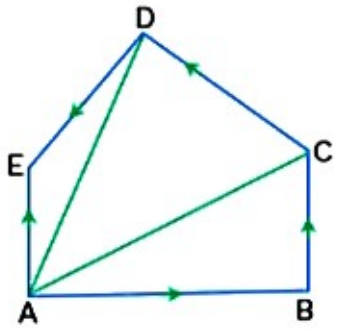
$(\overline{AB} + \overline{BC}) + \overline{CD} + \overline{DE} + \overline{AE} = \lambda\overline{AE}$

$(\overline{AC} + \overline{CD}) + \overline{DE} + \overline{AE} = \lambda\overline{AE}$

$(\overline{AD} + \overline{DE}) + \overline{AE} = \lambda\overline{AE}$

$\overline{AE} + \overline{AE} = \lambda\overline{AE}$

$\lambda = 2$



Q. **In triangle ABC, $\overline{AB} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\overline{BC} = 4\hat{i} + 5\hat{j} + 6\hat{k}$ then find the length of 3rd side and median?**

Sol. In $\Delta ABC \quad \overline{AB} + \overline{BC} + \overline{CA} = \vec{0}$

$(\hat{i}+2\hat{j}+3\hat{k})+(4\hat{i}+5\hat{j}+6\hat{k})+\overline{CA}=\vec{0}$

$(5\hat{i}+7\hat{j}+9\hat{k})+\overline{CA}=\vec{0}$

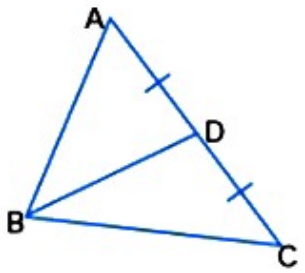
$\overline{CA}=-(5\hat{i}+7\hat{j}+9\hat{k})$

$|\overline{CA}|=\sqrt{25+49+81}=\sqrt{155}$

$\overline{BA}+\overline{BC}=2\overline{BD}$

$\overline{BD}=\frac{1}{2}(-\hat{i}-2\hat{j}-3\hat{k}+4\hat{i}+5\hat{j}+6\hat{k})=\frac{1}{2}(3\hat{i}+3\hat{j}+3\hat{k})$

$|\overline{BD}|=\frac{3}{2}\sqrt{1^2+1^2+1^2}=\frac{3\sqrt{3}}{2}$



Q. **In quadrilateral $\overline{ABCD}$ prove that figure formed by joining the mid-points of consecutive sides is a parallelogram.**

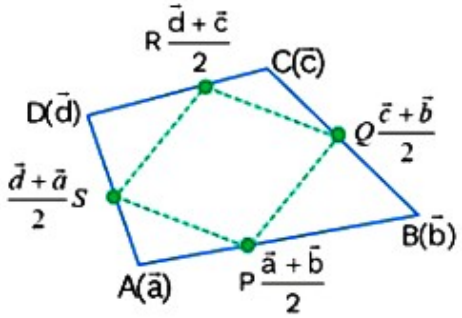
Sol. Let ABCD be a quadrilateral & p, q, r, s are midpoints of sides as shown-

$\overline{PQ}=\frac{\vec{c}+\vec{b}-(\vec{a}+\vec{b})}{2}=\frac{\vec{c}-\vec{a}}{2}$

$\overline{SR}=\frac{\vec{d}+\vec{c}-(\vec{d}+\vec{a})}{2}=\frac{\vec{c}-\vec{a}}{2}$

$\therefore \overline{PQ}=\overline{SR}\Rightarrow PQ=SR$ and $PQ \parallel SR$

$\Rightarrow$ PQRS is a parallelogram



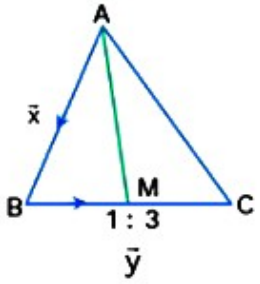
Q. **In triangle ABC, $\overline{AB}=\vec{x}$ and $\overline{BC}=\vec{y}$ then find $\overline{AM}$ in terms of $\vec{x}$ and $\vec{y}$, where M divides BC in 1 : 3.**

Sol. $\overline{BM}=\frac{\vec{y}}{4}$

$\overline{AB}+\overline{BM}+\overline{MA}=\vec{0}$

$\vec{x}+\frac{\vec{y}}{4}=-\overline{MA}=\overline{AM}$

$\therefore \overline{AM}=\vec{x}+\frac{\vec{y}}{4}$



RELATION BETWEEN TWO PARALLEL VECTORS :

1. If $\vec{a}$ and $\vec{b}$ be two parallel vectors, then there exists a non-zero scalar k such that $\vec{a} = k\vec{b}$ i.e. there exist two non-zero scalar quantities x and y so that $x\vec{a} + y\vec{b} = 0$
2. If a and be two non-zero non-parallel vectors then $x\vec{a} + y\vec{b} = 0 \Rightarrow x = 0$ and $y = 0$
3. If $x\vec{a} + y\vec{b} = 0 \Rightarrow \begin{cases} \vec{a} = 0, \vec{b} = 0 \\ \text{or} \\ x = 0, y = 0 \\ \text{or} \\ \vec{a} \parallel \vec{b} \end{cases}$

Point to Remember!!!

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then from the property of parallel vector, we have $\vec{a} \parallel \vec{b}$

$$\Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$$

Q. **The value of λ when $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{b} = 8\hat{i} + \lambda\hat{j} + 4\hat{k}$ are parallel is**

(A) 4 **(B) – 6** **(C) – 12** **(D) 1**

Sol. **(C)**

Since $\vec{a} \parallel \vec{b} \Rightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$

$\therefore \frac{2}{8} = -\frac{3}{\lambda} = \frac{1}{4} \Rightarrow \lambda = -12$

Q. **Find λ if $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{b} = 8\hat{i} + \lambda\hat{j} + 4\hat{k}$ are parallel.**

Sol. $\vec{a} = \mu\vec{b}$

$(2\hat{i} - 3\hat{j} + \hat{k}) = \mu(8\hat{i} + \lambda\hat{j} + 4\hat{k})$

$2\hat{i} - 3\hat{j} + \hat{k} = 8\mu\hat{i} + \mu\lambda\hat{j} + 4\hat{k})$

Comparing, $2 = 8\mu$, $-3 = \mu\lambda$, $1 = 4\mu$

$\mu = \frac{1}{4}, -3 = \frac{1}{4}(\lambda)$

$\Rightarrow \lambda = -12$

Q. **If $A \equiv (2\hat{i}+3\hat{j})$, $B \equiv (p\hat{i}+q\hat{j})$ and $C \equiv (\hat{i}-\hat{j})$ are collinear, find p and q.**

Sol. $\overrightarrow{AB} = (p-2)\hat{i}+(q-3)\hat{j}$

$\overrightarrow{AC} = -\hat{i}-4\hat{j}$

For collinearity, $\overrightarrow{AB} = \lambda(\overrightarrow{AC})$

$(p-2)\hat{i}+(q-3)\hat{j} = \lambda(-\hat{i}-4\hat{j})$

$p-2 = -\lambda \quad q-3 = -4\lambda$

$p = 2-\lambda,\ q = 3-4\lambda$ (Infinite values of p & q)

Linear combination :

A vector $\vec{r}$ is said to be a linear combination of vectors $\vec{a}_1, \vec{a}_2 \ldots, \vec{a}_n$ if there exist scalars $m_1, m_2, \ldots, m_n$ such that $\vec{r} = m_1\vec{a}_1 + m_2\vec{a}_2 + \ldots + m_n\vec{a}_n$.

Linearly Independent :

A system of vectors $\vec{a}_1, \vec{a}_2, \ldots, \vec{a}_n$ is said to be linearly independent if

$m_1\vec{a}_1 + m_2\vec{a}_2 + \ldots\ldots + m_n\vec{a}_n = \vec{0}$

$\Rightarrow m_1 = m_2 = \ldots\ldots = m_n = 0$

Point to Remember!!!

(i) A pair of non-collinear vector is linearly independent.
If $\vec{a}, \vec{b}$ non zero non-collinear vectors such that $x\vec{a} + y\vec{b} = \vec{0}$
$\Rightarrow x = y = 0$
Vector $\vec{a}$ and $\vec{b}$ are called as base vector.

(ii) A triad (set of three) of non-coplanar vector is linearly independent.

Q. **Can $\hat{i}-\hat{j}$ and $2\hat{i}-2\hat{j}$ be treated as the base vectors in xy-plane ?**

Sol. $(\hat{i}-\hat{j})$ and $(2\hat{i}-2\hat{j})$ are collinear vectors, so they cannot be treated as base vectors.

Q. **Can $\hat{i}-\hat{j}$ and $i+\hat{j}$ be treated as the base vectors in xy-plane ?**

Sol. $x(\hat{i}-\hat{j}) + y(\hat{i}+\hat{j}) = \vec{0}$

$(x+y)\hat{i} + (-x+y)\hat{j} = \vec{0}$

$$\Rightarrow x + y = 0 \quad \&$$
$$+ \quad -x + y = 0$$
$$y = 0 \,\&\, x = 0$$

As x = 0 = y so given vectors can be treated as base vector

Result - 1:

If $\vec{a}$ and $\vec{b}$ are two non-collinear vectors, then every vector $\vec{r}$ coplanar with $\vec{a}$ and $\vec{b}$ can be expressed in one and only one way as a linear combination $x\vec{a} + y\vec{b}$; x and y being scalars. Here $\vec{a}$ and $\vec{b}$ are termed as base vectors.

Know the facts



Any vector $\vec{r}$ in xy plane can be taken as $\vec{r} = x\hat{i} + y\hat{j}$.

GEOMETRICAL RESULTS WITH VECTORS:

(i) Straight line joining the mid points of two non parallel sides of a trapezium is parallel to the parallel sides and half of their sum.

(ii) Four diagonals of any parallelopiped (A prism whose base is a||gm) and the join of the mid point of each pair of opposite edges are concurrent and are bisected at the point concurrency. This point is called the centre of the parallelopiped with position vector $\frac{\vec{a}+\vec{b}+\vec{c}}{2}$ i.e. $\frac{\overrightarrow{OA}+\overrightarrow{OB}+\overrightarrow{OC}}{2}$

Point to Remember!!!

Tetrahedron (a pyramid on a triangular base)

(a) Lines joining the vertices of a tetrahedron to the centroids of the opposite faces are concurrent and this point (P) of concurrency with position vector $\vec{g} = \frac{\vec{a}+\vec{b}+\vec{c}+\vec{d}}{4}$ is called the centre of the tetrahedron (say G).

(b) In a tetrahedron straight lines joining the mid points of each pair of opposite edges are also concurrent at the centre of the tetrahedron.

Q. **Centre of the parallelopiped formed by $\overline{PA} = \hat{i} + 2\hat{j} + 2\hat{k}, \overline{PB} = (4\hat{i}, -3\hat{j}, 1\hat{k})$, $\overline{PC} = (3\hat{i}, 5\hat{j}, -1\hat{k})$ is (7, 6, 2). Find position vector of 'P'.**

Sol. $P(\alpha, \beta, \gamma)$

$A(1 + \alpha, 2 + \beta, 2 + \gamma)$

$B(4 + \alpha, -3 + \beta, 1 + \gamma)$

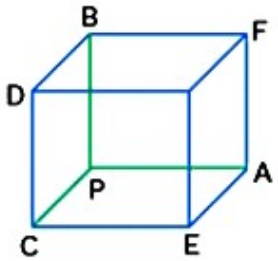
$C(3 + \alpha, 5 + \beta, -1 + \gamma)$

$D(7 + \alpha, 2 + \beta, \gamma)$

$F(5 + \alpha, -1 + \beta, 3 + \gamma)$

$G(8 + \alpha, 4 + \beta, 2 + \gamma)$

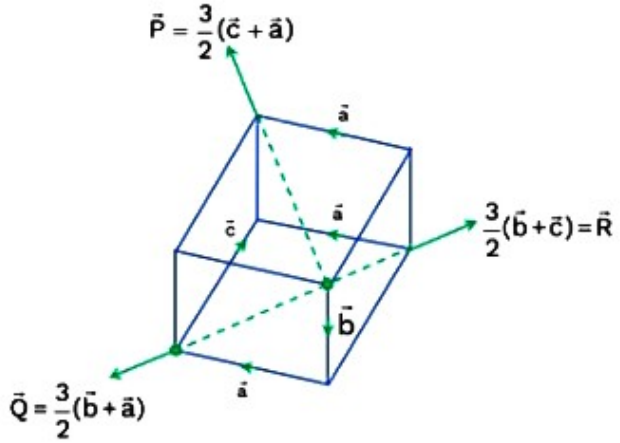
$E(4 + \alpha, 7 + \beta, 1 + \gamma)$



Average $\left(\frac{32 + 8\alpha}{8}, \frac{16 + 8\beta}{8}, \frac{8 + 8\gamma}{8}\right) = (7, 6, 2) \Rightarrow \alpha = 3, \beta = 4, \gamma = 1$

Q. **The diagonals of the three faces of the parallelopiped drawn from the same vertex are prolonged half their lengths. Show that the three points thus obtained are coplanar with the opposite vertex.**

Sol. Diagonal vector is $(\vec{a} + \vec{b} + \vec{c}) = \vec{d}$.



$\vec{d} - \vec{P} = \vec{b} - \frac{\vec{c}}{2} - \frac{\vec{a}}{2}$

$\vec{d} - \vec{Q} = \vec{c} - \frac{\vec{a}}{2} - \frac{\vec{b}}{2}$

$\vec{d} - \vec{R} = \vec{a} - \frac{\vec{b}}{2} - \frac{\vec{c}}{2}$

$\vec{b} - \frac{\vec{c}}{2} - \frac{\vec{a}}{2} = x\left(\vec{c} - \frac{\vec{a}}{2} - \frac{\vec{b}}{2}\right) + y\left(\vec{a} - \frac{\vec{b}}{2} - \frac{\vec{c}}{2}\right)$

$1 = -\frac{x}{2} - \frac{y}{2} \quad \frac{-1}{2} = x - \frac{y}{2} \quad \frac{-1}{2} = \frac{x}{-2} + y$

$x + y = -2 - 0 \quad ...(i) \quad x = \frac{y - 1}{2} \quad ...(ii) \quad y = \frac{x}{2} - \frac{1}{2} \quad ...(iii)$

$\frac{y - 1}{2} + y = -2 \qquad x = \frac{-1 - 1}{2}$

$y - 1 + 2y = -4 \qquad x = -1$

$3y = -3$

$y = -1$

Values of x & y from (i) & (ii) satisfy (iii)

Hence three points thus obtained are co-planar with the opposite vertex 1

VECTOR EQUATION OF A STRAIGHT LINE:

Vector equation of a straight line passing through a given point $A(\vec{a})$ and parallel to a given vector $\vec{b}$:

Let O be the origin. Let the line pass through a given point A whose position vector is $\vec{a}$, then $\overline{OA} = \vec{a}$.

Let the given line be parallel to vector $\vec{b}$

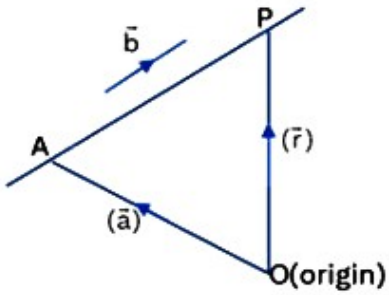
Let $\vec{r}$ be the position vector of any point P on the line, then

$\overline{OP} = \vec{r}$

Since $\overline{AP}$ is parallel to $\vec{b}$ $\therefore \overline{AP} = t\vec{b}$, where t is a scalar.

Now $\overline{OP} = \overline{OA} + \overline{AP}$ $\therefore \vec{r} = \vec{a} + t\vec{b}$ ... (i)

Since for different values of t, we get different positions of point P on the line, hence (i) is the vector equation of the required straight line.



Vector equation of straight line passing through two given point $A(\vec{a})$ and $B(\vec{b})$:

Let O be the origin. Let the line pass through two given point A and B whose position vectors referred to O be $\vec{a}$ and $\vec{b}$ respectively,

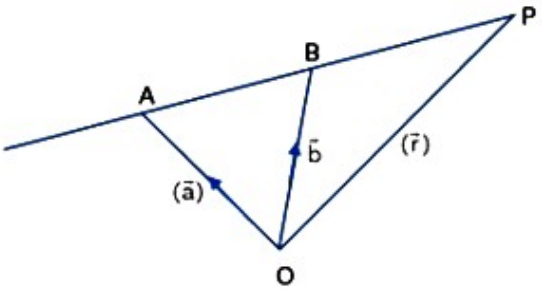
then $\overline{OA} = \vec{a}$ and $\overline{OB} = \vec{b}$

$\therefore \overline{AB} = \overline{OB} - \overline{OA} = \vec{b} - \vec{a}$

Clearly, the required line passes through $A(\vec{a})$ and is parallel to the vector $(\vec{b} - \vec{a})$.

Hence the vector equation of the required line is,

$\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$ or $\vec{r} = (1-t)\vec{a} + t\vec{b}$



Point to Remember!!!

(i) Two lines in a plane are either intersecting or parallel conversely two intersecting or parallel lines must be in the same plane

(ii) However in space we can have two neither parallel nor intersecting lines. Such non coplanar lines are known as skew lines. If two lines are parallel and have a common point then they are coincident.

Q. **Find the p.v. of the point of intersection of the lines**

(i) and $\left.\begin{array}{l}\vec{r} = \hat{i} - \hat{j} - 10\hat{k} + \lambda(2\hat{i} - 3\hat{j} + 8\hat{k}) \\ \vec{r} = 4\hat{i} - 3\hat{j} - \hat{k} + \mu(\hat{i} - 4\hat{j} + 7\hat{k})\end{array}\right]$

(ii) and $\left.\begin{array}{l}\vec{r} = -3\hat{i} + 6\hat{j} + \lambda(-4\hat{i} - 3\hat{j} + 2\hat{k}) \\ \vec{r} = -2\hat{i} - 7\hat{k} + \mu(-4\hat{i} - \hat{j} + \hat{k})\end{array}\right]$

(iii) and $\left.\begin{array}{l}\vec{r} = t(3\hat{i} - \hat{j} + \hat{k}) \\ \vec{r} = -2\hat{i} - s(-6\hat{i} - 2\hat{j} - 2\hat{k})\end{array}\right]$

(iv) and $\left.\begin{array}{l}\vec{r} = 2\hat{k} + \lambda(3\hat{i} + 2\hat{j} + \hat{k}) \\ \vec{r} = -3\hat{i} - 2\hat{j} + 3\hat{k} + \mu(6\hat{i} + 4\hat{j} + 2\hat{k})\end{array}\right]$

Sol. **(i)** $L_1 : \vec{r} = (2\lambda + 1)\hat{i} - (1 + 3\lambda)\hat{j} + (8\lambda - 10)\hat{k}$

$L_2 : \vec{r} = (4 + \mu)\hat{i} - (4\mu + 3)\hat{j} + (7\mu - 1)\hat{k}$

The given lines are not parallel. For coplanarity, the lines must intersect.

$\therefore \quad (2\lambda + 1)\hat{i} - (1 + 3\lambda)\hat{j} + (8\lambda - 10)\hat{k} = (4 + \mu)\hat{i} - (4\mu + 3)\hat{j} + (7\mu - 1)\hat{k}$

$2\lambda + 1 = 4 + \mu$ ...(i)

$1 + 3\lambda = 4\mu + 3$ ...(ii)

$8\lambda - 10 = 7\mu - 1$ ...(iii)

Solving (i) & (ii), $\lambda = 2$, $\mu = 1$ and $\lambda = 2$, $\mu = 1$ satisfies equation (iii) Given lines are intersecting & hence coplanar.

put $\mu = 1$ in $L_2 : \vec{r} = 5\hat{i} - 7\hat{j} + 6\hat{k}$

(ii) $-3\hat{i} + 6\hat{j} - 4\lambda\hat{i} - 3\lambda\hat{j} + 2\lambda\hat{k} = -2\hat{i} - 7\hat{k} + (-4\mu)\hat{i} - \mu\hat{j} + \mu\hat{k}$

$-3 - 4\lambda = -2 - 4\mu$

$4(\mu - \lambda) = 1$

$\mu = \frac{1}{4} + \lambda$ ...(i)

$6 - 3\lambda = -\mu$ ...(ii)

$6 - 3\lambda = -\frac{1}{4} - \lambda$

$6 + \frac{1}{4} = 2\lambda$

$\lambda = \frac{25}{8} \qquad \mu = \frac{1}{4} + \frac{25}{8} = \frac{27}{8}$

Also

$2\lambda = -7 + \mu$ ...(iii)

values of λ & μ from (i) & (ii) does not satisfy (iii)

Hence they are not intersecting

(iii) $3t\hat{i} - t\hat{j} + t\hat{k} = -2\hat{i} + 6s\hat{i} + 2s\hat{j} + 2s\hat{k}$

$3t = (-2 + 6s)$...(i) $\quad -t = 2s$ ...(ii) $\quad t = 2s$ ...(iii)

$3(-2s) = -2 + 6s$

$2 = 12s$

$s = \frac{1}{6} \quad \therefore\ t = -2\left(\frac{1}{6}\right)$

$t = -\frac{1}{3}$

value of s & t from (i) & (ii) does not satisfy (iii)

So lines are not coplanar

(iv) lines are parallel

Q. **Find the vector equation of the line through the point $2\hat{i} + \hat{j} - 3\hat{k}$ and parallel to the vector $\hat{i} + 2\hat{j} + \hat{k}$.**

Sol. Let the given point be $A(\vec{a})$ and given vector be $\vec{b}$ and O be the origin.

Then, $\vec{a} = \overrightarrow{OA} = 2\hat{i} + \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

Now vector equation of the line through A and parallel to $\vec{b}$ is

$\vec{r} = \vec{a} + t\vec{b}$, where t is a scalar.

or ; $\vec{r} = 2\hat{i} + \hat{j} - 3\hat{k} + t\left(\hat{i} + 2\hat{j} + \hat{k}\right)$

Q. **In a triangle ABC, D divides BC in the ratio 3 : 2 and E divides CA in the ratio 1 : 3. The lines AD and BE meet at H and CH meets AB in F. Find the ratio in which F divides AB.**

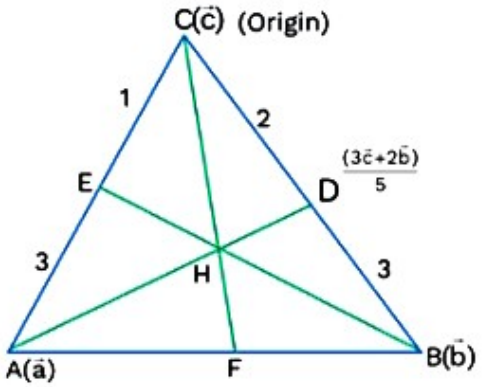
Sol. Take C as the origin of reference.

Let- $\vec{a}$, $\vec{b}$ be the position vectors of the points A and B respectively. The equation of AD and BE are

$\vec{r} = \vec{a} + k\left(\frac{2}{5}\vec{b} - \vec{a}\right)$ and $\vec{r} = \vec{b} + t\left(\frac{1}{4}\vec{a} - \vec{b}\right)$

respectively. For the point of intersection H, we have

$\vec{a} + k\left(\frac{2}{5}\vec{b} - \vec{a}\right) = b + t\left(\frac{1}{4}\vec{a} - \vec{b}\right)$ ...(i)



$\Rightarrow \quad 1-k=\frac{t}{4}, \frac{2}{5}k=1-t \Rightarrow k=\frac{5}{6}; t=\frac{2}{3}$

On putting these values of k and t in the equations of the lines AD and BE, we see that the point H is $\frac{1}{6}a+\frac{1}{3}b$. The equations of CH and AB being $\vec{r}=\vec{t}\left(\frac{\vec{a}}{6}+\frac{\vec{b}}{3}\right)$ and $r=\vec{a}+k(\vec{b}-\vec{a})$ respectively.

For the point of intersection F, $t\left(\frac{\vec{a}}{6}+\frac{\vec{b}}{3}\right)=\vec{a}+\vec{k}(\vec{b}-\vec{a})$

$\Rightarrow \quad \left(\frac{t}{6}-1+k\right)=0, \left(\frac{t}{3}-k\right)=0$

These give $t = 2; \qquad k=\frac{2}{3}$

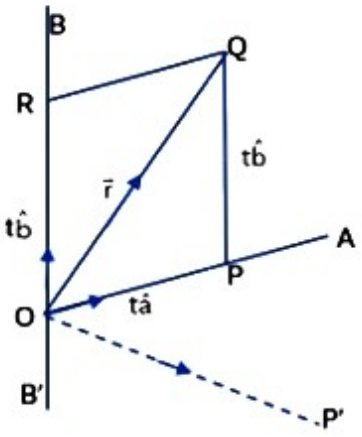
Thus, the point F is $\frac{\vec{a}+2\vec{b}}{3}$

So, that F divides AB in the ratio 2 : 1

Vector equation of bisectors of angle between two straight lines :

Let OA and OB be the given straight line parallel to unit vectors $\hat{a}$ and $\hat{b}$ respectively.

Take the point O as origin, and let Q be a point on the internal bisector of the angle AOB. From Q draw QR parallel to OA cutting OB at R.



Now $\therefore \angle AOQ = \angle BOQ$ (as OQ is the bisector)

and $\angle BOQ = \angle OQP$ (alternative angles)

$\therefore \angle AOQ = \angle OQP \qquad \therefore OP = QP = t$ (say),

where t is a scalar.

$\therefore \overline{OP} = t\hat{a}$ and $\overline{PQ} = t\hat{b}$

$\overline{OQ} = \vec{r}$

Let $\overline{OQ} = \overline{OP} + \overline{PQ}$

or $\vec{r} = t\hat{a} + t\hat{b}$

$\vec{r} = t\left(\hat{a}+\hat{b}\right)$

$$\vec{r} = t\left(\frac{\vec{a}}{a} + \frac{\vec{b}}{b}\right)$$

where $|\vec{a}| = a, |\vec{b}| = b$

This is the equation of internal bisector of $\angle AOB$.

Corollary:

If the lines intersect at E having position vector $\vec{\alpha}$, then the above equations becomes

$\vec{r} = \vec{\alpha} + t\left(\hat{a} + \hat{b}\right)$ and $\vec{r} = \vec{\alpha} + t\left(\hat{a} - \hat{b}\right)$

respectively.

Point to Remember!!!

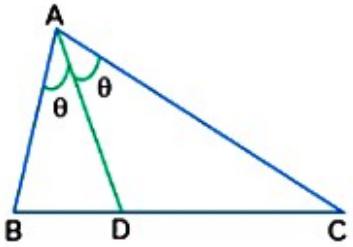
Equation of external Bisector:

If OP' be the external bisector of $\angle OAB$, then OP' may be regarded as the internal bisector of the angle between the lines which are parallel to $\hat{a}$ and $-\hat{b}$ Hence its equation is

$\vec{r} = t\left(\hat{a} - \hat{b}\right)$ or, $\vec{r} = t\left(\frac{\vec{a}}{a} - \frac{\vec{b}}{b}\right)$

Q. **Use vectors to prove that the internal (external) bisectors of a triangle divide the opposite base internally (externally)in the ratio of the side containing the angle.**

Sol. Let's suppose $\angle BAD = \angle CAD = \theta$, where θ is acute,



and

$|\overline{BD}| : |\overline{CD}| = r : s$. Using the section formula,

$$\overline{AD} = \frac{r}{r+s}\overline{AC} + \frac{s}{r+s}\overline{AB}.$$

Then by dot product,

$$\overline{AD}\cdot\overline{AC} = |\overline{AD}||\overline{AC}|\cos\theta = \frac{r}{r+s}|\overline{AC}|^2 + \frac{s}{r+s}\overline{AB}\cdot\overline{AC}$$

$$\Rightarrow |\overline{AD}|\cos\theta = \frac{r}{r+s}|\overline{AC}| + \frac{s}{r+s}|\overline{AB}|\cos 2\theta$$ Similarly, by considering $\overline{AD}\cdot\overline{AB}$, we have

$$|\overline{AD}|\cos\theta = \frac{r}{r+s}|\overline{AC}|\cos 2\theta + \frac{s}{r+s}|\overline{AB}|$$

Combining 1 and 2 gives

$$\frac{r}{r+s}|\overline{AC}| + \frac{s}{r+s}|\overline{AB}|\cos 2\theta = \frac{r}{r+s}|\overline{AC}|\cos 2\theta + \frac{s}{r+s}|\overline{AB}|$$

$$\frac{r}{r+s}|\overline{AC}|(1-\cos 2\theta) = \frac{s}{r+s}|\overline{AB}|(1-\cos 2\theta)$$

$$\frac{|\overline{AB}|}{|\overline{AC}|} = \frac{r}{s} = \frac{|\overline{BD}|}{|\overline{CD}|}.$$

Q. **Prove, by vector method that the internal bisectors of the angles of a triangle are concurrent.**

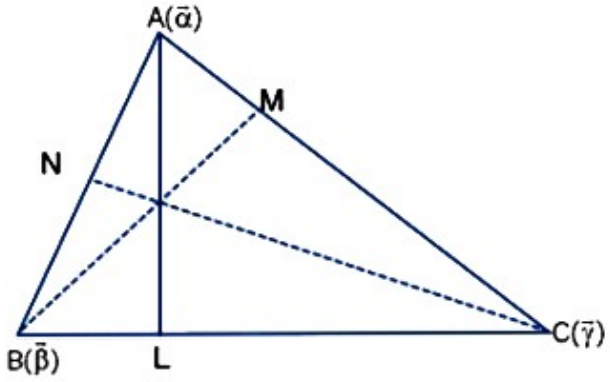
Sol. Let $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ be the position vectors of vertices A, B, C respectively of ΔABC.

Let AL be the bisectors of $\angle BAC$

Then $\frac{BL}{LC} = \frac{BA}{AC} = \frac{c}{b}$

Thus L divides BC internally in the ratio c : b

P.V. of L = $\frac{b\vec{\beta} + c\vec{\gamma}}{b + c}$



$\therefore$ Now P.V. of the point which divide AL internally in the ratio b + c : a will be $\frac{a\vec{\alpha} + b\vec{\beta} + c\vec{\gamma}}{a + b + c}$

Here we have divided AL in the ratio b + c : a because b + c occurs in the denominator of P.V. of L and there is $b\vec{\beta}$ and $c\vec{\gamma}$ therefore, there should also be $a\vec{\alpha}$

Similarly, we can show that the position vectors of the points which divide bisectors BM of $\angle ABC$ and CN of $\angle ACB$ in the ratio c + a : b and a + b : c respectively will be each $\frac{a\vec{\alpha} + b\vec{\beta} + c\vec{\gamma}}{a + b + c}$. Thus the point having position vector $\frac{a\vec{\alpha} + b\vec{\beta} + c\vec{\gamma}}{a + b + c}$ lies on the three internal bisectors of the angles of the triangle ABC and hence internal bisectors of angles of a triangle are concurrent and the position vector of incentre of ΔABC will be $\frac{a\vec{\alpha} + b\vec{\beta} + c\vec{\gamma}}{a + b + c}$.

COLLINEARITY OF THREE POINTS :

1. If $\vec{a}, \vec{b}, \vec{c}$ be position vectors of three points A, B and C respectively and x, y, z be three scalars so that all are not zero, then the necessary and sufficient conditions for three points to be collinear is that
   $x\vec{a} + y\vec{b} + z\vec{c} = 0$ where $x + y + z = 0$
2. Three points A, B and C are collinear, if $\overrightarrow{AB} = \lambda\overrightarrow{BC}$.

Q. **If $2\vec{a}-3\vec{b},\vec{b}$ and $\vec{a}-\vec{b}$ are position vectors of three points A, B and C then they are**

(A) Collinear **(B) Non- collinear**

(C) Can't say anything **(D) None of these**

Sol. **(A)**

$\because 1\left(2\vec{a}-3\vec{b}\right)+1\left(\vec{b}\right)-2\left(\vec{a}-\vec{b}\right)=0$ and $1+1-2=0$

So the given vectors are collinear.

Q. **If $A\equiv\left(2\hat{i}+3\hat{j}\right),B\equiv\left(p\hat{i}+9\hat{j}\right)$ and $C\equiv\left(\hat{i}-\hat{j}\right)$ are collinear, then the value of p is**

(A) 1 / 2 **(B)** 3 / 2 **(C)** 7 / 2 **(D)** 5 / 2

Sol. **(C)**

$\overline{AB}=(p-2)\hat{i}+6\hat{j},\overline{AC}=-\hat{i}-4\hat{j}$

Now A,B,C are collinear $\Leftrightarrow \overline{AB}\parallel\overline{AC}\Leftrightarrow\frac{p-2}{-1}=\frac{6}{-4}\Leftrightarrow p=7/2$

Q. **Find whether the following points are collinear or not**

(i) $2\hat{i}+5\hat{j}-4\hat{k};\hat{i}+4\hat{j}-3\hat{k};4\hat{i}+7\hat{j}-6\hat{k}$

(ii) $3\hat{i}-4\hat{j}+3\hat{k};-4\hat{i}+5\hat{j}-6\hat{k};4\hat{i}-7\hat{j}+6\hat{k}$

Sol. **(i)** $(2\hat{i}+5\hat{j}-4\hat{k})-(\hat{i}+4\hat{j}-3\hat{k})=\lambda(4\hat{i}+7\hat{j}-6\hat{k}-(\hat{i}+4\hat{j}-3\hat{k}))$

$\hat{i}+\hat{j}-\hat{k}=\lambda(3\hat{i}+3\hat{j}-3\hat{k})$ $\quad 1=3\lambda\Rightarrow\lambda=\frac{1}{3}\Rightarrow$ collinear

(ii) $[(3\hat{i}-4\hat{j}+3\hat{k})-(-4\hat{i}+5\hat{j}-6\hat{k})]=\lambda(4\hat{i}-7\hat{j}+6\hat{k}-(-4\hat{i}+5\hat{j}-6\hat{k})]$

$(7\hat{i}-9\hat{j}+9\hat{k})=\lambda(8\hat{i}-12\hat{j}+12\hat{k})$

Value of λ can't be calculated uniquely $\Rightarrow$ not colinear

Q. **Given $\left(\sin^2\theta\right)\vec{a}+(2\sin\theta)\vec{b}+\vec{c}=\vec{0}$ if $A(\vec{a}),B(\vec{b}),c(\vec{c})$ are collinear. Find θ.**

Sol. $\sin^2\theta+2\sin\theta+1=0$

$(\sin\theta+1)^2=0\Rightarrow\sin\theta=-1$

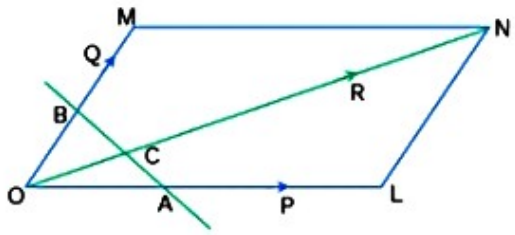
$\theta=n\pi+(-1)^n\left(-\frac{\pi}{2}\right),n\in z$

Q. **Vectors $\vec{P}, \vec{Q}$ act at 'O' (origin) have a resultant $\vec{R}$. If any transversal cuts their line of action at A, B, C respectively, then show that $\frac{P}{OA} + \frac{Q}{OB} = \frac{R}{OC}$.**

Sol. $\therefore \vec{P} + \vec{Q} = \vec{R}$ ...(1)

Again let $\vec{P} = \overline{OL} = m\overline{OA}$, $\vec{Q} = \overline{OM} = n\overline{OB}$

and $R = \overline{ON} = t \cdot \overline{OC}$



$\therefore \quad m = \frac{P}{OA}; n = \frac{Q}{OB}; t = \frac{R}{OC}$ ...(2)

Hence from (1), we get

$m\cdot\overline{OA} + n\cdot\overline{OB} = t\cdot\overline{OC}$

or $m \cdot \overline{OA} + n \cdot \overline{OB} - t \cdot \overline{OC} = 0$ ...(3)

But A,B,C are collinear and we know that if there exists a relation of the form $x\mathbf{a} + y\mathbf{b} + z\mathbf{c} = 0$ between the P.V's of three collinear points then $x + y + z = 0$.

Hence from (3) we must have

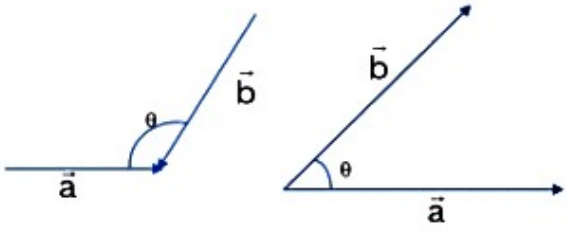
$m + n - t = 0 \quad \therefore m + n = t.$

Now putting the value of m, n and t from (2) in $m + n = t$ we get $\frac{P}{OA} + \frac{Q}{QB} = \frac{R}{OC}$.

PRODUCT OF VECTORS:

Product of two vectors is done by two methods when the product of two vectors results in a scalar quantity then it is called scalar product. It is also called as dot product because this product is represented by putting a dot.

When the product of two vectors results in a vector quantity then this product is called Vector Product. This product is represented by (x) sign so that it is also called as cross product.



Scalar or dot product of two vectors:

Definition :

If $\vec{a}$ and $\vec{b}$ are two vectors and θ be the angle between their tails or heads, then their scalar product (or dot product) is defined as the number $|\vec{a}||\vec{b}|\cos\theta$ where $|\vec{a}|$ and $|\vec{b}|$ are

moduli of $\vec{a}$ and $\vec{b}$ respectively and $0 \le \theta \le \pi$. It is denoted by $\vec{a} \cdot \vec{b}$.
Thus $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$

Note:

(i) $\vec{a} \cdot \vec{b} \in R$

(ii) $\vec{a} \cdot \vec{b} \le |\vec{a}||\vec{b}|$

(iii) $\vec{a} \cdot \vec{b} > 0 \Rightarrow$ angle between $\vec{a}$ and $\vec{b}$ say $\theta \in \left[0, \frac{\pi}{2}\right)$.

$\vec{a} \cdot \vec{b} < 0 \Rightarrow$ angle between $\vec{a}$ and $\vec{b}$ say $\theta \in \left(\frac{\pi}{2}, \pi\right]$

$\vec{a} \cdot \vec{b} = 0 \Rightarrow$ angle between $\vec{a}$ and $\vec{b}$ say $\theta = \frac{\pi}{2}$ or atleast one of $\vec{a}$ and $\vec{b}$ is zero vector.

(iv) The dot product of a zero and non-zero vector is a scalar zero i.e. $\vec{0} \cdot \vec{a} = 0$.

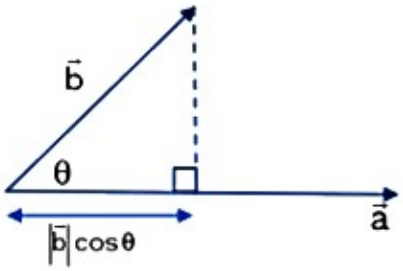
(v) If θ be angle between any two non zero vector $\vec{a}$ and $\vec{b}$ then $\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$.

Geometrical Interpretation:

Geometrically, the scalar product of two vectors is equal to the product of the magnitude of one and the projection of second in the direction of first vector i.e. $\vec{a} \cdot \vec{b} = |\vec{a}|\left(|\vec{b}|\cos\theta\right) = |\vec{a}|$ (projection of $\vec{b}$ in the direction of $\vec{a}$)

Similarly $\vec{a} \cdot \vec{b} = |\vec{b}|\left(|\vec{a}|\cos\theta\right)$

$= |\vec{b}|$ (projection of $\vec{a}$ in the direction of $\vec{b}$)



Here projection of $\vec{b}$ on $\vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|}$

Projection of $\vec{a}$ on $\vec{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

Q. **If angle between $\vec{a}$ and $\vec{b}$ is 120° and their magnitudes are respectively 2 and $\sqrt{3}$, then $\vec{a}\cdot\vec{b}$ equals**

(A) 3 **(B) $-\sqrt{3}$** **(C) $\sqrt{3}$** **(D) – 3**

Sol. **(B)**

We know that $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$

$= 2\sqrt{3}\cos 120^\circ$

$= 2\sqrt{3}(-1/2) = -\sqrt{3}$

Q. **The projection of vector $\hat{i}+\hat{j}+\hat{k}$ on the vector $\hat{i}-\hat{j}+\hat{k}$ is**

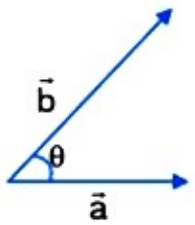
(A) $\sqrt{3}$ **(B) $1/\sqrt{3}$** **(C) $2/\sqrt{3}$** **(D) $2\sqrt{3}$**

Sol. **(B)**

$$\text{Projection} = \frac{(\hat{i}+\hat{j}+\hat{k})\cdot(\hat{i}-\hat{j}+\hat{k})}{|\hat{i}-\hat{j}+\hat{k}|} = \frac{1-1+1}{\sqrt{1+1+1}} = \frac{1}{\sqrt{3}}$$

Scalar product in particular cases :

1. $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}$
2. $\vec{a}\cdot(\vec{b}+\vec{c}) = \vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}$
3. $(ma)\cdot\vec{b} = m(\vec{a}\cdot\vec{b}) = \vec{a}\cdot(m\vec{b})$
4. If $\theta = 0 \Rightarrow \vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|$ (like vectors)
5. If $\theta = \pi \Rightarrow \vec{a}\cdot\vec{b} = -|\vec{a}||\vec{b}|$ (unlike vectors)
6. If $\hat{a}$ and $\hat{b}$ are unit vectors then $\hat{a}\cdot\hat{b} = \cos\theta$ (where θ is angle between them).
7. $\vec{a}\cdot\vec{a} = |\vec{a}|^2 \Rightarrow |\vec{a}| = \sqrt{\vec{a}\cdot\vec{a}}$
8. If $\vec{a}\perp\vec{b} \Rightarrow \vec{a}\cdot\vec{b} = 0$ but $\vec{a}\cdot\vec{b} = 0 \Rightarrow \vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a}\perp\vec{b}$.
9. If $\hat{i},\hat{j}$ and $\hat{k}$ are unit vectors along the rectangular coordinate OX, OY and OZ then $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1, \hat{i}\cdot\hat{j} = \hat{j}\cdot\hat{k} = \hat{k}\cdot\hat{i} = 0$



10. $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c} \not\Rightarrow \vec{b}=\vec{c}$

Infact $\vec{a}\cdot\vec{b}=\vec{a}\cdot\vec{c}\Rightarrow\vec{a}\cdot(\vec{b}-\vec{c})=0\Rightarrow\vec{a}=0$ or $\vec{b}=\vec{c}$ or $\vec{a}\perp(\vec{b}-\vec{c})$

11. $(\vec{a}\cdot\vec{b})\cdot\vec{c}$ is meaningless.

Point to Remember!!!

(a) $(\vec{a}\cdot\vec{b})\cdot\vec{b}$ is not defined

(b) $(\vec{a}+\vec{b})^2=|\vec{a}|^2+2\vec{a}\cdot\vec{b}+|\vec{b}|^2$

(c) $(\vec{a}-\vec{b})^2=|\vec{a}|^2-2\vec{a}\cdot\vec{b}+|\vec{b}|^2$

(d) $(\vec{a}+\vec{b})\cdot(\vec{a}-\vec{b})=|\vec{a}|^2-|\vec{b}|^2$

(e) $|\vec{a}+\vec{b}|=|\vec{a}|+|\vec{b}|\Rightarrow\vec{a}\parallel\vec{b}$

(f) $|\vec{a}+\vec{b}|^2=|\vec{a}|^2+|\vec{b}|^2\Rightarrow\vec{a}\perp\vec{b}$

(g) $|\vec{a}+\vec{b}|=|\vec{a}-\vec{b}|\Rightarrow\vec{a}\perp\vec{b}$

Q. **If the sum of two unit vectors is a unit vector then find the magnitude of their difference and the angle between $\hat{a}$ and $\hat{b}$.**

Sol. $|\hat{a}+\hat{b}|=1\Rightarrow|\hat{a}+\hat{b}|^2=1$

$|\hat{a}|^2+|\hat{b}|^2+2\hat{a}\cdot\hat{b}=1$

$1+1+2|\hat{a}||\hat{b}|\cos\theta=1$

(θ = Angle between $\hat{a}+\hat{b}$)

$1+2(1)(1)\cos\theta=0$

$\cos\theta=-\frac{1}{2}\Rightarrow\theta=\frac{2\pi}{3}.$

$|\hat{a}-\hat{b}|=\sqrt{|\hat{a}|^2+|\hat{b}|^2-2\hat{a}\cdot\hat{b}}$

$=\sqrt{(1)^2+(1)^2-2|\hat{a}||\hat{b}|\cos\theta}$

$=\sqrt{1+1-(1)(1)(-1)}$

$=\sqrt{1+1+1}=\sqrt{3}$ units.

Q. **Determine the values of c such that for all x (real) the vectors $cx\hat{i}-6\hat{j}+3\hat{k}$ and $x\hat{i}+2\hat{j}+2cx\hat{k}$ make an obtuse angle with each other.**

Sol. If θ is the angle between the given vectors then

$$\cos\theta = \frac{cx^2 - 12 + 6cx}{\sqrt{c^2x^2 + 45}\sqrt{x^2 + 4c^2x^2 + 4}}$$

If θ is obtuse then $\cos\theta < 0 \Rightarrow cx^2 + 6cx - 12 < 0 \forall x \in R$

Which is possible if $c < 0$ and $36c^2 + 48c < 0 \Rightarrow c < 0$ and $12c(3c + 4) < 0$

$\Rightarrow c < 0$ and $-\frac{4}{3} < c < 0$ (but for $c = 0, cx^2 + 6cx - 12 < 0 \ \forall x \in R$)

Hence $-\frac{4}{3} < c \leq 0$

Q. **If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors such that $\vec{a}$ is perpendicular to the plane of $\vec{b}$ and $\vec{c}$ then find $|\vec{a} + \vec{b} + \vec{c}|$ when the angle between $\vec{b}$ and $\vec{c}$ is $\pi / 3$.**

Sol. We have $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1, \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$ and $\vec{b} \cdot \vec{c} = \cos\frac{\pi}{3} = \frac{1}{2}$

Now $|\vec{a} + \vec{b} + \vec{c}|^2 = (\vec{a} + \vec{b} + \vec{c}) \cdot (\vec{a} + \vec{b} + \vec{c}) = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2\vec{b} \cdot \vec{c} \ \ (\because \vec{a} \cdot \vec{c} = \vec{a} \cdot \vec{b} = 0)$

$$= 1 + 1 + 1 + 2 \cdot \frac{1}{2} = 4 \Rightarrow |\vec{a} + \vec{b} + \vec{c}| = 2$$

Scalar product in terms of components:

Let $\vec{a}$ and $\vec{b}$ be two vectors such that

$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

Then $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

In particular

$\vec{a} \cdot \vec{a} = |\vec{a}|^2 = a_1^2 + a_2^2 + a_3^2$

For any vector $\vec{a}$

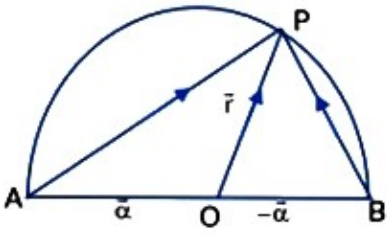
$\vec{a} = (\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k}$

Q. **If $\vec{a} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 5\hat{k}$ then find $\vec{a} \cdot \vec{b}$.**

Sol. $\vec{a} \cdot \vec{b} = (3)(1) + (2)(-2) + (1)(5) = 3 - 4 + 5 = 4$

Q. **Prove that the angle in a semi circle is a right angle.**

Sol. Let O be the centre of the semi circle with AOB as its diameter. Let P be a point on the semi-circle, so that $\angle APB$ is an angle in the semi circle. Join OP. Let O be taken as origin. Let the position vectors of A, B and P be $\vec{\alpha}, -\vec{\alpha}$ and $\vec{r}$ respectively.



Clearly, $OA = OB = OP$

Now $\overrightarrow{AP} = (\vec{r} - \vec{\alpha})$ and $\overrightarrow{BP} = (\vec{r} + \vec{\alpha})$

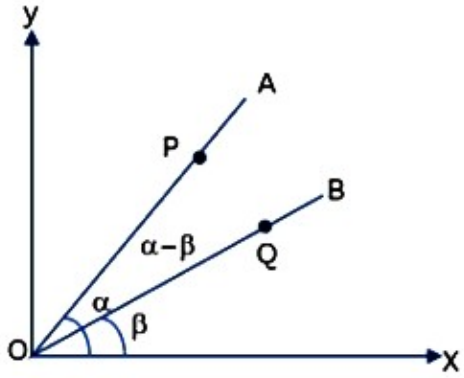
$\therefore \quad \overrightarrow{AP} \cdot \overrightarrow{BP} = (\vec{r} - \vec{\alpha}) \cdot (\vec{r} + \vec{\alpha}) = r^2 - \alpha^2 = OP^2 - OA^2 = 0$

$[\because OP = OA]$

$\therefore$ AP $\perp$ BP i.e. $\angle APB = 90°$.

Q. **Prove by vector method the following formula of plane trigonometry**
$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

Sol. Let the unit vectors along OX and OY be $\hat{i}$ and $\hat{j}$ respectively. If OA and OB be any two lines in the same plane making angles α and β respectively, with OX then, $\angle AOB = \alpha - \beta$



Again, let $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ represent unit vectors along OA and OB respectively, so that their dot product is the cosine of the angle between their directions.

Now, $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 1 \cdot 1\cos(\alpha - \beta) = \cos(\alpha - \beta)$ ...(1)

Since $\overrightarrow{OP}$ makes an angle α with x-axis.

$\therefore \quad \overrightarrow{OP} = \cos\alpha\hat{i} + \sin\alpha\hat{j}$

Similarly, $\overrightarrow{OQ} = \cos\beta\hat{i} + \sin\beta\hat{j}$

$\therefore \overrightarrow{OP} \cdot \overrightarrow{OQ} = \left[\cos\alpha\hat{i} + \sin\alpha\hat{j}\right] \cdot \left[\cos\beta\hat{i} + \sin\beta\hat{j}\right] = \cos\alpha \cdot \cos\beta + \sin\alpha \cdot \sin\beta \quad \ldots(2)$

From (1) and (2), we get $\cos(\alpha - \beta) = \cos\alpha \cdot \cos\beta + \sin\alpha \cdot \sin\beta$

Q. **In any ΔABC, prove that $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$ with the help of vectors.**

Sol. In ΔABC, $\overline{BC} + \overline{CA} + \overline{AB} = \vec{0}$

$\Rightarrow \overline{AB} = -\left(\overline{BC} + \overline{CA}\right)$

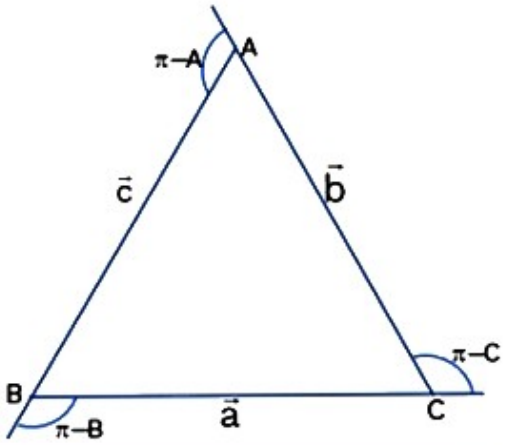
$\Rightarrow \overline{AB} \cdot \overline{AB} = \left(\overline{BC} + \overline{CA}\right) \cdot \left(\overline{BC} + \overline{CA}\right)$

$\Rightarrow \vec{c} \cdot \vec{c} = \left(\vec{a} + \vec{b}\right) \cdot \left(\vec{a} + \vec{b}\right)$

$\Rightarrow c^2 = a^2 + b^2 + 2\vec{a} \cdot \vec{b}$

$\Rightarrow c^2 = a^2 + b^2 + 2ab\cos(\pi - C)$

$\Rightarrow c^2 = a^2 + b^2 - 2ab\cos C$



Q. **Prove by vector method that $\left(a_1b_1 + a_2b_2 + a_3b_3\right)^2 \le \left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$**

Sol. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

Then $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$ ...(1)

Also $|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$ and $|\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$ ...(2)

If θ be the angle between the vectors $\vec{a}$ and $\vec{b}$, then

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} \quad \because \cos^2\theta \le 1 \quad \therefore \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}\right)^2 \le 1$$

or $(\vec{a} \cdot \vec{b})^2 \le |\vec{a}|^2 |\vec{b}|^2$

$\therefore \quad \left(a_1b_1 + a_2b_2 + a_3b_3\right)^2 \le \left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$

Angle between two vectors in terms of components :

If $\vec{a}$ and $\vec{b}$ be two vectors such that $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $b = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and θ be the angle between them, then

$$\cos\theta = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}}$$

Point to Remember!!!

If $\vec{a}$ and $\vec{b}$ are perpendicular to each other then

$a_1b_1 + a_2b_2 + a_3b_3 = 0$

Q. **Find the angle between the vectors $4\hat{i}+\hat{j}+3\hat{k}$ and $2\hat{i}+2\hat{j}-\hat{k}$.**

Sol. Let the required angle is θ.

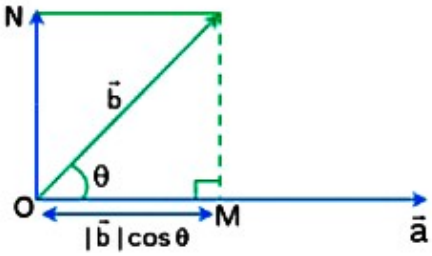
$$\therefore \theta = \cos^{-1}\left(\frac{4.2+1.2+3(-1)}{\sqrt{16+1+9}\sqrt{4+4+1}}\right) = \cos^{-1}\left(\frac{7}{3\sqrt{26}}\right)$$

Components of $\vec{b}$ along and perpendicular to $\vec{a}$:

1. Component of $\vec{b}$ along $\vec{a} = \overrightarrow{OM}$

$$= OM\hat{a} = (b\cos\theta)\hat{a}$$

$$= \frac{(ab\cos\theta)}{a}\hat{a} = \frac{(\vec{a}\cdot\vec{b})}{a^2}\vec{a} = \frac{(\vec{a}\cdot\vec{b})}{|a|^2}\vec{a}$$



2. Component perpendicular to $\vec{a} = \overrightarrow{ON}$

$\because \vec{b} = \overrightarrow{ON} + \overrightarrow{OM}$

$\therefore \overrightarrow{ON} = \vec{b} - \overrightarrow{OM}$

$$\overrightarrow{ON} = \vec{b} - \frac{(\vec{a}\cdot\vec{b})}{a^2}\vec{a}$$

Q. **Find the vector components of a vector $2\hat{i}+3\hat{j}+6\hat{k}$ along and perpendicular to non-zero vector $2\hat{i}+\hat{j}+2\hat{k}$.**

Sol. Let $\vec{a} = 2\hat{i}+3\hat{j}+6\hat{k}$ and $\vec{b} = 2\hat{i}+\hat{j}+2\hat{k}$

Now vector component of $\vec{a}$ along $\vec{b}$

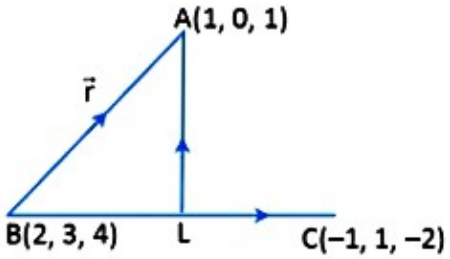
$$= \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b} = \frac{4+3+12}{9}(2\hat{i}+\hat{j}+2\hat{k}) = \frac{19}{9}(2\hat{i}+\hat{j}+2\hat{k})$$

and vector component of $\vec{a}$ perpendicular to $\vec{b}$

$$= \vec{a} - \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b} = (2\hat{i}+3\hat{j}+6\hat{k}) - \frac{19}{9}(2\hat{i}+\hat{j}+2\hat{k}) = \frac{1}{9}(-20\hat{i}+8\hat{j}+16\hat{k})$$

Q. **Find the perpendicular distance of the point $A(1,0,1)$ to the line through the points $B(2,3,4)$ and $C(-1,1,-2)$.**

Sol. $\vec{r} = \overline{BA} = -\hat{i} - 3\hat{j} - 3\hat{k}$ and $\vec{a} = \overline{BC} = -3\hat{i} - 2\hat{j} - 6\hat{k}$



Now $\overline{BL}$ = Projection vector of $\vec{r}$ on $\vec{a}$

$$= \frac{\vec{r}\cdot\vec{a}}{|\vec{a}|^2}\vec{a} = \frac{3+6+18}{49}(-3\hat{i} - 2\hat{j} - 6\hat{k})$$

$$= \frac{-27}{49}(3\hat{i} + 2\hat{j} + 6\hat{k})$$

$$\overline{LA} = \overline{LB} + \overline{BA} = \overline{BA} - \overline{BL}$$

$$= \vec{r} - \frac{\vec{r}\cdot\vec{a}}{|\vec{a}|^2}\cdot\vec{a} = (-\hat{i} - 3\hat{j} - 3\hat{k}) + \frac{27}{49}(3\hat{i} + 2\hat{j} + 6\hat{k}) = \frac{1}{49}(32\hat{i} - 93\hat{j} + 15\hat{k})$$

$$\therefore \text{ LA} = \left|\overline{LA}\right| = \frac{\sqrt{9898}}{49}$$

More Illustrations on Dot Product

Q. **If** $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$ and $\vec{b} = -2\hat{i} - 5\hat{j} - 14\hat{k}$

If $\lambda = \dfrac{\text{Projection of } \vec{a} \text{ on } \vec{b}}{\text{Projection of } \vec{b} \text{ on } \vec{a}}$**. Find** λ**.**

Sol. $$\frac{\vec{a}\cdot\hat{b}}{\vec{b}\cdot\hat{a}} = \frac{(\vec{a}\cdot\vec{b})}{|\vec{b}|\frac{(\vec{b}\cdot\vec{a})}{(|\vec{a}|)}} = \frac{|\vec{a}|}{|\vec{b}|} = \frac{\sqrt{4+9+36}}{\sqrt{4+25+196}} = \frac{7}{15}$$

Q. **Express the vector** $\vec{a} = 5\hat{i} - 2\hat{j} + 5\hat{k}$ **as the sum of two vectors such that one is parallel to** $\vec{b} = 3\hat{i} + \hat{k}$ **and the other perpendicular to** $\vec{b}$**.**

Sol. $\vec{a} = 5\hat{i} - 2\hat{j} + 5\hat{k}$

$\vec{b} = 3\hat{i} + \hat{k}$

Consider, $\vec{a} = \overline{\beta_1} + \overline{\beta_2}$

Where $\overline{\beta_1}$ is parallel to $\vec{b}$ and $\overline{\beta_2}$ is perpendicular to $\vec{b}$

Since, $\overline{\beta_1}$ is parallel to $\vec{b}$ -

Then, $\overline{\beta_1} = \lambda\vec{b} = \lambda(3\hat{i} + \hat{k}) = 3\lambda\hat{i} + \lambda\hat{k}$ ...(1)

Now, $\vec{a} = \overline{\beta_1} + \overline{\beta_2}$

$\overrightarrow{\beta_2} = \vec{a} - \overrightarrow{\beta_1}$

$\overrightarrow{\beta_2} = (5\hat{i} - 2\hat{j} + 5\hat{k}) - (3\lambda\hat{i} + \lambda\hat{k})$

$\overrightarrow{\beta_2} = (5 - 3\lambda)\hat{i} - 2\hat{j} + (5 - \lambda)\hat{k}$ ...(2)

Since, $\overrightarrow{\beta_2}$ is perpendicular to $\vec{b}$

$\overrightarrow{\beta_2}.\vec{b} = 0$

Then, $3(5 - 3\lambda) + 0(-2) + (5 - \lambda) = 0$

$\lambda = 2$

Therefore,

$\overrightarrow{\beta_1} = 6\hat{i} + 2\hat{k}$ from (1)

$\overrightarrow{\beta_2} = -\hat{i} - 2\hat{j} + 3\hat{k}$ ...(3)

Hence, $\vec{a} = (6\hat{i} + 2\hat{k}) + (-\hat{i} - 2\hat{j} + 3\hat{k})$

Q. **If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}, \vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, find the angle between $\vec{a}$ and $\vec{a} + 2\vec{b}$.**

Sol. $\vec{a} + 2\vec{b} = (5\hat{i} - \hat{j} - 3\hat{k}) + 2(\hat{i} + 3\hat{j} - 5\hat{k}) = 7\hat{i} + 5\hat{j} - 13\hat{k}$

$$\cos\theta = \frac{(\vec{a} + 2\vec{b})\cdot\vec{a}}{|\vec{a}||\vec{a} + 2\vec{b}|} = \frac{(7\hat{i} + 5\hat{j} - 13\hat{k})\cdot(5\hat{i} - \hat{j} - 3k)}{\sqrt{25 + 1 + 9}\sqrt{49 + 25 + 169}} = \frac{35 - 5 + 39}{\sqrt{35}\sqrt{243}} = \frac{69}{\sqrt{359}\sqrt{3}} = \frac{13}{3\sqrt{105}}$$

$$\theta = \cos^{-1}\left(\frac{13}{3\sqrt{105}}\right)$$

Q. **Find the angle between $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ if $|\vec{a}| = 2, |\vec{b}| = 1$ & $\vec{a} \wedge \vec{b} = \frac{\pi}{3}$.**

Sol. If angle between $(\vec{a} + \vec{b})$ and $(\vec{a} - \vec{b})$ is θ Then $(\vec{a} + \vec{b})\cdot(\vec{a} - \vec{b}) = |(\vec{a} + \vec{b})||(\vec{a} - \vec{b})|\cos\theta$

$$\Rightarrow \cos\theta = \frac{(\vec{a} + b)\cdot(\vec{a} - \vec{b})}{|(\vec{a} + \vec{b})||\vec{a} - \vec{b}|}$$

$$\Rightarrow \cos\theta = \frac{a^2 - a\cdot\vec{b} + \vec{b}\cdot\vec{a} - b^2}{\sqrt{a^2 + b^2 + 2ab\cos\phi}\sqrt{a^2 + b^2 - 2ab\cos\phi}}$$

$$\Rightarrow \cos\theta = \frac{a^2 - b^2}{\sqrt{a^2 + b^2 + 2ab\cos\phi}\sqrt{a^2 + b^2 - 2ab\cos\phi}}$$

$$\Rightarrow \cos\theta = \frac{2^2 - 1^2}{\sqrt{2^2 + 1^2 + 2\times 2\times 1\cos\frac{\pi}{3}}\sqrt{2^2 + 1^2 - 2\times 2\times 1\cos\frac{\pi}{3}}}$$

$$\Rightarrow \cos\theta = \frac{3}{\sqrt{7}\times\sqrt{3}}$$

$$\Rightarrow \cos\theta = \sqrt{\frac{3}{7}}$$

$$\Rightarrow \quad \theta = \cos^{-1}\sqrt{\frac{3}{7}}$$

$$\Rightarrow \quad \theta = 49.11°$$

Q. **Find $\vec{p}\wedge\vec{q}$ if $2\vec{p}+\vec{q}=\hat{i}+\hat{j}$, $\vec{p}+2\vec{q}=\hat{i}-\hat{j}$.**

Sol. $(2\vec{p}+\vec{q})+(\vec{p}+2\vec{q}) = \hat{i}+\hat{j}+\hat{i}-\hat{j}$

$\Rightarrow 3(\vec{p}+\vec{q}) = 2\hat{i} \Rightarrow \vec{p}+\vec{q} = \frac{2}{3}\hat{i}$ ...(1)

Also, $(2\vec{p}+\vec{q})-(\vec{p}+2\vec{q}) = \hat{i}+\hat{j}-(\hat{i}-\hat{j})$

$\therefore \vec{p}-\vec{q} = 2\hat{j}$ ...(2)

From (1) and (2)

$2\vec{p} = \frac{2}{3}\hat{i}+2\hat{j}$ ...(3)

$2\vec{q} = \frac{2}{3}\hat{i}-2\hat{j}$ ...(4)

$\therefore 4\vec{p}\cdot\vec{q} = \frac{2}{3}\cdot\frac{2}{3}+2\cdot(-2) = \frac{4}{9}-4 = -\frac{32}{9}$

$\therefore \quad \vec{p}\cdot\vec{q} = -\frac{8}{9}$

Also from (3), $2|\vec{p}| = \sqrt{\left(\frac{2}{3}\right)^2+(2)^2} = \frac{2}{3}\sqrt{10}$

$\therefore |\vec{p}| = \frac{1}{3}\sqrt{10}$, and from (4)

$2|\vec{q}| = \sqrt{\left(\frac{2}{3}\right)^2+(-2)^2} = \frac{2}{3}\sqrt{10}$, $\therefore |\vec{q}| = \frac{1}{3}\sqrt{10}$

$$\therefore \cos\theta = \frac{\vec{p}\cdot\vec{q}}{|\vec{p}||\vec{q}|} = \frac{-\frac{8}{9}}{\frac{1}{3}\sqrt{10}\cdot\frac{1}{3}\sqrt{10}} = -\frac{4}{5}$$

$$\therefore \theta = \cos^{-1}\left(-\frac{4}{5}\right)$$

Hence $\vec{p} \wedge \vec{q} = \cos^{-1}\left(-\frac{4}{5}\right)$

Q. $|\vec{a}| = 11, |\vec{b}| = 23$ and $|\vec{a} - \vec{b}| = 30$ **find** $|\vec{a} + \vec{b}|$

Sol. Given, $|\vec{a}| = 11, |\vec{b}| = 23, |\vec{a} - \vec{b}| = 30$

We know that, $|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - 2|\vec{a}||\vec{b}|\cos\theta$

$30^2 = 11^2 + 23^2 - 2 \times 11 \times 23 \cdot \cos\theta$

$2 \times 11 \times 23 \cdot \cos\theta = 11^2 + 23^2 - 30^2 = -250$

Now, $|\vec{a} + \vec{b}|^6 = |\vec{a}| + |\vec{b}| + 2|\vec{a}||\vec{b}|\cos\theta = 11^2 + 23^2 - 250 = 400$

$|\vec{a} + \vec{b}| = \sqrt{400} = 20$

Q. **If** $\vec{a} + \vec{b} + \vec{c} = 0, |\vec{a}| = 3; |\vec{b}| = 1$ **and** $|\vec{c}| = 4$. Find $\sum(\vec{a}\vec{b})$

Sol. We know, $(\vec{a} + \vec{b} + \vec{c})^2 = (\vec{a} + \vec{b} + \vec{c})\cdot(\vec{a} + \vec{b} + \vec{c})$

$\Rightarrow 0 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

(Given $\vec{a} + \vec{b} + \vec{c} = 0$)

$\Rightarrow 0 = (3)^2 + (1)^2 + (4)^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

(Given $|\vec{a}| = 3, |\vec{b}| = 1, |\vec{c}| = 4$)

$\Rightarrow 0 = 9 + 1 + 16 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

$\Rightarrow 0 = 26 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

$\Rightarrow \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = -\frac{26}{2} = -13$

Hence the value asked in the question is

$\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} = -13$

Q. **A vector of magnitude 25 is collinear with $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and makes obtuse angle with negative z-axis find the vector.**

Sol. $|\vec{b}| = 25$ and $\vec{b} = \lambda\vec{a}$ ...(i)

$\vec{b} = \lambda(2\hat{i} - 3\hat{j} + 4\hat{k})$

$|\vec{b}| = \lambda\sqrt{4+9+16}$

$\Rightarrow \lambda = \frac{25}{\sqrt{29}}$ So, $\vec{b} = \frac{25}{\sqrt{29}}(2\hat{i} - 3\hat{j} + 4\hat{k})$

Q. **Prove that $|\hat{a} + \hat{b}| = \sqrt{2 + 2\cos\theta} = 2\cos\left(\frac{\theta}{2}\right)$ and $|\hat{a} - \hat{b}| = \sqrt{2 - 2\cos\theta} = 2\sin\left(\frac{\theta}{2}\right)$, where θ is the angle between the vectors $\vec{a}$ & $\vec{b}$.**

Sol. Let $|\hat{a} + \hat{b}| = k$

$\Rightarrow |\hat{a} + \hat{b}|^2 = k^2$

$\Rightarrow (\hat{a} + \hat{b})\cdot(\hat{a} + \hat{b}) = k^2$

$\Rightarrow 1 + 1 + 2\cos\theta = k^2$

$\Rightarrow k = \sqrt{2 + 2\cos\theta}$ $\because 1 + \cos\theta = 2\cos^2\frac{\theta}{2}$

Similarly $|\hat{a} - \hat{b}| = \sqrt{2 - 2\cos\theta} = 2\sin\frac{\theta}{2}$.

Q. **Let $\bar{u}$ be a vector on rectangular coordinate system with sloping angle 60°. Suppose that $|\bar{u} - \hat{i}|$ is geometric mean of $|\bar{u}|$ and $|\bar{u} - 2\hat{i}|$ where $\hat{i}$ is the unit vector along x-axis then $|\bar{u}|$ has the value equal to $\sqrt{a} - \sqrt{b}$ where a, b $\in$ N, find the value $(a + b)^3 + (a - b)^3$.**

Sol. Let use consider $|\bar{u}| = \lambda$

$\Rightarrow \bar{u} = \frac{\lambda}{2}(\hat{i} + \sqrt{3}\,\hat{j})$

Also $|\bar{u} - \hat{i}| = \sqrt{|\bar{u}||\bar{u} - 2\hat{i}|}$

$\Rightarrow \left|\frac{\lambda}{2}(\hat{i} + \sqrt{3}\,\hat{j}) - \hat{i}\right|^2 = \lambda\left|\frac{\lambda}{2}(\hat{i} + \sqrt{3}\hat{j}) - 2\hat{i}\right|$

$$\Rightarrow \left(\left(\frac{\lambda}{2}-1\right)^2+\frac{3\lambda^2}{4}\right)^2=\lambda^2\left(\left(\frac{\lambda-4}{2}\right)^2+\frac{3\lambda^2}{4}\right)$$

$$\Rightarrow (4\lambda^2-4\lambda+4)^2=16\lambda^2(\lambda^2-2\lambda+4)$$

$$\Rightarrow \left(\lambda^2-\lambda+1\right)^2=\lambda^2\left(\lambda^2-2\lambda+4\right)$$

$$\Rightarrow \lambda=\frac{-2\pm\sqrt{4+4}}{2}=-1\pm\sqrt{2}$$

$\because \lambda > 0 \Rightarrow \lambda = \sqrt{2}-1$

$\therefore$ a = 2 and b = 1

$(a+b)^3+(a-b)^3=(3)^3+(1)^3=27+1=28$

Hence, the value of $(a+b)^3+(a-b)^3$ is 28.

Q. **Prove that** $\left(\frac{\vec{a}}{|\vec{a}|^2}-\frac{\vec{b}}{|\vec{b}|^2}\right)^2=\left(\frac{\vec{a}-\vec{b}}{|\vec{a}||\vec{b}|}\right)^2$

Sol. Given $|\vec{a}|=a$ and $|\vec{b}|=b$

We need to prove that $\left(\frac{\vec{a}}{a^2}-\frac{\vec{b}}{b^2}\right)^2=\left(\frac{\vec{a}-\vec{b}}{ab}\right)^2$

$$\text{LHS}=\left(\frac{\vec{a}}{a^2}-\frac{\vec{b}}{b^2}\right)^2$$

$$=\left(\frac{\vec{a}}{a^2}-\frac{\vec{b}}{b^2}\right)\left(\frac{\vec{a}}{a^2}-\frac{\vec{b}}{b^2}\right)$$

$$=\frac{|\vec{a}|^2}{a^2\cdot a^2}-\frac{\vec{a}\cdot\vec{b}}{a^2\cdot b^2}-\frac{\vec{b}\cdot\vec{a}}{b^2\cdot a^2}+\frac{|\vec{b}|^2}{b^2\cdot b^2}$$

$$=\frac{a^2}{a^2\cdot a^2}-\frac{\vec{a}\cdot\vec{b}}{a^2\cdot b^2}-\frac{\vec{a}\cdot\vec{b}}{a^2\cdot b^2}+\frac{b^2}{b^2\cdot b^2}$$

$$=\frac{1}{a^2}-\frac{2\vec{a}\cdot\vec{b}}{a^2\cdot b^2}+\frac{1}{b^2} \qquad \text{...(i)}$$

$$\text{RHS}=\left(\frac{\vec{a}-\vec{b}}{ab}\right)^2=\left(\frac{\vec{a}-\vec{b}}{ab}\right)\left(\frac{\vec{a}-\vec{b}}{ab}\right)$$

$$=\frac{1}{a^2b^2}(\vec{a}\cdot\vec{a}-\vec{a}\cdot\vec{b}-\vec{b}\cdot\vec{a}+\vec{b}\cdot\vec{b})$$

$$=\frac{1}{a^2b^2}\left(|\vec{a}|^2-\vec{a}\cdot\vec{b}-\vec{a}\cdot\vec{b}+|\vec{b}|^2\right)$$

$$= \frac{1}{a^2b^2}\left(a^2 - 2\vec{a}\cdot\vec{b} + b^2\right)$$

$$= \frac{1}{b^2} - \frac{2\vec{a}\cdot\vec{b}}{a^2b^2} + \frac{1}{a^2}$$

$$= \frac{1}{a^2} - \frac{2\vec{a}\cdot\vec{b}}{a^2b^2} + \frac{1}{b^2} \qquad \text{...(ii)}$$

From (i) and (ii), we get LHS = RHS **Hence proved.**

Q. **A point P moves in space such that $P(\vec{A})\cdot P(\vec{B}) < 0$ then the locus of the point P is the interior of the sphere with AB as diameter.**

Sol. Let origin is $\vec{A}$.

Let $\vec{P} = x\hat{i} + y\hat{j} + z\hat{k}$

$\vec{B} = a\hat{i} + b\hat{j} + c\hat{k}$

[fixed point]

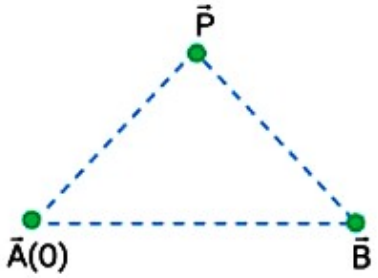
$\overrightarrow{PA} = -(x\hat{i} + y\hat{j} + z\hat{k})$

$\overrightarrow{PB} = (a-x)\hat{i} + (b-y)\hat{j} + (c-z)\hat{k}$

Now, $\overrightarrow{PA}\cdot\overrightarrow{PB} < 0$

$\Rightarrow x^2 + y^2 + z^2 - ax - by - cz < 0$

Which represents interior portion of sphere.



Q. **Find vector equation of an ellipse/a Hyperbola whose foci are the position vectors and $-\vec{c}$ and length of major/Transversal Axis is 2a.**

Sol. If $\vec{r}$ be any point on Ellipse/Hyperbola

then $|\vec{r} + \vec{c}| + |\vec{r} - \vec{c}| = 2a$ for Ellipse ...(i)

and $|\vec{r} + \vec{c}| - |\vec{r} - \vec{c}| = 2a$ for Hyperbola ...(ii)

$|\vec{r} + \vec{c}|^2 = (2a - |\vec{r} - \vec{c}|)^2$ [from (i)]

$r^2 + c^2 + 2\vec{r}\cdot\vec{c} = 4a^2 + r^2 + c^2 - 2\vec{r}\cdot\vec{c} - 4a|\vec{r} - \vec{c}|$

$a^2 - \vec{r}.\vec{c} = a|\vec{r} - \vec{c}|$

Similarly for Hyperbola $\vec{r}\cdot\vec{c} - a^2 = a|\vec{r} - \vec{c}|$

Q. **Arc AC of the quadrant of a circle with centre as origin and radius unity subtends a right angle at the origin. Point B divides the arc AC in the ratio 1 : 2. Express the vector $\vec{c}$ in terms of $\vec{a}$ and $\vec{b}$.**

Sol. Let the radius of the arc be r. Now, B divided the arc in ratio 1 : 2, so AB would subtend angle of 30° at the origin O and similarly, BC would subtend an angled of 60° origin. Let $\vec{a} = r\hat{i}, \vec{c} = r\hat{j}, \vec{b} = r\left(\cos 30^\circ\right)\hat{i} + r(\sin 30^\circ)\hat{j} = \frac{r}{2}\hat{j} + \frac{r\sqrt{3}}{2}\hat{i}$

$\therefore \vec{b} = \frac{1}{2}\vec{c} + \frac{\sqrt{3}}{2}\vec{a} \Rightarrow \vec{c} = 2\vec{b} - \sqrt{3}\vec{a}$

Q. **Two adjacent sides of a parallelogram ABCD are given by $\overline{AB} = 2\hat{i} + 10\hat{j} + 11\hat{k}$ and $\overline{AD} = -\hat{i} + 2\hat{j} + 2\hat{k}$. The side AD is rotated by an actue angle α in the plane of the parallelogram so that AD becomes AD', If AD' makes a right angle with the side AB, then the cosine of the angle α is given by [JEE 2010]**

(A) $\frac{8}{9}$ **(B)** $\frac{\sqrt{17}}{9}$ **(C)** $\frac{1}{9}$ **(D)** $\frac{4\sqrt{5}}{9}$

Sol. We have

$$\cos\phi = \frac{-2 + 20 + 22}{\sqrt{2^2 + 10^2 + 11^2}\sqrt{1^2 + 2^2 + 2^2}} = \frac{40}{15 \times 3} = \frac{8}{9}$$

We have $\frac{\sqrt{17}}{9} = \sin\phi$ then $\cos\alpha = \sin\phi = \frac{\sqrt{17}}{9}$



Q. **Given that $\vec{a} = \hat{i} - \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j}$ are two vectors. Find a unit vector coplanar with $\vec{a}$ and $\vec{b}$ and perpendicular to $\vec{a}$.**

Sol. Let r is co-planar vector with $\vec{a}$ & $\vec{b}$

$\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} \perp \vec{a} \Rightarrow \vec{r}\cdot\vec{a} = 0$

So, $\vec{r}\cdot\vec{a} = \vec{a}\cdot\vec{a} + \lambda\vec{a}\cdot\vec{b}$

$\Rightarrow 0 = 2 + \lambda(-1)$

$\Rightarrow \lambda = 2$

$\vec{r} = \vec{a} + 2\vec{b} = \hat{i} - \hat{j} + 2\hat{i} + 4\hat{j} = 3\hat{i} + 3\hat{j}$

$\hat{r} = \frac{\hat{i} + \hat{j}}{\sqrt{2}} = \frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$

Q. **If $\vec{a},\vec{b},\vec{c}$ are coplanar vectors, prove that** $\begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \end{vmatrix} = 0$.

Sol. As $\vec{a},\vec{b}$ and $\vec{c}$ are coplanar, they are linearly dependent.

So, $\lambda\vec{a} + \mu\vec{b} + \nu\vec{c} = 0$ ...(1)

for some scalars λ, μ, ν.

Taking dot product with $\vec{a}$ and $\vec{b}$ respectively,

$$\lambda\vec{a}\cdot\vec{a} + \mu\vec{a}\cdot\vec{b} + \nu\vec{a}\cdot\vec{c} = 0$$

and $$\lambda\vec{b}\cdot\vec{a} + \mu\vec{b}\cdot\vec{b} + \nu\vec{b}\cdot\vec{c} = 0$$

$$\therefore \quad \frac{\lambda}{(\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{c}) - (\vec{b}\cdot\vec{b})(\vec{a}\cdot\vec{c})}$$

$$= \frac{\mu}{(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{a}) - (\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{c})}$$

$$= \frac{\nu}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{a})}$$

$\therefore$ from (1),

$$\vec{a}\{(\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{c}) - (\vec{b}\cdot\vec{b})(\vec{a}\cdot\vec{c})\} + \vec{b}[(\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{a}) - (\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{c})] + \vec{c}[(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})(\vec{b}\cdot\vec{a})] = 0$$

$$\therefore \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \\ \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \end{vmatrix} = 0$$

Q. **If two pairs of opposite edges of a tetrahedron are at right angles, then show that the third pair is also at right angle. Further show that for such a tetrahedron, the sum of the squares of each pair of opposite edges is the same.**

Sol. Let ABCD be the tetrahedron.

Take D as the origin.

Let the position vectors of A, B, C be $\vec{a},\vec{b},\vec{c}$ respectively.

Then $\overline{AB} = \vec{b} - \vec{a}, \overline{BC} = \vec{c} - \vec{b}, \overline{CA} = \vec{a} - \vec{c}$

Let DA $\perp$ BC and DB $\perp$ CA.

(i) **To prove:** DC $\perp$ AB.

Since DA $\perp$ BC,

$\therefore \quad \overline{DA}\cdot\overline{BC} = 0$ i.e. $(\vec{a}-\vec{0})\cdot(\vec{c}-\vec{b}) = 0$

$\Rightarrow \quad \vec{a}\cdot\vec{c} - \vec{a}\cdot\vec{b} = 0 \qquad ...(1)$

Since DB $\perp$ CA.

$\therefore \ \overline{DB}\cdot\overline{CA} = 0$ i.e. $(\vec{b}-\vec{0})\cdot(\vec{a}-\vec{c}) = 0$

$\Rightarrow \vec{b}\cdot\vec{a} - \vec{b}\cdot\vec{c} = 0 \qquad ...(2)$

Adding (1) and (2), $\vec{a}\cdot\vec{c} - \vec{b}\cdot\vec{c} = 0 \qquad [\because \vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a}]$

$\Rightarrow \quad (\vec{a}-\vec{b})\cdot\vec{c} = 0$

$\Rightarrow \quad \vec{c}\cdot(\vec{a}-\vec{b}) = 0$

$\Rightarrow (\vec{c}-\vec{0})\cdot(\vec{a}-\vec{b}) = 0$

$\Rightarrow \overline{DC}\cdot\overline{BA} = 0 \Rightarrow \overline{DC}\cdot\overline{AB} = 0 \quad \therefore DC \perp AB$

Hence, the first part.

(ii) Now $(DA)^2 + (BC)^2 = (\vec{a}-\vec{0})^2 + (\vec{c}-\vec{b})^2$

$= \vec{a}^2 + \vec{c}^2 + \vec{b}^2 - 2\vec{c}\cdot\vec{b}$

$(DB)^2 + (CA)^2 = (\vec{b}-\vec{0})^2 + (\vec{a}-\vec{c})^2$

$= \vec{b}^2 + \vec{a}^2 + \vec{c}^2 - 2\vec{a}\cdot\vec{c}$

and $(DC)^2 + (AB)^2 = (\vec{c}-\vec{0})^2 + (\vec{b}-\vec{a})^2$

$= \vec{c}^2 + \vec{b} + \vec{a}^2 - 2\vec{a}\cdot\vec{b}$

But $\vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{c} = \vec{c}\cdot\vec{a}$ [From (1) and (2)]

$\therefore (DA)^2 + (BC)^2 = (DB)^2 + (CA)^2 = (DC)^2 + (AB)^2$

Hence, the second part.

Q. **If $\vec{a},\vec{b}$ and $\vec{c}$ are unit vectors, then $|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2$ does NOT exceed**

(A) 4 **(B) 9** **(C) 8** **(D) 6**

Sol. **(B)**

Given that,

$|\vec{a}-\vec{b}|^2 + |\vec{b}-\vec{c}|^2 + |\vec{c}-\vec{a}|^2$

$= |\vec{a}|^2 + |\vec{b}|^2 - 2(\vec{a}\cdot\vec{b}) + |\vec{b}|^2 + |\vec{c}|^2 - 2(\vec{b}\cdot\vec{c}) + |\vec{c}|^2 + |\vec{a}|^2 - 2(\vec{c}\cdot\vec{a})$

$= 2\left[|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 - (\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})\right]$

$= 2[3 - (\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})]$

$= 6 - 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a})$

So, $|\vec{a} + \vec{b} + \vec{c}|^2 \geq 0$

$\Rightarrow |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) \geq 0$

$\Rightarrow 3 + 2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) \geq 0$

$\Rightarrow \vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a} \geq -\frac{3}{2}$

$\Rightarrow -2(\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{a}) \leq 3$

By using (i) and (ii), it will get

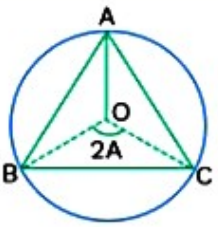
$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 \leq 6 + 3$

$\Rightarrow |\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2 \leq 9$

Therefore, the correct option is (B).

Q. **Use vectors to prove that in a ΔABC cos 2A + cos 2B + cos 2C > − 3/2. Also prove that the distance between the circumcentre and the centroid is $\sqrt{R^2 - \frac{1}{9}(a^2 + b^2 + c^2)}$.**

Sol. Consider a circle with centre O and a ΔABC inscribed in it.



$\Rightarrow \quad |\overline{OA}| = |\overline{OB}| = |\overline{OC}| = R$ (circum-radius)

$\because \quad |\overline{OA} + \overline{OB} + \overline{OC}|^2 \geq 0$

$\Rightarrow (\overline{OA} + \overline{OB} + \overline{OC})\cdot(\overline{OA} + \overline{OB} + \overline{OC}) \geq 0$

$\Rightarrow |\overline{OA}|^2 + |\overline{OB}|^2 + |\overline{OC}|^2 + 2$

$(\overline{OA}\cdot\overline{OB} + \overline{OB}\cdot\overline{OC} + \overline{OC}\cdot\overline{OA}) \geq 0$

$\Rightarrow 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C) \geq 0$

$\Rightarrow 2(\cos 2A + \cos 2B + \cos 2C) + 3 \geq 0$

$\therefore \cos 2A + \cos 2B + \cos 2C \geq -\frac{3}{2}$

and

Take "O" as the origin. Let $\overline{OA} = \vec{\alpha}, \overline{OB} = \vec{\beta}$ and $\overline{OC} = \vec{\gamma}$. Therefore

$(\vec{\beta}, \vec{\gamma}) = 2A, (\vec{\gamma}, \vec{\alpha}) = 2B$ and $(\vec{\alpha}, \vec{\beta}) = 2C$

It is known that $\overline{OG} = \frac{\vec{\alpha} + \vec{\beta} + \vec{\gamma}}{3}$ (see Corollary)

Therefore

$9|\overline{OG}|^2 = |\vec{\alpha}+\vec{\beta}+\vec{\gamma}|^2$

$= |\vec{\alpha}|^2 + |\vec{\beta}|^2 + |\vec{\gamma}|^2 + 2\sum(\vec{\beta}\cdot\vec{\gamma})$

$= 3R^2 + 2R^2(\cos 2A + \cos 2B + \cos 2C)$

$= 3R^2 + 2R^2\left[1-2\sin^2 A + 1 - 2\sin^2 R + 1 - 2\sin^2 C\right]$

$= 9R^2 - 4R^2\sin^2 A - 4R^2\sin^2 B - 4R^2\sin^2 C$

$= 9R^2 - a^2 - b^2 - c^2 \left(\because \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R\right)$

Hence $|\overline{OG}|^2 = R^2 - \frac{1}{9}\left(a^2+b^2+c^2\right)$

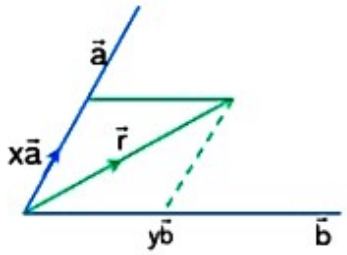
Linear Combination of Vectors :

A vector $\vec{r}$ is said to be a linear combination of the vectors $\vec{a},\vec{b},\vec{c}$..... if $\exists$ scalars $x, y, z, \ldots$ such that

$\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + \ldots$

Theorem in plane :

If $\vec{a}$ and $\vec{b}$ are two non zero non collinear vectors then any vector $\vec{r}$ coplanar with them can be expressed as a linear combination $\vec{r} = x\vec{a} + y\vec{b}$. (Explain using a sketch)



Q. **Arc AC of the quadrant of a circle with centre as origin and radius unity subtends a right angle at the origin. Point B divides the arc AC in the ratio 1 : 2. Express the vector $\vec{c}$ in terms of $\vec{a}$ and $\vec{b}$.**

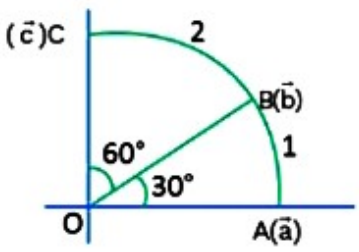
Sol. $\because \vec{c} = x\vec{a} + y\vec{b}$ ...(i)

Taking dot product with $\vec{a}$ in (i)

$\vec{c}\cdot\vec{a} = x\vec{a}\cdot\vec{a} + y\vec{a}\cdot\vec{b}$

$0 = x + y\frac{\sqrt{3}}{2}$ ...(ii)

Taking dot product with $\vec{c}$ in (i)



$\vec{c}\cdot\vec{c} = x\vec{a}\cdot\vec{c} + y\vec{b}\cdot\vec{c}$

$1 = 0 + \frac{y}{2} \Rightarrow y = 2$

from (ii), $x = -\sqrt{3} \therefore \vec{c} = -\sqrt{3}\vec{a} + 2\vec{b}$

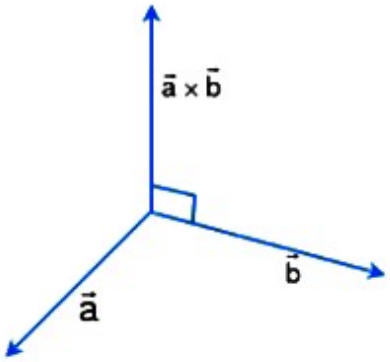
VECTOR OR CROSS PRODUCT OF TWO VECTORS :

Definition :

If $\vec{a}$ and $\vec{b}$ be two vectors and $\theta(0 \le \theta \le \pi)$ be the angle between them, then their vector (or cross) product is defined to be a vector whose magnitude is $ab\sin\theta$ and whose direction is perpendicular to the plane of $\vec{a}$ and $\vec{b}$ such that $\vec{a},\vec{b}$ and $\vec{a}\times\vec{b}$ form a right handed system.

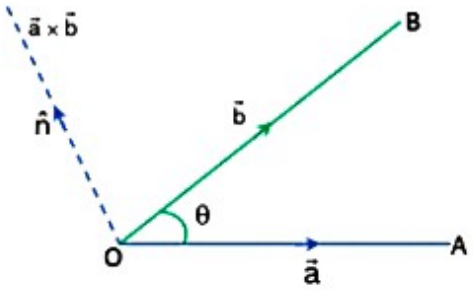
$\therefore \vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\hat{n}$

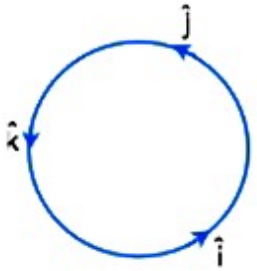
Where $\hat{n}$ is a unit vector perpendicular to the plane of $\vec{a}$ and $\vec{b}$ such that $\vec{a},\vec{b}$ and $\hat{n}$ form a right handed system.



Properties of Vector Product :

1. In general, $\vec{a}\times\vec{b} \ne \vec{b}\times\vec{a}$. In fact $\vec{a}\times\vec{b} = -\vec{b}\times\vec{a}$.
2. For scalar $m, m\vec{a}\times\vec{b} = m(\vec{a}\times\vec{b}) = \vec{a}\times m\vec{b}$.
3. $\vec{a}\times(\vec{b}\pm\vec{c}) = \vec{a}\times\vec{b} \pm \vec{a}\times\vec{c}$
4. If $\vec{a}\parallel\vec{b}$ then $\theta = 0$ or $\pi \Rightarrow \vec{a}\times\vec{b} = \vec{0}$ (but $\vec{a}\times\vec{b} = \vec{0} \Rightarrow \vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$ or $\vec{a}\parallel\vec{b}$). In particular $\vec{a}\times\vec{a} = \vec{0}$
5. If $\vec{a}\perp\vec{b}$ then $\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\hat{n}$ (or $|\vec{a}\times\vec{b}| = |\vec{a}||\vec{b}|$)
6. $\hat{i}\times\hat{i} = \hat{j}\times\hat{j} = \hat{k}\times\hat{k} = 0$ and $\hat{i}\times\hat{j} = \hat{k}, \hat{j}\times\hat{k} = \hat{i}$ and $\hat{k}\times\hat{i} = \hat{j}$ (use cyclic system)
7. Unit vector perpendicular to $\vec{a}$ and $\vec{b}$ is given by $\pm\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$





8. If θ is angle between $\vec{a}$ and $\vec{b}$ then

$$\sin\theta = \frac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$$

9. If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2b_3 - a_3b_2)\hat{i} + (a_3b_1 - a_1b_3)\hat{j} + (a_1b_2 - a_2b_1)\hat{k}$$

Q. **If $\vec{a} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = 6\hat{i} - 3\hat{j} + 2\hat{k}$ then $\vec{a}\times\vec{b}$ equals**

(A) $2\hat{i} - 2\hat{j} - \hat{k}$ **(B)** $\hat{i} - 10\hat{j} - 18\hat{k}$ **(C)** $\hat{i} + \hat{j} + \hat{k}$ **(D)** $6\hat{i} - 3\hat{j} + 2\hat{k}$

Sol. **(B)**

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 6 & -3 & 2 \end{vmatrix} = \hat{i}(4-3) - \hat{j}(4+6) + \hat{k}(-6-12) = \hat{i} - 10\hat{j} - 18\hat{k}$$

Q. **If angle between $\hat{i} - 2\hat{j} + 3\hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$ is θ then $\sin\theta$ equals**

(A) $5/\sqrt{7}$ **(B) 5/21** **(C)** $5/2\sqrt{7}$ **(D)** $3/\sqrt{14}$

Sol. **(C)**

We know that $\sin\theta = \dfrac{|\vec{a}\times\vec{b}|}{|\vec{a}||\vec{b}|}$

Now $\vec{a}\times\vec{b} = -5\hat{i} + 5\hat{j} + 5\hat{k}$

$\therefore \quad |\vec{a}\times\vec{b}| = \sqrt{(5)^2 + (5)^2 + (5)^2} = \sqrt{75} = 5\sqrt{3}$

$$|\vec{a}| = \sqrt{1+4+9}, |\vec{b}| = \sqrt{4+1+1}$$

$$\therefore \quad \sin\theta = \frac{5\sqrt{3}}{\sqrt{1+4+9}\sqrt{4+1+1}} = \frac{5\sqrt{3}}{\sqrt{14}\sqrt{6}} = \frac{5}{\sqrt{28}} = \frac{5}{2\sqrt{7}}$$

Q. **Find equation of a line which passes through the point with p.v. $\vec{a}$ and perpendicular to the lines $\vec{r} = \vec{b} + \lambda\vec{p}$ and $\vec{r} = \vec{c} + \mu\vec{q}$.**

Sol. Equation $\vec{r} = \vec{a} + \lambda\vec{d}$ for $\vec{d} = \vec{p} \times \vec{q}$

So required equation of line is $\vec{r} = \vec{a} + \lambda(\vec{p} \times \vec{q})$

Q. **Prove that for any vector**

$|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2 = 2\vec{a}^2$

Sol. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\therefore \vec{a} \times \hat{i} = \left(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}\right) \times \hat{i} = a_1(\hat{i} \times \hat{i}) + a_2(\hat{j} \times \hat{i}) + a_3(\hat{k} \times \hat{i})$

$= a_3\hat{j} - a_2\hat{k}$

$\therefore |\vec{a} \times \hat{i}| = \sqrt{a_3^2 + a_2^2}$

Similarly, $|\vec{a} \times \hat{j}| = \sqrt{a_1^2 + a_3^2}$ and $|\hat{a} \times \hat{k}| = \sqrt{a_1^2 + a_2^2}$

L.H.S. $= |\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$

$= a_2^2 + a_3^2 + a_1^2 + a_3^2 + a_1^2 + a_2^2$

$= 2\left(a_1^2 + a_2^2 + a_3^2\right) = 2|\vec{a}|^2 = \text{R.H.S.}$

Q. **Find the equation of the line through the point with p.v. $2\hat{i} + 3\hat{j}$ and perpendicular to the vectors $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{j} + 53\hat{k}$**

Sol. $\vec{a} = 2\hat{i} + 3\hat{j}$

Equation of line $\vec{r} = \vec{a} + \lambda\vec{b}$

$\vec{b} = \vec{A} \times \vec{B}$

$$\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & 4 & 53 \end{vmatrix} = \hat{i}(106 - 12) - \hat{j}(53 - 9) + \hat{k}(4 - 6) = 94\hat{i} - 44\hat{j} - 2\hat{k}$$

$\vec{r} = (2\hat{i} + 3\hat{j}) + \lambda\left(94\hat{i} - 44\hat{j} - 2\hat{k}\right).$

Q. **If $(2\hat{i}+6\hat{j}+27\hat{k})\times(\hat{i}+\lambda\hat{j}+\mu\hat{k})=0$, find λ and μ.**

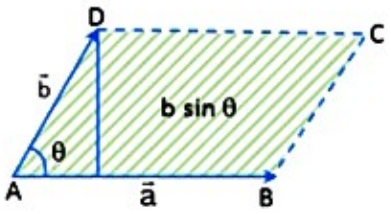
Sol. $\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 6 & 27\\ 1 & \lambda & \mu\end{vmatrix}=\hat{i}(6\mu-27\lambda)-\hat{j}(2\mu-27)+\hat{k}(2\lambda-6)=0$

So, $2\lambda-6=0\Rightarrow\lambda=3$

and $2\mu-27=0$

$\Rightarrow \mu=\frac{27}{2}$

Geometrical interpretation of vector product :



The vector product of the vectors $\vec{a}$ and $\vec{b}$ represents a vector whose modulus is equal to the area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ and $\vec{b}$.

Area of parallelogram = base × height $= ab\sin\theta=|\vec{a}\times\vec{b}|$

Area of quadraliteral if its diagonal vectors are $\vec{d}_1$ & $\vec{d}_2$ is given by $=\frac{1}{2}|\vec{d}_1\times\vec{d}_2|$

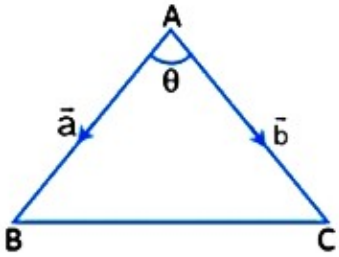
Q. **Find the area of a parallelogram whose two adjacent sides are represented by $\vec{a}=3\hat{i}+\hat{j}+2\hat{k}$ and $b=2\hat{i}-2\hat{j}+4\hat{k}$**

Sol. Area of parallelogram $=|\vec{a}\times\vec{b}|$

Now $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 3 & 1 & 2\\ 2 & -2 & 4\end{vmatrix}=8\hat{i}-8\hat{j}-8\hat{k}$

$\therefore$ Area $=|8\hat{i}-8\hat{j}-8\hat{k}|=8\sqrt{3}$ units

Area of a triangle :



1. Area of triangle $ABC=\frac{1}{2}ab\sin\theta=\frac{1}{2}|\vec{a}\times\vec{b}|=\frac{1}{2}|\overrightarrow{AB}\times\overrightarrow{AC}|$

2. If $\vec{a},\vec{b},\vec{c}$ are position vectors of vertices of a ΔABC then its

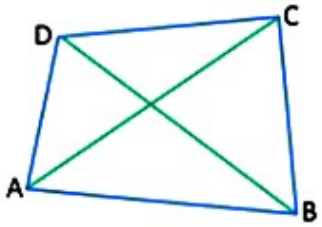
Area $=\frac{1}{2}|(\vec{a}\times\vec{b})+(\vec{b}\times\vec{c})+(\vec{c}\times\vec{a})|$

Q. **Find the area of ΔABC if position vectors of its vertices A, B, C are $\hat{i}+\hat{j}, \hat{j}+\hat{k}$ and $\hat{k}+\hat{i}$ respectively.**

Sol. $\overline{AB} = (\hat{j}+\hat{k}) - (\hat{i}+\hat{j}) = \hat{k} - \hat{i}$

$\overline{AC} = (\hat{k}+\hat{i}) - (\hat{i}+\hat{j}) = \hat{k} - \hat{j}$

$$\therefore \overline{AB} \times \overline{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{vmatrix} = \hat{i}+\hat{j}+\hat{k}$$

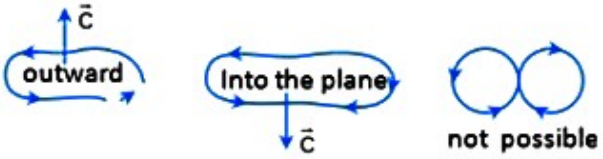


$$\therefore \text{ Area of } \Delta ABC = \frac{1}{2}\left|\overline{AB} \times \overline{AC}\right| = \frac{1}{2}\sqrt{1+1+1} = \sqrt{3}/2$$

INTERPRETATION OF VECTOR PRODUCT AS VECTOR AREA :

1. Vector area of plane figures:

With every closed bound surface which has been described in a certain specific manner and whose boundaries do not cross, it is possible to associate a directed $\vec{c}$ such that



(i) $|\vec{c}|$ = no. of units of area enclosed by the plane figure

(ii) The support of $\vec{c}$ is perpendicular to the area and

(iii) The sense of description of the boundaries and the direction of $\vec{c}$ is in accordance with the R.H. screw rule.

2. Vector area of a plane Δ (Triangle) :

Vector area of ΔOAB is $\vec{\Delta} = \frac{1}{2}(\vec{a} \times \vec{b})$



If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors then the vector area of ΔABC is

$$= \frac{1}{2}\left[(\vec{c}-\vec{b}) \times (\vec{a}-\vec{b})\right]$$

$$= \frac{1}{2}\left((\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})\right)$$

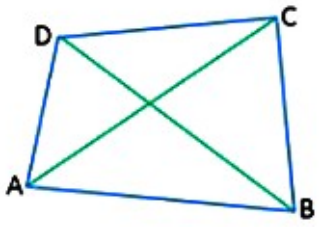


Note :

(i) If 3 points with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are collinear then $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=0$

(ii) Unit vector perpendicular to the plane of the ΔABC when $\vec{a}, \vec{b}, \vec{c}$ are the p.v. of its angular point is $\hat{n}=\pm\dfrac{\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}}{2\Delta}$, where $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of the angular points of the triangle ABC.

(iii) Vector Area of a quadrilateral ABCD = Vector area of ΔABC + vector area of ΔACD



$$=\frac{1}{2}\left(\overline{AB}\times\overline{AC}\right)+\frac{1}{2}\left(\overline{AC}\times\overline{AD}\right)$$

$$=\frac{1}{2}\left(\overline{AB}\times\overline{AC}-\overline{AD}\times\overline{AC}\right)=\frac{1}{2}\left(\overline{AB}-\overline{AD}\right)\times\overline{AC}$$

$$=\frac{1}{2}\left(\overline{AB}+\overline{DA}\right)\times\overline{AC}=\frac{1}{2}\overline{DB}\times\overline{AC}$$

$\therefore$ Area of Quadrilateral $ABCD=\dfrac{1}{2}\left|\overline{DB}\times\overline{AC}\right|=\dfrac{1}{2}\left|\overline{AC}\times\overline{BD}\right|$

Q. **Using vector method, show that the points $A(2,-1,3), B(4,3,1)$ and $C(3,1,2)$ are collinear**

Sol. Let O be the origin

Given

$A\equiv(2,-1,3) \quad \therefore \quad \overline{OA}=2\hat{i}-\hat{j}+3\hat{k}$

$B\equiv(4,3,1) \quad \therefore \quad \overline{OB}=4\hat{i}+3\hat{j}+\hat{k}$

$C\equiv(3,1,2) \quad \therefore \quad \overline{OC}=3\hat{i}+\hat{j}+2\hat{k}$

Now $\overline{AB}=(\text{P.V. of B})-(\text{P.V. of A})=(4\hat{i}+3\hat{j}+\hat{k})-(2\hat{i}-\hat{j}+3\hat{k})=(2\hat{i}+4\hat{j}-2\hat{k})$

And $\overline{AC}=(\text{P.V. of C})-(\text{P.V. of A})=(3\hat{i}+\hat{j}+2\hat{k})-(2\hat{i}-\hat{j}+3\hat{k})=(\hat{i}+2\hat{j}-\hat{k})$

$$\therefore \overline{AB}\times\overline{AC}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&4&-2\\1&2&-1\end{vmatrix}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&2&-1\\1&2&-1\end{vmatrix}=\vec{0} \quad [\because R_2 \text{ and } R_3 \text{ are identical }]$$

Thus, $\overline{AB}$ and $\overline{AC}$ are parallel vectors, having a common point A.

Hence, the points A, B, C are collinear.

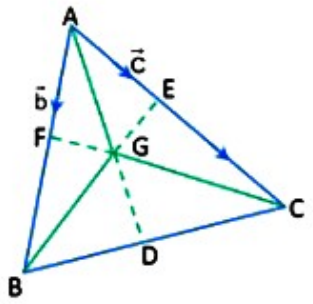
Q. **AD, BE and CF are the medians of a triangle ABC intersecting in G. Show that**

$\Delta AGB = \Delta BGC = \Delta CGA = \frac{1}{3}\Delta ABC$

Sol. Let $\vec{b}, \vec{c}$ be the position vectors of B and C with respect to A as the origin of reference.

Therefore, the position vectors of D, E, F are $\frac{1}{2}(\vec{b}+\vec{c}), \frac{1}{2}\vec{c}, \frac{1}{2}\vec{b}$ respectively. Also the position vector of the point G, the centroid, is

$\frac{1}{3}(0+\vec{b}+\vec{c}) = \frac{1}{3}(\vec{b}+\vec{c})$



Therefore, area of $\Delta AGB = \frac{1}{2}(\overrightarrow{AB} \times \overrightarrow{AG})$

$= \frac{1}{2}\left|\vec{b} \times \frac{1}{3}(\vec{b}+\vec{c})\right| = \frac{1}{6}|\vec{b}\times\vec{b} + \vec{b}\times\vec{c}|$

$= \frac{1}{6}|\vec{b}\times\vec{c}| = \frac{1}{3}$ area of ΔABC

Similarly, we can show that area of $\Delta BGC = \frac{1}{3}$ area of ΔABC and area of $\Delta CGA = \frac{1}{3}$ area of ΔABC

Q. **For a non zero vector $\vec{a}$, if $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}$ and $\vec{a}\times\vec{b} = \vec{a}\times\vec{c}$. Prove that $\vec{b} = \vec{c}$.**

Sol. $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c}$

$\vec{a}\cdot(\vec{b}-\vec{c}) = 0 \Rightarrow \vec{a} \perp \vec{b}-\vec{c}$ and $\vec{a}\times\vec{b} = \vec{a}\times\vec{c}$

$a\times(\vec{b}-\vec{c}) = 0 \Rightarrow \vec{a} \parallel (\vec{b}-\vec{c})$

As a is not zero

$\Rightarrow (\vec{b}-\vec{c}) = 0$

$\vec{b} = \vec{c}$

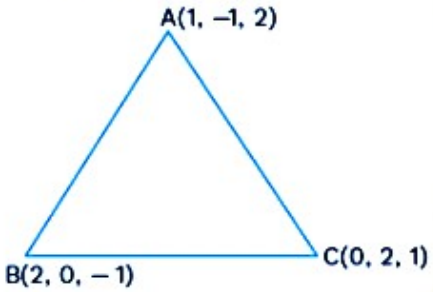
Q. **Find : (i) A vector of magnitude $\sqrt{6}$ perpendicular to the plane ABC**

(ii) Area of triangle ABC

(iii) Length of altitude from A

(iv) Perpendicular distance of a point P from a line using cross product.

$d = \left|\frac{(\vec{a}-\vec{b})\times\vec{c}}{|\vec{c}|}\right|$.

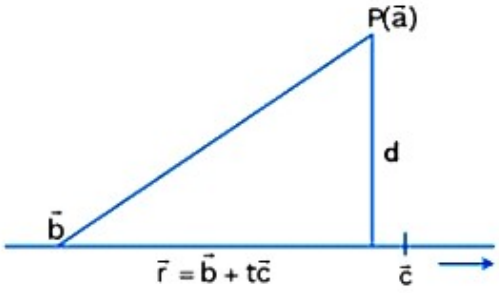


Sol. $\overline{AB} = 2\hat{i} - \hat{k} - \hat{i} + \hat{j} - 2\hat{k}$

$= \hat{i} + \hat{j} - 3\hat{k}$

$\overline{BC} = 2\hat{j} + \hat{k} - 2\hat{i} + \hat{k}$

$= -2\hat{i} + 2\hat{j} + 2\hat{k}$



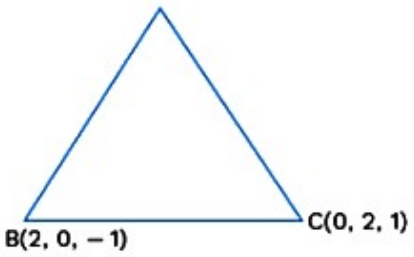
(i) $\overline{AB} \times \overline{BC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -3 \\ -2 & 2 & 2 \end{vmatrix}$

$= (2+6)\hat{i} - \hat{j}(2-6) + \hat{k}(2+2)$

$= 8\hat{i} + 4\hat{j} + 4\hat{k}$

Required vector $= \sqrt{6}\dfrac{8\hat{i} + 4\hat{j} + 4\hat{k}}{4\sqrt{6}}$

$= 2\hat{i} + \hat{j} + \hat{k}$



(ii) Area $= \dfrac{1}{2}|\overline{AB} \times \overline{BC}|$

$= \dfrac{1}{2} 4\sqrt{6} = 2\sqrt{6}$

(iii) $\dfrac{1}{2} AD \times BC =$ Area of ΔABC, where AD is altitude

$\Rightarrow \quad AD = \dfrac{2 \times 2\sqrt{6}}{2\sqrt{3}} = 2\sqrt{2}$

(iv) $|(\vec{a} - \vec{b}) \times \vec{c}|$

$= |\vec{c}| \quad \underbrace{|\vec{a} - \vec{b}| \sin\theta}_{\text{(Perpendicular distance d)}}$

$\Rightarrow \quad d = \dfrac{|(\vec{a} - \vec{b}) \times \vec{c}|}{|\vec{c}|}$

Q. **Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$; $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find the vector $\vec{d}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$ and satisfy $\vec{c} \cdot \vec{d} = 15$.**

Sol. We have,

$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ 3 & -2 & 7 \end{vmatrix} = 32\hat{i} - \hat{j} - 14\hat{k}$

Since $\vec{d}$ -is perpendicular to both $\vec{a}$ and $\vec{b}$ and $\vec{a}\times\vec{b}$ is also perpendicular to both $\vec{a}$ and $\vec{b}$.

$\Rightarrow \vec{d}$ is parallel or collinear to $\vec{a}\times\vec{b}$

$\Rightarrow \vec{d}=\lambda\left(\vec{a}\times\vec{b}\right)$ for some scalar λ

$\Rightarrow \quad \vec{d}=32\lambda\hat{i}-\lambda\hat{j}-14\lambda\hat{k}$

Given $\vec{c}\cdot\vec{d}=15$

$\Rightarrow \quad 2.32\lambda+(-1)(-\lambda)+4(-14\lambda)=15$

$\Rightarrow \quad 9\lambda=15 \Rightarrow \lambda=\frac{5}{3}$

$\Rightarrow \quad \vec{d}=\frac{5}{3}(32\hat{i}-\hat{j}-14\hat{k})$

Q. **Four vertices O, A, B, C of a tetrahedron satisfy** $\overrightarrow{OA}\times\overrightarrow{OB}=\hat{i}-\hat{j}+\hat{k}$; $\overrightarrow{OB}\times\overrightarrow{OC}=\hat{i}$; $\overrightarrow{OC}\times\overrightarrow{OA}=-\hat{i}+\hat{j}$ **find** $\overrightarrow{CA}\times\overrightarrow{CB}$.

Sol. The concept here is the sum of all normal vectors in a tetrahedron is zero.

$\overrightarrow{OA}\times\overrightarrow{OB}+\overrightarrow{OB}\times\overrightarrow{OC}+\overrightarrow{OC}\times\overrightarrow{OA}+\overrightarrow{CB}\times\overrightarrow{CA}=0$

$\overrightarrow{CB}\times\overrightarrow{CA}=-\left(\left(\hat{i}-\hat{j}+\hat{k}\right)+\left(\hat{i}\right)+\left(-\hat{i}+\hat{j}\right)\right)$

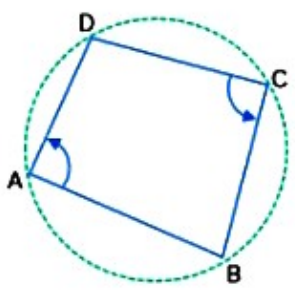
$\overrightarrow{CA}\times\overrightarrow{CB}=-\hat{i}-\hat{k}$

Q. **If** $\vec{a},\vec{b},\vec{c},\vec{d}$ **are position vectors of the vertices of a cyclic quadrilateral ABCD prove that :** $\frac{|\vec{a}\times\vec{b}+\vec{b}\times\vec{d}+\vec{d}\times\vec{a}|}{(\vec{b}-\vec{a})\cdot(\vec{d}-\vec{a})}+\frac{|\vec{b}\times\vec{c}+\vec{c}\times\vec{d}+\vec{d}\times\vec{b}|}{(\vec{b}-\vec{c})\cdot(\vec{d}-\vec{c})}=0$

Sol. $\frac{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{d}+\vec{d}\times\vec{a}\right|}{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{d}-\vec{a}\right)}=\frac{\left|\left(\vec{a}-\vec{d}\right)\times\left(\vec{b}-\vec{a}\right)\right|}{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{d}-\vec{a}\right)}=\frac{\sin A}{\cos A}=\tan A$

$\frac{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{d}+\vec{d}\times\vec{b}\right|}{\left(\vec{b}-\vec{c}\right)\cdot\left(\vec{d}-\vec{c}\right)}$

$=\frac{\left|\left(\vec{b}-\vec{c}\right)\times\left(\vec{c}-\vec{d}\right)\right|}{\left(\vec{b}-\vec{c}\right)\cdot\left(\vec{d}-\vec{c}\right)}=\frac{\sin C}{\cos C}=\tan C$



As it is a cyclic quadrilateral, we have
$A = 180° - C$ or $\tan A = \tan(180° - C)$ or $\tan A + \tan C = 0$

$$\frac{\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{d}+\vec{d}\times\vec{a}\right|}{\left(\vec{b}-\vec{a}\right)\cdot\left(\vec{d}-\vec{a}\right)}+\frac{\left|\vec{b}\times\vec{c}+\vec{c}\times\vec{d}+\vec{d}\times\vec{b}\right|}{\left(\vec{b}-\vec{c}\right)\cdot\left(\vec{d}-\vec{c}\right)}=0$$

Q. **Find the unknown vector $\vec{R}$ satisfying $\vec{R}\times\vec{b}=\vec{c}\times\vec{b}$ and $\vec{R}.\vec{a}=0$ where $\vec{a}=2\hat{i}+\hat{k},\vec{b}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{c}=4\hat{i}-3\hat{j}+7\hat{k}$.**

Sol. $\vec{R}\times\vec{b}=\vec{c}\times\vec{b}\Rightarrow\left(\vec{R}-\vec{c}\right)\times\vec{b}=\vec{0}\Rightarrow\vec{R}=\vec{c}+t\vec{b}$

Since $\vec{R}\cdot\vec{a}=0\therefore\vec{c}\cdot\vec{a}+t\vec{b}\cdot\vec{a}=0\Rightarrow t=-\frac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}$

$$\therefore\vec{R}=\vec{c}-\frac{\vec{c}\cdot\vec{a}}{\vec{b}\cdot\vec{a}}\vec{b}=(4,-3,7)-\frac{15}{3}(1,1,1)$$

$= (4,-3, 7)-(5, 5, 5)$

$=(-1,-8,2)=-\hat{i}-8\hat{j}+2\hat{k}$

Q. **Prove that 3 points with position vectors $\vec{a}+\vec{b},\vec{a}-\vec{b}$ and $\vec{a}+\lambda\vec{b}$ are collinear for $\forall\ \lambda\in$ R.**

Sol. Let the position vectors of points A, B and C be $\vec{a}+\vec{b},\vec{a}-\vec{b},\vec{a}+\lambda\vec{b}$ respectively.

Then, $\overline{AB}$ = P.V. of B - P.V. of A.

$=\vec{a}-\vec{b}-\vec{a}-\vec{b}$

$=-2\vec{b}$

$\overline{BC}$ = P.V. of C - P.V. of B.

$=\vec{a}+\lambda\vec{b}-\vec{a}+\vec{b}$

$=\vec{b}(\lambda+1)$

$\overline{CA}$ = P.V. of A - P.V. of C.

$=\vec{a}+\vec{b}-\vec{a}-\lambda\vec{b}$

$=\vec{b}(1-\lambda)$

Now, the points A,B,C are collinear if and only if, $\overline{AB}$ and $\overline{CA}$ are multiples of $\overline{BC}$

Let, $\overline{AB}=\beta\overline{BC}$

$\Rightarrow -2\vec{b}=\beta\vec{b}(\lambda+1)$

$\Rightarrow -2=\beta(\lambda+1)$

$\Rightarrow \beta = -\frac{2}{\lambda+1}$

Hence, for real values of λ, the given position vectors are parallel.

Q. **Prove the identity $(\vec{a}-\vec{b})\times(\vec{a}+\vec{b}) = 2(\vec{a}\times\vec{b})$ and give its geometrical interpretation.**

Sol. L.H.S $= (\vec{a}-\vec{b})\times(\vec{a}+\vec{b})$

$= \vec{a}\times\vec{a} + \vec{a}\times\vec{b} - \vec{b}\times\vec{a} - \vec{b}\times\vec{b}$ [By distributive law]

$= \vec{a}\times\vec{b} - \vec{b}\times\vec{a}$

$[\because \vec{a}\times\vec{a} = 0$ and $\vec{b}\times\vec{b} = 0]$

$= \vec{a}\times\vec{b} + \vec{a}\times\vec{b}$

$= 2(\vec{a}\times\vec{b})$

If $\vec{a}$ and $\vec{b}$ are the adjacent sides of a parallelogram, then its diagonal are $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$

Now, $(\vec{a}-\vec{b})\times(\vec{a}+\vec{b}) = 2(\vec{a}\times\vec{b})$

$\Rightarrow \vec{a}\times\vec{b} = \frac{1}{2}(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})$

$\Rightarrow \left|\vec{a}\times\vec{b}\right| = \frac{1}{2}\left|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})\right|$

$\Rightarrow$ Area of the parallelogram

$= \frac{1}{2}$ (Area of the parallelogram whose adjacent side are diagonal of the given parallelogram)

Previous Year's Question

If $\vec{a},\vec{b},\vec{c}$ and $\vec{d}$ are unit vectors such that $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = 1$ and $\vec{a}\cdot\vec{c} = \frac{1}{2}$, then

(A) $\vec{a},\vec{b},\vec{c}$ are non-coplanar

(B) $\vec{b},\vec{c},\vec{d}$ are non-coplanar **[JEE-2009]**

(C) $\vec{b},\vec{d}$ are non-parallel

(D) $\vec{a},\vec{d}$ are parallel and $\vec{b},\vec{c}$ -are parallel

Q. **Let $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 10\vec{a} + 2\vec{b}$ and $\overrightarrow{OC} = \vec{b}$ where O, A & C are non-collinear points. Let 'p' denote the area of the quadrilateral OABC, and let 'q' denote the area of the parallelogram with OA and OC as adjacent sides. If p = kq. Find k.**

Sol. Here $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = 10\vec{a} + 2\vec{b}, \overrightarrow{OC} = \vec{b}$

q = Area of parallelogram with OA and OC as adjacent sides.

$\therefore\ q = |\vec{a} \times \vec{b}|$ ...(i)

p = Area of quadrilateral OABC

= Area of ΔOAB + Area of ΔOBC

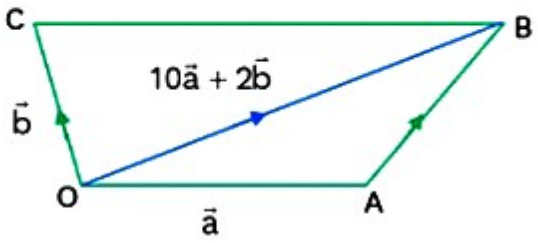
$$= \frac{1}{2}\left|\vec{a} \times \left(10\vec{a} + 2\vec{b}\right)\right| + \frac{1}{2}\left|\left(10\vec{a} + 2\vec{b}\right) \times \vec{b}\right|$$

$$= |\vec{a} \times \vec{b}| + 5|\vec{a} \times \vec{b}|$$

$$\therefore\ p = 6|\vec{a} \times \vec{b}|$$

or p = 6q [From Eq. (i)]

$\therefore$ k = 6



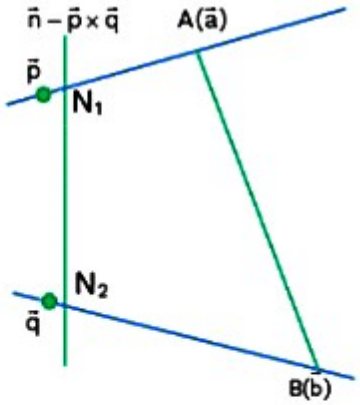
Shortest distance between 2 skew lines :

Note:

(i) 2 lines in a plane if not || must intersect and 2 lines in a plane if not intersecting must be parallel. Convertely 2 intersecting or parallel lines must be coplanar.

(ii) In space, however we come across situation when two lines neither intersect nor ||, Two such lines (like the flight paths of two planes) in space are known as skew lines or non coplanar lines.

(iii) S.D. between two such skew lines is the segment intercepted between the two lines and perpendicular to both.



Method I: Two ways to determine the S.D.

$L_1 : \vec{r} = \vec{a} + \lambda\vec{p}$

$L_2 : \vec{r} = \vec{b} + \mu\vec{q}$

$\vec{n} = \vec{p} \times \vec{q}$

$\overrightarrow{AB} = (\vec{b} - \vec{a})$

$$\text{S.D.} = \left|\text{Projection of } \overline{AB} \text{ on } \vec{n}\right| = \left|\frac{\overline{AB}\cdot\vec{n}}{|\vec{n}|}\right| = \left|\frac{(\vec{b}-\vec{a})\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right|$$

If S.D. = 0 $\Rightarrow$ lines are intersecting and hence coplanar.

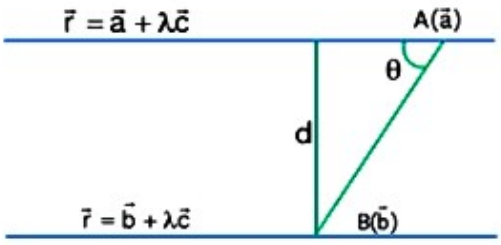
Method II: p.v. of $N_1 = \vec{a} + \lambda\vec{p}$; p.v. of $N_2 = \vec{b} + \mu\vec{q}$

$\vec{N}_2 - \vec{N}_1 = (\vec{b}-\vec{a}) + (\mu\vec{q} - \lambda\vec{p})$

Now $(\vec{N}_1 - \vec{N}_2).\vec{p} = 0$ and $(\vec{N}_1 - \vec{N}_2).\vec{q} = 0$ (two linear equations to get the unique values of λ and μ.) One p.v's of N_1 and N_2 are known we can also determine the equation to the line of shortest distance and the S.D.

Shortest Distance between two parallel lines :

$$d = |\vec{a}-\vec{b}|\sin\theta = \left|\frac{(\vec{a}-\vec{b})\times\vec{c}}{|\vec{c}|}\right|$$



Q. **Find the shortest distance between the two lines whose vector equations are given by :** $\vec{r} = \hat{i} + 2\hat{j} + 3\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 4\hat{k})$ and $\vec{r} = 2\hat{i} + 4\hat{j} + 5\hat{k} + \mu(3\hat{i} + 4\hat{j} + 5\hat{k})$

Sol. If the equations of the lines are $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$, then shortest distance 'd' between them is given by

$$d = \left|\frac{(\vec{a}_2 - \vec{a}_1)\cdot(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right| \quad \ldots(i)$$

Here $\vec{a}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b}_1 = 2\hat{i} + 3\hat{j} + 4\hat{k}$

$\vec{a}_2 = 2\hat{i} + 4\hat{j} + 5\hat{k}$, $\vec{b}_2 = 3\hat{i} + 4\hat{j} + 5\hat{k}$

Now, $\vec{a}_2 - \vec{a}_1 = (2\hat{i} + 4\hat{j} + 5\hat{k}) - (\hat{i} + 2\hat{j} + 3\hat{k})$

$= \hat{i} + 2\hat{j} + 2\hat{k}$ ...(ii)

$$\vec{b}_1\times\vec{b}_2 = \begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & 3 & 4\\ 3 & 4 & 5\end{vmatrix}$$

$= (15-16)\hat{i} - (10-12)\hat{j} + (8-9)\hat{k}$

$= -\hat{i} + 2\hat{j} - \hat{k}$ ...(iii)

$\left|\vec{b}_1 \times \vec{b}_2\right| = \sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{6}$ ...(iv)

and $\left(\vec{a}_2 - \vec{a}_1\right)\cdot\left(\vec{b}_2 \times \vec{b}_1\right) = (\hat{i} + 2\hat{j} + 2\hat{k})\cdot(-\hat{i} + 2\hat{j} - \hat{k})$

$= 1\times(-1) + 2\times 2 + 2\times(-1) = 1$ ...(v)

Substituting the values from (iv) and (v) in (i), we get $d = \left|\frac{1}{\sqrt{6}}\right| = \frac{1}{\sqrt{6}}$

Q. **Determine whether the following pair of lines intersect $\vec{r} = \hat{i} - \hat{j} + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k})$.**

Sol. The shortest distance between the given pair of lines is given by

$$d = \left|\frac{\left(\vec{a}_2 - \vec{a}_1\right)\cdot\left(\vec{b}_1 \times \vec{b}_2\right)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right|$$

The two lines will intersect if and only if d = 0.

i.e. $\left(\vec{a}_2 - \vec{a}_1\right)\cdot\left(\vec{b}_1 \times \vec{b}_2\right) = 0$

Here, equation of first line is $\vec{r} = \hat{i} - \hat{j} + \lambda(2\hat{i} + \hat{k}) = \vec{a}_1 + \lambda\vec{b}_1$

where $\vec{a}_1 = \hat{i} - \hat{j}$ and $\vec{b}_1 = 2\hat{i} + \hat{k}$

Also equation of second line is $\vec{r} = 2\hat{i} - \hat{j} + \mu(\hat{i} + \hat{j} - \hat{k}) = \vec{a}_2 + \mu\vec{b}_2$

where $\vec{a}_2 = 2\hat{i} - \hat{j}$ and $\vec{b}_2 = \hat{i} + \hat{j} - \hat{k}$

Now, $\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -\hat{i} + 3\hat{j} + 2\hat{k}$ and $\vec{a}_2 - \vec{a}_1 = 2\hat{i} - \hat{j} - (\hat{i} - \hat{j}) = \hat{i}$

Since $\left(\vec{a}_2 - \vec{a}_1\right)\cdot\left(\vec{b}_1 \times \vec{b}_2\right) = \hat{i}\cdot(-\hat{i} + 3\hat{j} + 2\hat{k}) = (-1)(1) + 3(0) + 2(0) = -1 \neq 0$

Hence the given lines do not intersect.

Q. **Find shortest distance between**

$\vec{r} = \hat{i} + \hat{j} + \lambda(2\hat{i} - \hat{j} + \hat{k})$,

$\vec{r} = 2\hat{i} + \hat{j} - \hat{k} + \mu(3\hat{i} - 5\hat{j} + 2\hat{k})$

Sol. Here $\vec{a}_1 = \hat{i} + \hat{j}$, $\vec{b}_1 = 2\hat{i} - \hat{j} + \hat{k}$

$\vec{a}_2 = 2\hat{i} + \hat{j} - \hat{k}, \vec{b}_2 = 3\hat{i} - 5\hat{j} + 2\hat{k}$

$\therefore \quad \vec{a}_2 - \vec{a}_1 = \hat{i} - \hat{k}$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 3 & -5 & 2 \end{vmatrix} = 3\hat{i} - \hat{j} - 7\hat{k}$$

$\therefore |\vec{b}_1 \times \vec{b}_2| = \sqrt{9 + 1 + 49} = \sqrt{59}$

$\therefore$ The shortest distance between the given lines is

$$d = \left| \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

$$= \frac{|3 - 0 + 7|}{\sqrt{59}} = \frac{10}{\sqrt{59}}$$

Q. **Vector equation of two lines are**

$\vec{r} = (1 - t)\hat{i} + (t - 2)\hat{j} + (3 - 2t)\hat{k}$,

$\vec{r} = (s + 1)\hat{i} + (2s - 1)\hat{j} - (2s + 1)\hat{k}$

Find shortest distance.

Sol. $\vec{r} = (1 - t)\hat{i} + (t - 2)\hat{j} + (3 - 2t)\hat{k} = (\hat{i} - 2\hat{j} + 3\hat{k}) + t(-\hat{i} + \hat{j} - 2\hat{k}) = \vec{a}_1 + t\vec{b}_1$

where $\vec{a}_1 = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b}_1 = -\hat{i} + \hat{j} - 2\hat{k}$

Again, $\vec{r} = (s + 1)\hat{i} + (2s - 1)\hat{j} - (2s + 1)\hat{k} = (\hat{i} - \hat{j} - \hat{k}) + s(\hat{i} + 2\hat{j} - 2\hat{k})$

$\vec{r} = \vec{a}_2 + s\vec{b}_2$

where $\vec{a}_2 = \hat{i} - \hat{j} - \hat{k}$ and $\vec{b}_2 = \hat{i} + 2\hat{j} - 2\hat{k}$

$$\text{Now, } \vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & -2 \\ 1 & 2 & -2 \end{vmatrix}$$

$= \hat{i}(-2+4) - \hat{j}(2+2) + \hat{k}(-2-1) = 2\hat{i} - 4\hat{j} - 3\hat{k}$

$\therefore \left|\vec{b}_1 \times \vec{b}_2\right| = \sqrt{(2)^2 + (-4)^2 + (-3)^2} = \sqrt{4+16+9} = \sqrt{29}$

$\vec{a}_2 - \vec{a}_1 = \hat{i} - \hat{j} - \hat{k} - (\hat{i} - 2\hat{j} + 3\hat{k}) = \hat{j} - 4\hat{k}$

Now $\left(\vec{b}_1 \times \vec{b}_2\right) \cdot \left(\vec{a}_2 - \vec{a}_1\right) = (2\hat{i} - 4\hat{j} - 3\hat{k}) \cdot (\hat{j} - 4\hat{k})$

$= 2(0) + (-4)(1) + (-3)(-4) = 0 - 4 + 12 = 8$

$\therefore \; d = \dfrac{\left|\left(\vec{b}_1 \times \vec{b}_2\right) \cdot \left(\vec{a}_2 - \vec{a}_1\right)\right|}{\left|\vec{b}_1 \times \vec{b}_2\right|} = \dfrac{|8|}{\sqrt{29}} = \dfrac{8}{\sqrt{29}} = \dfrac{8\sqrt{29}}{29}$ units

Q. **Find the distance between the lines given by**

$\vec{r} = (\hat{i} + 2\hat{j} - 4\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

$\vec{r} = (3\hat{i} + 3\hat{j} - 5\hat{k}) + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$

Sol. Here $\vec{a}_1 = \hat{i} + 2\hat{j} - 4\hat{k}, \vec{a}_2 = 3\hat{i} + 3\hat{j} - 5\hat{k}$

and $\vec{b} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ [$\because$ both lines are parallel]

$\therefore \; \vec{a}_2 - \vec{a}_1 = (3\hat{i} + 3\hat{j} - 5\hat{k}) - (\hat{i} + 2\hat{j} - 4\hat{k})$

$= 2\hat{i} + \hat{j} - \hat{k}$

$|\vec{b}| = \sqrt{(2)^2 + (3)^2 + (6)^2}$

$= \sqrt{4+9+36} = 7$

and $\left(\vec{a}_2 - \vec{a}_1\right) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 2 & 3 & 6 \end{vmatrix}$

$= \hat{i}(6+3) - \hat{j}(12+2) + \hat{k}(6-2)$

$= 9\hat{i} - 14\hat{j} + 4\hat{k}$

$\therefore \quad \left|\left(a_2 - a_1\right) \times \vec{b}\right| = \sqrt{(9)^2 + (-14)^2 + (4)^2} = = \sqrt{81+196+16} = \sqrt{293}$

$\therefore$ Shortest distance between the lines is

S.D. $d = \left|\dfrac{\left(\vec{a}_2 - \vec{a}_1\right) \times \vec{b}}{|\vec{b}|}\right| = \left|\dfrac{\sqrt{293}}{7}\right| = \dfrac{\sqrt{293}}{7}$

PRODUCT OF THREE OR MORE VECTORS :

Scalar triple product :

Definition :

If $\vec{a},\vec{b},\vec{c}$ are three vectors, then their scalar triple product is defined as the dot product of two vectors $\vec{a}$ and $\vec{b}\times\vec{c}$. It is generally denoted by $\vec{a}\cdot(\vec{b}\times\vec{c})$ or $[\vec{a}\ \vec{b}\ \vec{c}]$. It is read as box product of $\vec{a},\vec{b},\vec{c}$. Similarly other scalar triple products can be defined as $(\vec{b}\times\vec{c})\cdot\vec{a},(\vec{c}\times\vec{a})\cdot\vec{b}$

Know the facts

Scalar triple product always results in a scalar quantity (number).

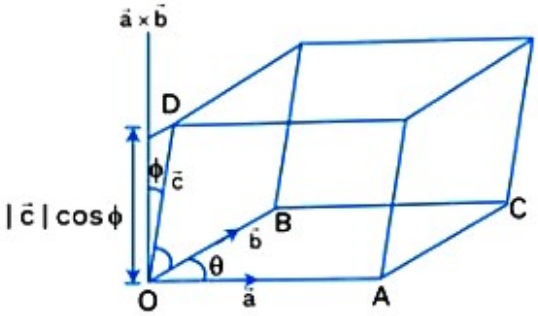
Geometrical Interpretation :

$(\vec{a}\times\vec{b})\cdot\vec{c} = |\vec{a}||\vec{b}|\sin\theta\hat{n}\cdot\vec{c} = |\vec{a}||\vec{b}||\vec{c}|\sin\theta\cos\phi$

where $\theta = \vec{a}\wedge\vec{b};\phi = \vec{n}\wedge\vec{c}$

But

$|\vec{a}||\vec{b}|\sin\theta =$ area of $\|^{gm}$ OACB and $|\vec{c}|\cos\phi = h$



There absolute value of scalar triple product of three vectors is equal to the volume of the parallelopiped whose three coterminous edges are represented by the given vectors.

Therefore $|(\vec{a}\times\vec{b})\cdot\vec{c}| = |[\vec{a}\ \vec{b}\ \vec{c}]|$ = Volume of the parallelopiped whose coterminous edges are $\vec{a},\vec{b}$ and $\vec{c}$.

Point to Remember!!!

Formula for scalar Triple Product :

If $\vec{a} = a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$,

$\vec{b} = b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$ and $\vec{c} = c_1\hat{i}+c_2\hat{j}+c_3\hat{k}$,

then

$$[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Properties of Scalar Triple product :

1. The position of $(.)$ and $(\times)$ can be interchanged i.e. $\vec{a}\cdot(\vec{b}\times\vec{c}) = (\vec{a}\times\vec{b})\cdot\vec{c}$
2. $[\vec{a}\ \vec{b}\ \vec{c}]=[\vec{b}\ \vec{c}\ \vec{a}]=[\vec{c}\ \vec{a}\ \vec{b}]$

   Therefore if we don't change the cyclic order of a, b and c then the value of scalar triple product is not changed.
3. If the cyclic order of vectors is changed, then sign of scalar triple product is changed i.e. $\vec{a}\cdot[\vec{b}\times\vec{c}] = -\vec{a}\cdot(\vec{c}\times\vec{b})$ or $[\vec{a}\ \vec{b}\ \vec{c}]= -[\vec{a}\ \vec{c}\ \vec{b}]$

   From (ii) and (iii) we have

$$\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\left[\vec{b}\ \vec{c}\ \vec{a}\right]=\left[\vec{c}\ \vec{a}\ \vec{b}\right]=-\left[\vec{a}\ \vec{c}\ \vec{b}\right]=-\left[\vec{b}\ \vec{a}\ \vec{c}\right]=-\left[\vec{c}\ \vec{b}\ \vec{a}\right]$$

4. The scalar triple product of three vectors when two of them are equal or parallel, is zero i.e. $\left[\vec{a}\ \vec{b}\ \vec{b}\right]=\left[\vec{a}\ \vec{b}\ \vec{a}\right]=0$

5. The scalar triple product of three mutually perpendicular unit vectors is ± 1. Thus
$\left[\hat{i}\ \hat{j}\ \hat{k}\right]=1, \left[\hat{i}\ \hat{k}\ \hat{j}\right]=-1$

6. If two of the three vectors $\vec{a},\vec{b},\vec{c}$ are parallel then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$

7. $\vec{a},\vec{b},\vec{c}$ are three coplanar vectors if $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=0$ i.e. the necessary and sufficient condition for three non-zero non-collinear vectors to be coplanar is $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=0$

8. For any vectors $\vec{a},\vec{b},\vec{c},\vec{d}$
$\left[\vec{a}+\vec{b}\ \ \vec{c}\ \ \vec{d}\right]=\left[\vec{a}\ \ \vec{c}\ \ \vec{d}\right]+\left[\vec{b}\ \ \vec{c}\ \ \vec{d}\right]$

9. For right handed system, $\left[\vec{a}\ \vec{b}\ \vec{c}\right]>0$ and for left handed system, $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]<0$; where $\vec{a},\vec{b},\vec{c}$ are non-coplanar.

10. $\left[\vec{a}+\vec{b},\ \vec{b}+\vec{c},\ \vec{c}+\vec{a}\right]=2\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$

11. $\left[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}\right]$ is always zero.

12. $\left[\vec{l}\ \vec{m}\ \vec{n}\right]\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\begin{vmatrix}\vec{l}\cdot\vec{a} & \vec{l}\cdot\vec{b} & \vec{l}\cdot\vec{c}\\ \vec{m}\cdot\vec{a} & \vec{m}\cdot\vec{b} & \vec{m}\cdot\vec{c}\\ \vec{n}\cdot\vec{a} & \vec{n}\cdot\vec{b} & \vec{n}\cdot\vec{c}\end{vmatrix}$, where $\vec{l},\vec{m},\vec{n}\ \&\ \vec{a},\vec{b},\vec{c}$ are non coplanar vectors.

13. $\left[\vec{a}\ \vec{b}\ \vec{c}\right](\vec{p}\times\vec{q})=\begin{vmatrix}\vec{p}\cdot\vec{a} & \vec{q}\cdot\vec{a} & \vec{a}\\ \vec{p}\cdot\vec{b} & \vec{q}\cdot\vec{b} & \vec{b}\\ \vec{p}\cdot\vec{c} & \vec{q}\cdot\vec{c} & \vec{c}\end{vmatrix}$

14. If $\vec{a}=a_1\vec{\ell}+a_2\vec{m}+a_3\vec{n}$, $\vec{b}=b_1\vec{\ell}+b_2\vec{m}+b_3\vec{n}$ and $\vec{c}=c_1\vec{\ell}+c_2\vec{m}+c_3\vec{n}$, then

$$\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}\left[\vec{\ell}\ \vec{m}\ \vec{n}\right]$$

Q. **If $\vec{a} = 2\hat{i} - 3\hat{j}, \vec{b} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 3\hat{i} - \hat{k}$ represent three coterminous edges of a parallelopiped, then the volume of that parallelopiped is-**

(A) 2 **(B) 4** **(C) 6** **(D) 10**

Sol. **(B)**

Volume $= \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$

$$= \left|\begin{vmatrix} 2 & -3 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix}\right| = \left|-2 + 9 - 3\right| = 4$$

Q. **For any three vectors $\vec{a}, \vec{b}, \vec{c}$ $\left[\vec{a} + \vec{b},\ \vec{b} + \vec{c},\ \vec{c} + \vec{a}\right]$ equals-**

(A) $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$ **(B)** $2\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]$ **(C)** $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]^2$ **(D) 0**

Sol. **(B)**

For any three vectors $\left[\vec{a} + \vec{b}\ \vec{b} + \vec{c}\ \ \vec{c} + \vec{a}\right] = \left(\vec{a} + \vec{b}\right) \cdot \left[\left(\vec{b} + \vec{c}\right) \times \left(\vec{c} + \vec{a}\right)\right]$

$= \left(\vec{a} + \vec{b}\right) \cdot \left[\left(\vec{b} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{c} + \vec{c} \times \vec{a}\right)\right]$

$= \left(\vec{a} + \vec{b}\right) \cdot \left(\vec{b} \times \vec{c} + \vec{b} \times \vec{a} + \vec{c} \times \vec{a}\right) \qquad \left[\because \vec{c} \times \vec{c} = 0\right]$

$= \left[\vec{a}\vec{b}\vec{c}\right] + \left[\vec{a}\vec{b}\vec{a}\right] + \left[\vec{a}\vec{c}\vec{a}\right] + \left[\vec{b}\vec{b}\vec{c}\right] + \left[\vec{b}\vec{b}\vec{a}\right] + \left[\vec{b}\vec{c}\vec{a}\right]$

$= \left[\vec{a}\vec{b}\vec{c}\right] + \left[\vec{b}\vec{c}\vec{a}\right] = 2\left[\vec{a}\vec{b}\vec{c}\right]$

Q. If $\vec{a}, \vec{b}, \vec{c}$ be three non- zero vectors, then $\left|\left(\vec{a} \times \vec{b}\right) \cdot \vec{c}\right| = \left|\vec{a}\right|\left|\vec{b}\right|\left|\vec{c}\right|$, if –

(A) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = 0$ **(B)** $\vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

(C) $\vec{c} \cdot \vec{a} = \vec{a} \cdot \vec{b} = 0$ **(D)** $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$

Sol. **(D)**

$\left|\left(\vec{a} \times \vec{b}\right) \cdot c\right| = \left|\vec{a}\right|\left|\vec{b}\right|\left|\vec{c}\right| \Leftrightarrow \left|\left(a \times \vec{b}\right)\right|\left|\vec{c}\right|\cos\theta\left|\vec{a}\right|\left|\vec{b}\right|\left|\vec{c}\right|$

(where θ is the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$) $= \left|\vec{a}\right|\left|\vec{b}\right|\left|\vec{c}\right|\sin\phi\cos\theta = \left|\vec{a}\right|\left|\vec{b}\right|\left|\vec{c}\right|$

(where ϕ is the angle between $\vec{a}$ and $\vec{b}$) $\Leftrightarrow \sin\phi = 1, \cos\theta = 1 \Leftrightarrow \phi = 90°, \theta = 0°$

$\Leftrightarrow \vec{a} \cdot \vec{b} = 0, \vec{a} \cdot \vec{c} = 0, \vec{b} \cdot \vec{c} = 0$

Q. **If vectors $a\hat{i}+\hat{j}+\hat{k}$, $\hat{i}+b\hat{j}+\hat{k}$ and $\hat{i}+\hat{j}+c\hat{k}\,(a\neq b\neq c\neq 1)$ are coplanar, then $\frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}$ equals**

(A) 1 **(B) 0** **(C) –1** **(D) None of these**

Sol. **(A)**

Since vectors are coplanar,

$$\therefore \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} a & 1 & 1 \\ 1-a & b-1 & 0 \\ 0 & 1-b & c-1 \end{vmatrix} = 0 \quad \left[\text{Using} R_2 - R_1, R_3 - R_2\right]$$

$$\Rightarrow a(b-1)(c-1)-(1-a)\left[(c-1)-(1-b)\right]=0$$

$$\Rightarrow a(1-b)(1-c)+(1-a)(1-c)+(1-a)(1-b)=0$$

$$\Rightarrow (a-1+1)(1-b)(1-c)+(1-a)(1-c)+(1-a)(1-b)=0$$

$$\Rightarrow (1-b)(1-c)+(1-a)(1-c)+(1-a)(1-b)=(1-a)(1-b)(1-c)$$

$$\Rightarrow \frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}=1$$

Q. **If $\vec{u}=2\hat{i}-\hat{j}+\hat{k}$, $\vec{v}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{w}$ is unit vector then find the maximum value of $[\vec{u}\ \vec{v}\ \vec{w}]$.**

Sol. $[\vec{u}\,\vec{v}\,\vec{w}] = (\vec{u}\times\vec{v})\cdot\vec{w}$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} \cdot \vec{w}$$

$$= [\hat{i}(-2)-\hat{j}(1)+\hat{k}(3)]\cdot\vec{w} \le \sqrt{4+1+9}$$

$\therefore$ Maximum value of $\left[\vec{u}\,\vec{v}\,\vec{w}\right] = \sqrt{14}$

Q. **Find the value of λ for which three points with p.v.'s A (1, 0, 3), B (–1, 3, 4), C (1, 2, 1) and D (λ, 2, 5) are in the same plane. Also find the equations of the plane through ABC.**

Sol. Equation of plane is

$$\begin{vmatrix} x-1 & y-0 & z-3 \\ -1-1 & 3-0 & 4-3 \\ 1-1 & 2-0 & 1-3 \end{vmatrix} = 0$$

$$(x-1)(-6-2) - y(4-0) + (z-3)(-4-0) = 0$$

$-8x + 8 - 4y - 4z + 12 = 0$

$2x + y + z = 5$

$\therefore 2(\lambda) + 2 + 5 = 5 \Rightarrow \lambda = -1$

Q. **Find the value of p for which the vectors $(p+1)\hat{i} - 3\hat{j} + p\hat{k}, p\hat{i} + (p+1)\hat{j} - 3\hat{k}$ and $-3\hat{i} + p\hat{j} + (p+1)\hat{k}$ are coplanar.**

Sol. The vectors are linearly dependent

$$\Rightarrow \begin{vmatrix} p+1 & -3 & p \\ p & p+1 & -3 \\ -3 & p & p+1 \end{vmatrix} = 0 \qquad c_1 \to c_1 + c_2 + c_3 \text{ and take } (2p-2) \text{ common}$$

$$\Rightarrow (2p-2)\begin{vmatrix} 1 & -3 & p \\ 1 & p+1 & -3 \\ 1 & p & p+1 \end{vmatrix} = 0$$

$$\Rightarrow 2(p-1)\begin{vmatrix} 1 & -3 & p \\ 0 & p+4 & -3-p \\ 0 & p+3 & 1 \end{vmatrix} = 0$$

$\Rightarrow 2(p-1)(p+4+(p+3)^2) = 0$

$\Rightarrow (p-1)(p^2+7p+13) = 0$

Roots of $p^2 + 7p + 13 = 0$ are (imaginary)

$\therefore p = 1$

Q. **Find the altitude of the parallelopiped whose coterminous edges are vectors, $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + 4\hat{j} - \hat{k}, \vec{c} = \hat{i} + \hat{j} + 3\hat{k}$ where $\vec{a}$ & $\vec{b}$ are base vectors.**

Sol. $$h = \frac{\text{Volume of parallelepiped}}{\text{Area of base}}$$

$$= \frac{[\vec{A}\vec{B}\vec{C}]}{|\vec{A} \times \vec{B}|} = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & -1 \\ 1 & 1 & 3 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 2 & 4 & -1 \end{vmatrix}} = \frac{4}{|-5\hat{i} + 3\hat{j} + 2\hat{k}|} = \frac{2\sqrt{38}}{19}$$

Q. **The vector $\overline{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ turns through a right angle, passing through the positive xaxis on the way. Find the vector in its new position.**

Sol. Given, $\overline{OP} = \hat{i} + 2\hat{j} + 2\hat{k}$ after rotation of $\overline{OP}$, let new vector is $\overline{OP'}$.

Now, $\overline{OP}, \hat{i}, \overline{OP'}$ will be coplanar

So, $\overline{OP'} = |\overline{OP}| \dfrac{(\overline{OP} \times \hat{i}) \times \overline{OP}}{|(\overline{OP} \times \hat{i}) \times \overline{OP}|} \left[\because |\overline{OP}| = \left|\overline{OP'}\right|\right]$

But, $(\overline{OP} \times \hat{i}) \times \overline{OP} = 8\hat{i} - 2\hat{j} - 2\hat{k}$

$\Rightarrow \quad \overline{OP'} = \dfrac{3(8\hat{i} - 2\hat{j} - 2\hat{k})}{2 \times 3\sqrt{2}}$

$\Rightarrow \quad \overline{OP'} = \dfrac{4}{\sqrt{2}}\hat{i} - \dfrac{1}{\sqrt{2}}\hat{j} - \dfrac{1}{\sqrt{2}}\hat{k}$

Q. **$2\bar{v} + \bar{v} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k}$ then find $3|\bar{v}|$.**

Sol. $2\bar{V} + \bar{V} \times (\hat{i} + 2\hat{j}) = (2\hat{i} + \hat{k})$ (i) or

$2\bar{V} \cdot (\hat{i} + 2\hat{j}) + 0 = (2\hat{i} + \hat{k}) \cdot (\hat{i} + 2\hat{j})$

or $2\bar{V} \cdot (\hat{i} + 2\hat{j}) = 2$ or

$|\bar{V} \cdot (\hat{i} + 2\hat{j})| = 1$ or

$|\bar{V}|^2 \cdot |\hat{i} + 2\hat{j}|^2 \cos^2\theta = 1$

(θ is the angle between $\bar{V}$ and $\hat{i} + 2\hat{j}$)

or $|\bar{V}|^2\, 5\left(1 - \sin^2\theta\right) = 1$ or

$\left|\bar{V}\right|^2 5\sin^2\theta = 5\left|\bar{V}\right|^2 - 1$ (ii)From

Eq. (i), we have

$|2\bar{V} + \bar{V} \times (\hat{i} + 2\hat{j})|^2 = |2\hat{i} + \hat{k}|^2$

or $\quad 4|\bar{V}|^2 + |\bar{V} \times (\hat{i} + 2\hat{j})|^2 = 5$

or $\quad 4|\bar{V}|^2 + |\bar{V}|^2 \cdot |\hat{i} + 2\hat{j}|^2 \sin^2\theta = 5$

or $\quad 4|\bar{V}|^2 + 5|\bar{V}|^2 \sin^2\theta = 5$

or $\quad 4\left|\bar{V}\right|^2 + 5\left|\bar{V}\right|^2 - 1 = 5$ or $9|\bar{V}|^2 = 6$ or $3\left|\bar{V}\right| = \sqrt{6}$

Q. **Show that :** $\vec{a}\times\vec{b}=[\hat{i}\vec{a}\vec{b}]\hat{i}+[\hat{j}\vec{a}\vec{b}]\hat{j}+[\hat{k}\vec{a}\vec{b}]\hat{k}$

Sol. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $\vec{b}=b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$

$$[\hat{i}\,\vec{a}\,\vec{b}]\hat{i}=\begin{vmatrix}1 & 0 & 0\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}\hat{i}$$

$$=(a_2b_3-b_2a_3)\hat{i}$$

Similarly, $[\hat{j}\,\vec{a}\,\vec{b}]\hat{j}=(a_3b_1-a_1b_3)\hat{j}$

$[\hat{k}\,\vec{a}\,\vec{b}]\hat{k}=(a_1b_2-a_2b_1)\hat{k}$

$\therefore[\hat{i}\,\vec{a}\,\vec{b}]\hat{i}+[\hat{j}\,\vec{a}\,\vec{b}]\hat{j}+[\hat{k}\,\vec{a}\,\vec{b}]\hat{k}=(a_2b_3-b_2a_3)\hat{i}+(a_3b_1-a_1b_3)\hat{j}+(a_1b_2-a_2b_1)\hat{k}$

$$=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\end{vmatrix}=\vec{a}\times\vec{b}$$

Previous Year's Question

If $\vec{u},\vec{v},\vec{w}$ are three non-coplanar unit vectors and α, β, γ are the angles between $\vec{u}$ and $\vec{v}$, $\vec{v}$ and $\vec{w}$, $\vec{w}$ and $\vec{u}$ respectively and $\vec{x},\vec{y},\vec{z}$ are unit vectors along the bisectors of the angles α, β, γ respectively. Prove that

$$\left[\vec{x}\times\vec{y}\quad\vec{y}\times\vec{z}\quad\vec{z}\times\vec{x}\right]=\frac{1}{16}\left[\vec{u}\quad\vec{v}\quad\vec{w}\right]^2\sec^2\frac{\alpha}{2}\sec^2\frac{\beta}{2}\sec^2\frac{\gamma}{2}$$ **[JEE 2003, 4 out of 60]**

Q. $[\vec{a}+\vec{b}\quad\vec{b}+\vec{c}\quad\vec{c}+\vec{a}]=2[\vec{a}\vec{b}\vec{c}]$ **. This identity can be geometrically interpreted as :**

(Volume of a cuboid whose three coterminous edges are the face diagonals of the cuboid is twice the volume of the cuboid, whose three coterminous edges are the vectors $\vec{a},\vec{b},\vec{c}$) . This is also conclusive that if $\vec{a},\vec{b},\vec{c}$ are coplanar then $\vec{a}+\vec{b},\vec{b}+\vec{c},\vec{c}+\vec{a}$, are also coplanar.

Sol. $\left[\vec{a}+\vec{b}\ \ \vec{b}+\vec{c}\ \ \vec{c}+\vec{a}\right]=\left(\vec{a}+\vec{b}\right)\cdot\left[\left(\vec{b}+\vec{c}\right)\times\left(\vec{c}+\vec{a}\right)\right]$

$=\left(\vec{a}+\vec{b}\right)\cdot\left[\vec{b}\times\vec{c}+\vec{c}\times\vec{c}+\vec{b}\times\vec{a}+\vec{c}\times\vec{a}\right]$

$=\left(\vec{a}+\vec{b}\right)\cdot\left[\vec{b}\times\vec{c}+0+\vec{b}\times\vec{a}+\vec{c}\times\vec{a}\right]$

$=\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)+\vec{a}\cdot\left(\vec{b}\times\vec{a}\right)+\vec{a}\cdot\left(\vec{c}\times\vec{a}\right)+\vec{b}\cdot\left(\vec{b}\times\vec{c}\right)+\vec{b}\cdot\left(\vec{b}\times\vec{a}\right)-\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)$

$=\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)+0+0+0+0+\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)$

$=\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)+\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)=2\left[\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)\right]$

$=2\left[\vec{a}\vec{b}\vec{c}\right]$

Q. **Prove that :** $[\vec{a}-\vec{b}\ \vec{b}-\vec{c}\ \vec{c}-\vec{a}]=0$

Sol. To Prove : $\left[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}\right]=0$

Solution: $\left[\vec{a}-\vec{b}\ \ \vec{b}-\vec{c}\ \ \vec{c}-\vec{a}\right]$

$\left(\vec{a}-\vec{b}\right)\cdot\left[\left(\vec{b}-\vec{c}\right)\times\left(\vec{c}-\vec{a}\right)\right]$

$\left(\vec{a}-\vec{b}\right)\cdot\left[\vec{b}\times\vec{c}-\vec{b}\times\vec{a}-\vec{c}\times\vec{c}+\vec{c}\times\vec{a}\right]$ By distributive law

$(\vec{a}-\vec{b}).\left[\vec{b}\times\vec{c}-\vec{b}\times\vec{a}+\vec{c}\times\vec{a}\right]$

$\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)-\vec{b}\cdot\left(\vec{b}\times\vec{c}\right)-\vec{a}\cdot\left(\vec{b}\times\vec{a}\right)+\vec{b}\cdot\left(\vec{b}\times\vec{a}\right)+\vec{a}\cdot\left(\vec{c}\times\vec{a}\right)-\vec{b}.(\vec{c}\times\vec{a})$

$\left[\vec{a}\vec{b}\vec{c}\right]-\left[\vec{b}\vec{b}\vec{c}\right]-\left[\vec{a}\vec{b}\vec{a}\right]+\left[\vec{b}\vec{b}\vec{a}\right]+\left[\vec{a}\vec{c}\vec{a}\right]-\left[\vec{b}\vec{c}\vec{a}\right]$

$\left[\vec{a}\vec{b}\vec{c}\right]-\left[\vec{b}\vec{c}\vec{a}\right]$

$\Rightarrow 0$ L.H.S = R.H.S **Hence proved.**

Previous Year's Question

Let $\vec{a}=\hat{i}+2\hat{j}+4\hat{k},\vec{b}=\hat{i}+\lambda\hat{j}+4\hat{k}$ and $\vec{c}=2\hat{i}+4\hat{j}+\left(\lambda^2-1\right)\hat{k}$ be coplanar vectors. Then the non- zero vector $\vec{a}\times\vec{c}$ is: **[JEE (Main) 2019]**

(A) $-10\hat{i}-5\hat{j}$ (B) $-14\hat{i}-5\hat{j}$ (C) $-14\hat{i}+5\hat{j}$ (D) $-10\hat{i}+5\hat{j}$

Q. **Prove that** $[\vec{l}\vec{m}\vec{n}][\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} \vec{l}\cdot\vec{a} & \vec{l}\cdot\vec{b} & \vec{l}\cdot\vec{c} \\ \vec{m}\cdot\vec{a} & \vec{m}\cdot\vec{b} & \vec{m}\cdot\vec{c} \\ \vec{n}\cdot\vec{a} & \vec{n}\cdot\vec{b} & \vec{n}\cdot\vec{c} \end{vmatrix}$.

Sol. Let $\vec{l} = l_1\hat{i} + l_2\hat{j} + l_3\hat{k}, \vec{m} = m_1\hat{i} + m_2\hat{j} + m_3\hat{k}$ and $\vec{n} = n_1\hat{i} + n_2\hat{j} + n_3\hat{k}$

$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$.

Therefore, $\vec{l}\cdot\vec{a} = l_1a_1 + l_2a_2 + l_3a_3 = \Sigma l_1a_1$

Similarly, $\vec{l}\cdot\vec{b} = \Sigma l_1b_1$, etc.

Now $[\vec{l}\vec{m}\vec{n}][\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

$$= \begin{vmatrix} \Sigma l_1a_1 & \Sigma l_1b_1 & \Sigma l_1c_1 \\ \Sigma m_1a_1 & \Sigma m_1b_1 & \Sigma m_1c_1 \\ \Sigma n_1a_1 & \Sigma n_1b_1 & \Sigma n_1c_1 \end{vmatrix}$$

$$= \begin{vmatrix} \vec{l}\cdot\vec{a} & \vec{l}\cdot\vec{b} & \vec{l}\cdot\vec{c} \\ \vec{m}\cdot\vec{a} & \vec{m}\cdot\vec{b} & \vec{m}\cdot\vec{c} \\ \vec{n}\cdot\vec{a} & \vec{n}\cdot\vec{b} & \vec{n}\cdot\vec{c} \end{vmatrix}$$

Q. **Prove that** $[\vec{a}\vec{b}\vec{c}]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$,

where $\vec{a}, \vec{b}, \vec{c}$ are non–coplanar vectors.

Sol. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, etc. $\quad \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k} \quad \vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$

$\therefore \quad \vec{a}.\vec{a} = \sum a_1^2, \vec{a}.\vec{b} = \sum a_1b_1$

$$\therefore \left[\vec{a}\,\vec{b}\,\vec{c}\right] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \Delta$$

$$\Delta^2 = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} \Sigma a_1^2 & \Sigma a_1 b_1 & \Sigma a_1 c_1 \\ \Sigma b_1 a_1 & \Sigma b_1^2 & \Sigma b_1 C_1 \\ \Sigma c_1 a_1 & \Sigma c_1 b_1 & \Sigma c_1^2 \end{vmatrix}$$

$$= \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$$

Q.

Prove that $\left[\vec{l}\ \vec{m}\ \vec{n}\right](\vec{a}\times\vec{b}) = \begin{vmatrix} \vec{l}\cdot\vec{a} & \vec{l}\cdot\vec{b} & \vec{l} \\ \vec{m}\cdot\vec{a} & \vec{m}\cdot\vec{b} & \vec{m} \\ \vec{n}\cdot\vec{a} & \vec{n}\cdot\vec{b} & \vec{n} \end{vmatrix}$.

Sol. Let $\vec{l} = l_1\hat{i} + l_2\hat{j} + l_3\hat{k}$ etc.

$$\therefore \left[\vec{l}\ \vec{m}\ \vec{n}\right] = \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix}$$

$$\left(\vec{a}\times\vec{b}\right) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\left[\vec{l}\ \vec{m}\ \vec{n}\right]\left(\vec{a}\times\vec{b}\right) = \begin{vmatrix} l_1 & l_2 & l_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{vmatrix} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= \begin{vmatrix} l_1\hat{i} + l_2\hat{j} + l_3\hat{k} & \Sigma l_1 a_1 & \Sigma l_1 b_1 \\ m_1\hat{i} + m_2\hat{j} + m_3\hat{k} & \Sigma m_1 a_1 & \Sigma m_1 b_1 \\ n_1\hat{i} + n_2\hat{j} + n_3\hat{k} & \Sigma n_1 a_1 & \Sigma n_1 b_1 \end{vmatrix}$$

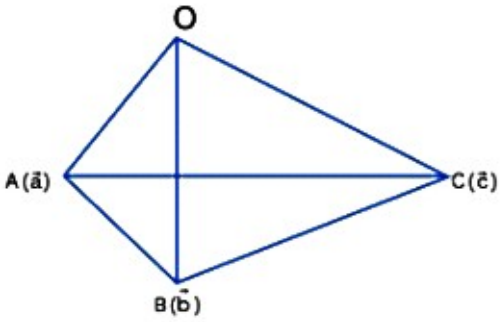
$$= \begin{vmatrix} \vec{l} & \vec{l}\cdot\vec{a} & \vec{l}\cdot\vec{b} \\ \vec{m} & \vec{m}\cdot\vec{a} & \vec{m}\cdot\vec{b} \\ \vec{n} & \vec{n}\cdot\vec{a} & \vec{n}\cdot\vec{b} \end{vmatrix}$$

$$= \begin{vmatrix} \vec{l}\cdot\vec{a} & \vec{l}\cdot\vec{b} & \vec{l} \\ \vec{m}\cdot\vec{a} & \vec{m}\cdot\vec{b} & \vec{m} \\ \vec{n}\cdot\vec{a} & \vec{n}\cdot\vec{b} & \vec{n} \end{vmatrix}$$

VOLUME OF TETRAHEDRON :

1. If $\vec{a},\vec{b},\vec{c}$ are position vectors of vertices A,B and C with respect to O, then volume of tetrahedron OABC represented by V is given by

$V = \frac{1}{3}$ Base area × height



Base Area $= \frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|$

Let $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\vec{n}$

$\therefore$ Base area $= \frac{1}{2}|\vec{n}|$

Height = projection of $\vec{a}$ on $\vec{n}$

$$= \frac{|\vec{a}\cdot\vec{n}|}{|\vec{n}|} = \frac{|\vec{a}\cdot(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a})|}{|\vec{n}|} = \frac{|[\vec{a}\,\vec{b}\,\vec{c}]|}{|\vec{n}|}$$

$$\therefore \quad V = \frac{1}{3}\cdot\frac{1}{2}|\vec{n}|\frac{\left|\left[\vec{a}\,\vec{b}\,\vec{c}\right]\right|}{|\vec{n}|} = \frac{1}{6}\left|\left[\vec{a}\,\vec{b}\,\vec{c}\right]\right|$$

2. If $\vec{a},\vec{b},\vec{c},\vec{d}$ are position vectors of vertices A,B,C,D of a tetrahedron ABCD, then

$$\text{Its volume} = \begin{cases} \frac{1}{6}|[\overline{AB}\ \overline{AC}\ \overline{AD}]| \\ \text{or} \\ \frac{1}{6}|[\vec{b}-\vec{a}\ \ \vec{c}-\vec{a}\ \ \vec{d}-\vec{a}]| \end{cases}$$

Q. **If the vertices of any tetrahedron be $\vec{a} = \hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} + \hat{k}$, $\vec{c} = 4\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{d} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ then find its volume.**

Sol. Let the p.v. of the vertices A, B, C, D with respect to 0 are $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ respectively then $\overline{AB} = \vec{b} - \vec{a} = 3\hat{i} - \hat{j} - \hat{k}$

$\overline{AC} = 4\hat{i} + 2\hat{j} + 4\hat{k}$ and $\overline{AD} = 2\hat{i} + 2\hat{j}$

Now volume of tetrahedron $= \frac{1}{6}\left[\overline{AB} \;\; \overline{AC} \;\; \overline{AD}\right] = \frac{1}{6}\begin{vmatrix} 3 & -1 & -1 \\ 4 & 2 & 4 \\ 2 & 2 & 0 \end{vmatrix} = -6$

$\therefore$ Required volume = 6 units

Previous Year's Question

Let the volume of a parallelopiped whose coterminous edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}, \vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos\theta$ can be : **[JEE Main 2020]**

(A) $\frac{7}{6\sqrt{6}}$ (B) $\frac{5}{7}$ (C) $\frac{7}{6\sqrt{3}}$ (D) $\frac{5}{3\sqrt{3}}$

VECTOR TRIPLE PRODUCT:

Definition:

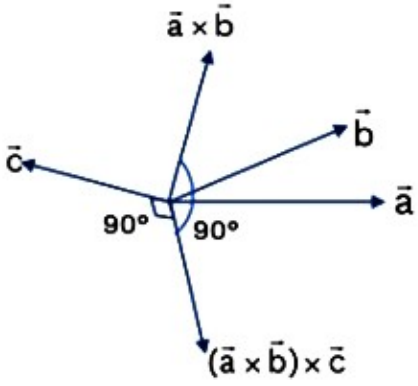
The vector triple product of three vectors $\vec{a}, \vec{b}, \vec{c}$ is defined as the vector product of two vectors $\vec{a}$ and $\vec{b} \times \vec{c}$. It is denoted by $\vec{a} \times (\vec{b} \times \vec{c})$

$(\vec{a} \times \vec{b}) \times \vec{c}$ is a vector which is coplanar with $\vec{a}$ and $\vec{b}$ and perpendicular to $\vec{c}$.

Hence $(\vec{a} \times \vec{b}) \times \vec{c} = x\vec{a} + y\vec{b}$ ...(1)

[linear combination of $\vec{a}$ and $\vec{b}$]

$\vec{c} \cdot (\vec{a} \times \vec{b}) \times \vec{c} = x(\vec{a} \cdot \vec{c}) + y(\vec{b} \cdot \vec{c})$



$0 = x(\vec{a}\cdot\vec{c}) + y(\vec{b}\cdot\vec{c})$ ...(2)

$\therefore \frac{x}{\vec{b}\cdot\vec{c}} = -\frac{y}{\vec{a}\cdot\vec{c}} = \lambda$

$\therefore x = \lambda(\vec{b}\cdot\vec{c})$ and $y = -\lambda(\vec{a}\cdot\vec{c})$

Substituting the values of x and y in $(\vec{a}\times\vec{b})\times\vec{c} = \lambda(\vec{b}\cdot\vec{c})\vec{a} - \lambda(\vec{a}\cdot\vec{c})\vec{b}$

This is an identity and must be true for all values of $\vec{a},\vec{b},\vec{c}$

Put $\vec{a} = \hat{i}; \vec{b} = \hat{j}$ and $\vec{c} = \hat{i}$

$(\hat{i}\times\hat{j})\times\hat{i} = \lambda(\hat{j}\cdot\hat{i})\hat{i} - \lambda(\hat{i}\cdot\hat{i})\hat{j}$

$\hat{j} = -\lambda\hat{j} \Rightarrow \lambda = -1$

hence $(\vec{a}\times\vec{b})\times\vec{c} = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{b}\cdot\vec{c})\vec{a}$

Properties:

1. Expansion formula for vector triple product is given by
   $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\cdot\vec{c})\vec{b} - (\vec{a}\cdot\vec{b})\vec{c}$
   $(\vec{b}\times\vec{c})\times\vec{a} = (\vec{b}\cdot\vec{a})\vec{c} - (\vec{c}\cdot\vec{a})\vec{b}$
2. $\left[\vec{a}\times\vec{b}\quad \vec{b}\times\vec{c}\quad \vec{c}\times\vec{a}\right] = \left[\vec{a}\,\vec{b}\,\vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$

   Note that if $\vec{a},\vec{b},\vec{c}$ are non coplanar vectors then $\vec{a}\times\vec{b},\vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$ will also be non coplanar vectors.
3. Vector triple product is a vector quantity.
4. $\vec{a}\times(\vec{b}\times\vec{c}) \neq (\vec{a}\times\vec{b})\times\vec{c}$
5. Unit vector coplanar with $\vec{a}$ & $\vec{b}$ and perpendicular to $\vec{c}$ is $\pm\frac{(\vec{a}\times\vec{b})\times\vec{c}}{\left|(\vec{a}\times\vec{b})\times\vec{c}\right|}$

Q. $\hat{i}\times(\hat{j}\times\hat{k}) + \hat{j}\times(\hat{k}\times\hat{i}) + \hat{k}\times(\hat{i}\times\hat{j})$ **equals**

(A) $\hat{i}$ **(B)** $\hat{j}$ **(C)** $\hat{k}$ **(D) 0**

Sol. **(D)**

$\hat{i}\times(\hat{j}\times\hat{k}) + \hat{j}\times(\hat{k}\times\hat{i}) + \hat{k}\times(\hat{i}\times\hat{j}) \Rightarrow \hat{i}\times\hat{i} + \hat{j}\times\hat{j} + \hat{k}\times\hat{k} = 0 + 0 + 0 = 0$

Q. If $\vec{a},\vec{b},\vec{c}$ are coplanar, then show that $\vec{a}\times\vec{b},\vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$ are also coplanar.

Sol. $\left[\vec{a}\,\vec{b}\,\vec{c}\right]$ are coplanar $\Rightarrow \left[\vec{a}\,\vec{b}\,\vec{c}\right]=0 \Rightarrow \left[\vec{a}\,\vec{b}\,\vec{c}\right]=0$

$\Rightarrow \left[\vec{a}\times\vec{b}\ \vec{b}\times\vec{c}\ \vec{c}\times\vec{a}\right]=0$

$\Rightarrow \vec{a}\times\vec{b},\vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$ are coplanar.

Q. **Let $\vec{a},\vec{b},\vec{c}$ such that $|\vec{a}|=1,|\vec{b}|=1$ and $|\vec{c}|=2$ and if $\vec{a}\times(\vec{a}\times\vec{c})+\vec{b}=0$ then acute angle between $\vec{a}$ and $\vec{c}$ is**

(A) $\frac{\pi}{3}$ **(B)** $\frac{\pi}{4}$ **(C)** $\frac{\pi}{6}$ **(D) None of these**

Sol. **(C)**

If angle between $\vec{a}$ and $\vec{c}$ is θ then

$\vec{a}\cdot\vec{c}=|\vec{a}||\vec{c}|\cos\theta$

$=1.2\cos\theta=2\cos\theta$

But $\vec{a}\times(\vec{a}\times\vec{c})+\vec{b}=0$

$\Rightarrow(\vec{a}\cdot\vec{c})\vec{a}-(\vec{a}\cdot\vec{a})\vec{c}+\vec{b}=0$

$\Rightarrow(2\cos\theta)\vec{a}-1\cdot\vec{c}=-\vec{b}$

$\Rightarrow[(2\cos\theta)\vec{a}-\vec{c}]^2=[-\vec{b}]^2$

$\Rightarrow 4\cos^2\theta|\vec{a}|^2-2\cdot(2\cos\theta)\vec{a}\cdot\vec{c}+|\vec{c}|^2=|\vec{b}|^2$

$\Rightarrow 4\cos^2\theta-4\cos\theta(2\cos\theta)+4=1$

$\Rightarrow 4(1-\cos^2\theta)=1 \quad [\because|\vec{a}|=1,|\vec{b}|=1]$

$\Rightarrow \sin\theta=1/2$

$\Rightarrow \theta=\frac{\pi}{6}$

Q. **Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$ if vector $\vec{c}$ is such that $\vec{a}\cdot\vec{c}=|\vec{c}|,|\vec{c}-\vec{a}|=2\sqrt{2}$ and angle between $(\vec{a}\times\vec{b})$ and $\vec{c}$ is the 30° then $|(\vec{a}\times\vec{b})\times\vec{c}|$ is equal to**

(A) $\frac{2}{3}$ **(B)** $\frac{3}{2}$ **(C) 2** **(D) 3**

Sol. **(B)**

$$|\vec{c}-\vec{a}|^2=(\vec{c}-\vec{a})\cdot(\vec{c}-\vec{a})=(2\sqrt{2})^2$$

$$\Rightarrow|\vec{c}|^2+|\vec{a}|^2-2\vec{c}\cdot\vec{a}=8$$

$$\Rightarrow|\vec{c}|^2+(4+1+4)-2\vec{c}\cdot\vec{a}=8$$

$$\Rightarrow|\vec{c}|^2+9-2|\vec{c}|=8 \quad [\because \vec{a}\cdot\vec{c}=|\vec{c}|]$$

$$\Rightarrow|\vec{c}|^2-2|\vec{c}|+1=0\Rightarrow|\vec{c}|^2-|2\vec{c}|+1=0$$

$$\Rightarrow(|\vec{c}|-1)^2=0\Rightarrow|\vec{c}|=1$$

$$|(\vec{a}\times\vec{b})\times\vec{c}|=\vec{a}\times\vec{b}||\vec{c}|\sin30^\circ=1\times\frac{1}{2}|\vec{a}\times\vec{b}|=\frac{1}{2}|\vec{a}\times\vec{b}|$$

$$\text{But } \vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&-2\\1&1&0\end{vmatrix}=2\hat{i}-2\hat{j}+\hat{k}$$

$$\therefore|\vec{a}\times\vec{b}|=\sqrt{4+4+1}=3$$

$$\therefore|(\vec{a}\times\vec{b})\times\vec{c}|=\frac{3}{2}$$

Q. **Prove that [a × b b × c c × a] = [a b c]²**

Sol. We have, [a × b b × c c × a]

$$=\{(a\times b)\times(b\times c)\}\cdot(c\times a)$$

$$=\{d\times(b\times c)\}\cdot(c\times a) \quad [\text{ where, } d=(a\times b)]$$

$$=[(d\cdot c)b-(d\cdot b)c]\cdot(c\times a)$$

$$=[\{(a\times b)\cdot c)b-(a\times b)\cdot b\}c]\cdot(c\times a)$$

$$=\{[a\ b\ c]b-0\}\cdot(c\times a) \quad [\because [a\ b\ b]=0]$$

$$=\{[a\ b\ c]\{b\cdot(c\times a)\}$$

$$=[a\ b\ c]\ [b\ c\ a] \qquad \{\because [a\ b\ c]=[b\ c\ a]\}$$

$$=[a\ b\ c]^2$$

Q. If $\left.\begin{array}{l}\bar{V}_1=\bar{a}\times(\bar{b}\times\bar{c})\\ \bar{V}_2=\bar{b}\times(\bar{c}\times\bar{a})\\ \bar{V}_3=\bar{c}\times(\bar{a}\times\bar{b})\end{array}\right]$ **then prove that**

(i) $\bar{V}_1,\bar{V}_2,\bar{V}_3$ **are coplanar**

(ii) $\bar{V}_1,\bar{V}_2,\bar{V}_3$ **from the sides of a triangle**

(iii) $\bar{V}_1+\bar{V}_2+\bar{V}_3$ **is a null vectors**

(iv) $\bar{V}_1+\bar{V}_2+\bar{V}_3$ **are linearly dependent.**

Sol. $\vec{a}\times(\bar{b}\times\vec{c})=(\bar{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}$

$\vec{b}\times(\vec{c}\times\vec{a})=(\bar{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}$

$\vec{c}\times(\bar{a}\times\vec{b})=(\bar{c}\cdot\vec{b})\vec{a}-(\bar{c}\cdot\vec{a})\vec{b}$

Adding, we get the result. $\vec{v}_1+\vec{v}_2+\vec{v}_3=\vec{0}$ which proves (iii) and (ii)

Now

Let $\vec{p}=\vec{a}\times(\vec{b}\times\vec{c}),\vec{q}=\vec{b}\times(\vec{c}\times\vec{a})$ and $\vec{r}=\vec{c}\times(\vec{a}\times\vec{b})$.

Then, $\vec{p}+\vec{q}+\vec{r}=\vec{a}\times(\vec{b}\times\vec{c})+\vec{b}\times(\vec{c}\times\vec{a})+\vec{c}\times(\vec{a}\times\vec{b})=0$

But, $\vec{p}+\vec{q}+\vec{r}=\vec{0}$ means that one of the vectors is a linear combination of other two.

This shows that $\vec{p},\vec{q},\vec{r}$ are coplanar.

Hence, $\vec{a}\times(\vec{b}\times\vec{c}),\vec{b}\times(\vec{c}\times\vec{a})$ and $\vec{c}\times(\vec{a}\times\vec{b})$ are coplanar. hence linearly dependent which Proves (i) & (iv).

Q. **If** $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{a}\cdot\vec{b}=1$ and $\vec{a}\times\vec{b}=\hat{j}-\hat{k}$**, then find** $\vec{b}$.

Sol. As$(\vec{a}\times\vec{b})\times\vec{a}=(\vec{a}\cdot\vec{a})\vec{b}-(\vec{a}\cdot\vec{b})\vec{a}$

$\vec{a}\cdot\vec{a}=|\bar{a}|^2=1+1+1=3$ and $\vec{a}\cdot\vec{b}=1$

$\therefore(\hat{j}-\hat{k})\times(\hat{i}+\hat{j}+\hat{k})=3(\vec{b})-(\hat{i}+\hat{j}+\hat{k})$

i.e. $\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\0&1&-1\\1&1&1\end{vmatrix}=3(\vec{b})-(\hat{i}+\hat{j}+\hat{k})$

$(2\hat{i}-\hat{j}-\hat{k})+(\hat{i}+\hat{j}+\hat{k})=3\vec{b}$

i.e. $\hat{i}=\vec{b}$

Q. **Find a vector $\bar{v}$ which is coplanar with the vectors $\hat{i}+\hat{j}-2\hat{k}$ & $\hat{i}-2\hat{j}+\hat{k}$ and is orthogonal to the vector $-2\hat{i}+\hat{j}+\hat{k}$ It is given that the projection of $\bar{v}$ along the vector $\hat{i}-\hat{j}+\hat{k}$ is equal to $6\sqrt{3}$.**

Sol. $\bar{v} = \alpha(-2\hat{i}+\hat{j}+\hat{k}) \times [(\hat{i}+\hat{j}-2\hat{k}) \times (\hat{i}-2\hat{j}+\hat{k})]$

$\bar{v} = \alpha(-2\hat{i}+\hat{j}+\hat{k}) \times [-3\hat{i}-3\hat{j}-3\hat{k}]$

$\bar{v} = \alpha[-9\hat{j}+9\hat{k}]$ ...(1)

$\frac{\bar{v}\cdot(\hat{i}-\hat{j}+\hat{k})}{\sqrt{3}} = 6\sqrt{3}$

$\Rightarrow \alpha(-9\hat{j}+9\hat{k})\cdot(\hat{i}-\hat{j}+\hat{k}) = 18$

$\Rightarrow \quad \alpha\,(9+9) = 18$

$\alpha = 1$

From (1), $\bar{v} = 9(-\hat{j}+\hat{k})$

Q. **If $\vec{a}$ and $\vec{b}$ are vectors in space given by $\vec{a} = \frac{\hat{i}-2\hat{j}}{\sqrt{5}}$ and $\vec{b} = \frac{2\hat{i}+\hat{j}+3\hat{k}}{\sqrt{14}}$, then find the value of $(2\vec{a}+\vec{b})\cdot[(\vec{a}\times\vec{b})\times(\vec{a}-2\vec{b})]$.**

Sol. From the given information, it is clear that

$\vec{a} = \frac{\hat{i}-2\hat{j}}{\sqrt{5}} \Rightarrow |\vec{a}| = 1,\ |\vec{b}| = 1,\ \vec{a}\cdot\vec{b} = 0$

Now, $(2\vec{a}+\vec{b})\cdot[(\vec{a}\times\vec{b})\times(\vec{a}-2\vec{b})] = (2\vec{a}+\vec{b})$

$[a^2\vec{b} - (\vec{a}\cdot\vec{b})\cdot\vec{a} + 2b^2\cdot\vec{a} - 2(\vec{b}\cdot\vec{a})\cdot b]$

$= [2\vec{a}+\vec{b}]\cdot[\vec{b}+2\vec{a}] = 4\vec{a}^2 + \vec{b}^2$ [as a . b = 0]

$= 4.1 + 1 = 5$

Q. **Prove that : $\hat{i}\times(\vec{a}\times\hat{i}) + \hat{j}\times(\vec{a}\times\hat{j}) + \hat{k}\times(\vec{a}\times\hat{k}) = 2\vec{a}$.**

Sol. Let $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\hat{i}\times(\vec{a}\times\hat{i}) = (\hat{i}\cdot\hat{i})\vec{a} - (\hat{i}\cdot\vec{a})\hat{i} = \vec{a} - a_1\hat{i}$ ...(1)

Similarly $\hat{j}\times(\vec{a}\times\hat{j}) = \vec{a} - a_2\hat{j}$ ...(2)

$\vec{k} \times (\vec{a} \times \vec{k}) = \vec{a} - a_3\vec{k}$ ...(3)

$(1) + (2) + (3) \Rightarrow \hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) = 3\vec{a} - \left(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}\right) = 3\vec{a} - \vec{a} = 2\vec{a}$

Previous Year's Question

If $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, then the value of $|\hat{i} \times (\vec{a} \times \hat{i})|^2 + |\hat{j} \times (\vec{a} \times \hat{j})|^2 + |\hat{k} \times (\vec{a} \times \hat{k})|^2$ is equal to____

[JEE Main 2020]

Q. **Given that $\vec{a}, \vec{b}, \vec{p}, \vec{q}$ are four vectors such that $\vec{a} + \vec{b} = \mu\vec{p}, \vec{b} \cdot \vec{q} = 0$ & $(\vec{b})^2 = 1$, where μ is a scalar then prove that $|(\vec{a} \cdot \vec{q})\vec{p} - (\vec{p} \cdot \vec{q})\vec{a}| = |\vec{p} \cdot \vec{q}|$**

Sol. Given,

$\vec{a} + \vec{b} = \mu\vec{p}$ ...(i)

$\Rightarrow \vec{a} \cdot \vec{q} + \vec{b} \cdot \vec{q} = \mu(\vec{p} \cdot \vec{q})$

$\Rightarrow \vec{a} \cdot \vec{q} = \mu(\vec{p} \cdot \vec{q})$ ...(ii)

Now., $|(\vec{a} \cdot \vec{q})\vec{p} - (\vec{p} \cdot \vec{q})\vec{a}| = |(\vec{p} \cdot \vec{q})||\mu\vec{p} - \vec{a}|$ from (i)

$= |(\vec{p} \cdot \vec{q})||\vec{b}|$ from (ii)

$= \left|(\vec{p} \cdot \vec{q})\right|$

Q. **If the vectors $\vec{b}, \vec{c}, \vec{d}$ are not coplanar, then prove that the vector $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{a} \times \vec{c}) \times (\vec{d} \times \vec{b}) + (\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c})$ is parallel to .**

Sol. We know that

$\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a}\,\vec{c}\,\vec{d}]\vec{b} - [\vec{b}\,\vec{c}\,\vec{d}]\vec{a}$ ...(i)

Similarly,for $(\vec{a} \times \vec{c}) \times (\vec{d} \times \vec{b}) = [\vec{a}\vec{d}\vec{b}]\vec{c} - [\vec{c}\vec{d}\vec{b}]\vec{a}$

Similarly, for $(\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c}) = [\vec{a}\vec{d}\vec{c}]\vec{b} - [\vec{a}\vec{d}\vec{b}]\vec{c}$

So adding all the three equations, we get

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) + (\vec{a} \times \vec{c}) \times (\vec{d} \times \vec{b}) + (\vec{a} \times \vec{d}) \times (\vec{b} \times \vec{c}) = -2[\vec{b}\vec{c}\vec{d}]\vec{a}$

Thus parallel to $\vec{a}$.

Scalar Product of four Vector :

(I) $(\vec{a}\times\vec{b})\cdot(\vec{c}\times\vec{d}) = (\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d}) - (\vec{a}\cdot\vec{d})(\vec{b}\cdot\vec{c}) = \begin{vmatrix}\vec{a}\cdot\vec{c} & \vec{a}\cdot\vec{d}\\ \vec{b}\cdot\vec{c} & \vec{b}\cdot\vec{d}\end{vmatrix}$

Proof: $:\underbrace{(\vec{a}\times\vec{b})}_{\vec{u}}\cdot(\vec{c}\times\vec{d}) = \vec{u}\cdot(\vec{c}\times\vec{d}) = (\vec{u}\times\vec{c})\cdot\vec{d}$ (Dot & cross are interchangeable in STP)

$$((\vec{a}\times\vec{b})\times\vec{c})\cdot\vec{d} = ((\vec{a}\cdot\vec{c})\vec{b} - (\vec{b}\cdot\vec{c})\vec{a})\cdot\vec{d} = ((\vec{a}\cdot\vec{c})(\vec{b}\cdot\vec{d}) - (\vec{b}\cdot\vec{c})(\vec{a}\cdot\vec{d})) = \begin{vmatrix}\vec{a}\cdot\vec{c} & \vec{a}\cdot\vec{d}\\ \vec{b}\cdot\vec{c} & \vec{b}\cdot\vec{d}\end{vmatrix}$$

$$(\vec{a}\times\vec{b})\cdot(\vec{a}\times\vec{b}) = (\vec{a}\times\vec{b})^2 = \begin{vmatrix}\vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b}\\ \vec{a}\cdot\vec{b} & \vec{b}\cdot\vec{b}\end{vmatrix} = (\vec{a})^2(\vec{b})^2 - (\vec{a}\cdot\vec{b})^2$$ which is lagrange's identity.

Q. **Prove that acute angle between the two plane faces of a regular tetrahedron is** $\cos^{-1}\frac{1}{3}$

Sol. Let edge length of regular tetrahedron = 1

$\vec{n}_1$ = normal vector to plane OAB = $\vec{a}\times\vec{b}$

$\vec{n}_2$ = normal vector to plane OBC = $\vec{b}\times\vec{c}$

$\therefore$ acute angle between plane focus OAB & OBC is given as

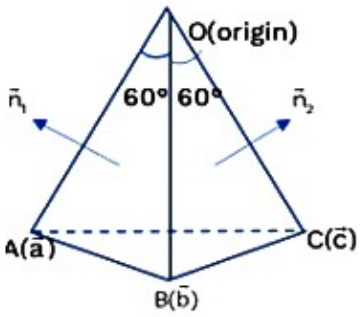
$$\cos\theta = \frac{|\vec{n}_1\vec{n}_2|}{|\vec{n}_1||\vec{n}_2|} = \frac{|(\vec{a}\times\vec{b})\cdot(\vec{b}\times\vec{c})|}{|(\vec{a}\times\vec{b})||(\vec{b}\times\vec{c})|}$$

$$= \frac{\begin{vmatrix}\vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c}\\ \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c}\end{vmatrix}}{\sin 60^\circ\cdot\sin 60^\circ}$$

$$= \frac{\begin{vmatrix}\cos 60^\circ & \cos 60^\circ\\ \cos 0^\circ & \cos 60^\circ\end{vmatrix}}{3/4}$$

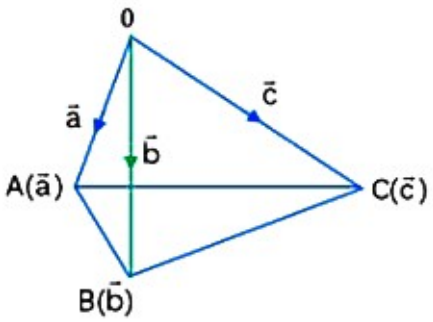
$$= \frac{\left|\frac{1}{4}-\frac{1}{2}\right|}{\frac{3}{4}} = \frac{1}{3}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{3}\right)$$



Q. **Prove that the angle between any edge and a face of a regular tetrahedral is** $\cos^{-1}\frac{1}{\sqrt{3}}$

Sol. Let us find angle between edge OA a face OBC, since tetrahedral is regular so each face is equilateral triangle required angle is angle between vector $\vec{a}$ and vector along angle bisector let $\vec{a},\vec{b},\vec{c}$ be unit vector Angle bisector of $\angle BOC$ in $\triangle OBC$ is $\vec{b}+\vec{c}$ $\therefore \cos\theta = \dfrac{\vec{a}\cdot(\vec{b}+\vec{c})}{|\vec{a}||\vec{b}+\vec{c}|}$



$$\Rightarrow \cos\theta = \frac{\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}}{|\vec{a}|\,|\vec{b}+\vec{c}|}$$

$$\Rightarrow \cos\theta = \frac{\vec{a}\cdot\vec{b}+\vec{a}\cdot\vec{c}}{|\vec{b}+\vec{c}|}$$

Now $|\vec{b}+\vec{c}|^2 = b^2 + c^2 + 2\vec{b}.\vec{c} = 1 + 1 + 2\cos 60°$

$|\vec{b}+\vec{c}| = \sqrt{3}$

$$\Rightarrow \cos\theta = \frac{1.1\times\cos 60° + 1.1\cdot\cos 60°}{\sqrt{3}}$$

$$\Rightarrow \theta = \cos^{-1}\frac{1}{\sqrt{3}}$$

Hence the statement is proved.

Q. **Prove that :** $\{(\vec{a}\times\vec{b})\times(\vec{a}\times\vec{c})\}\cdot\vec{d} = (\vec{a}\cdot\vec{d})[\vec{a}\vec{b}\vec{c}]$

Sol. $\vec{a}\times\vec{b} = \vec{\lambda}$

$\therefore [(\vec{a}\times\vec{b})\times(\vec{a}\times\vec{c})]\cdot\vec{d} = [\vec{\lambda}\times(\vec{a}\times\vec{c})]\cdot\vec{d} = [(\vec{\lambda}.\vec{c})\vec{a} - (-\vec{\lambda}.\vec{a})\vec{c}]\cdot\vec{d}$

$= [\{(\vec{a}\times\vec{b})\cdot\vec{c}\}\vec{a} - \{(\vec{a}\times\vec{b})\cdot\vec{a}\}\vec{c}]\cdot\vec{d} = (\vec{a}\cdot\vec{d})[\vec{a}\vec{b}\vec{c}]$

Q. **Prove that :** $\vec{d}\cdot[\vec{a}\times\{\vec{b}\times(\vec{c}\times\vec{d})\}] = (\vec{b}\cdot\vec{d})[\vec{a}\vec{c}\vec{d}]$.

Sol. $\left[\vec{b}\times(\vec{c}\times\vec{d})\right] = (\vec{b}\cdot\vec{d})(\vec{c}) - (\vec{b}\cdot\vec{c})(\vec{d})$

Now crossing with $\vec{a}$ from left , we have $\vec{a}\times\left[\vec{b}\times(\vec{c}\times\vec{d})\right]$

$$= \vec{a} \times \left[(\vec{b} \cdot \vec{d})(\vec{c}) - (\vec{b} \cdot \vec{c})(\vec{d}) \right]$$

$$= \vec{a} \times \left[(\vec{b} \cdot \vec{d})(\vec{c}) \right] - \vec{a} \times \left[(\vec{b} \cdot \vec{c})(\vec{d}) \right]$$

$$= (\vec{b} \cdot \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})(\vec{a} \times \vec{d})$$

$$\therefore \quad \vec{d} \cdot \left[\vec{a} \times \left[\vec{b} \times (\vec{c} \times \vec{d}) \right] \right] = (\vec{b} \cdot \vec{d}) \left[\vec{d} \cdot (\vec{a} \times \vec{c}) \right] - (\vec{b} \cdot \vec{c}) \left(d \cdot \left[\vec{a} \times \vec{d} \right] \right) = (\vec{b} \cdot \vec{d}) \left[\vec{d} \vec{a} \vec{c} \right]$$

$$= (\vec{b} \cdot \vec{d}) \left[\vec{a} \vec{c} d \right]$$

Vector Product of Four Vector :

$\vec{V} = (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$

$= \vec{u} \times (\vec{c} \times \vec{d}) \quad = [\vec{a}\,\vec{b}\,\vec{d}]\vec{c} - [\vec{a}\,\vec{b}\,\vec{c}]\vec{d}$ ...(1) (where $\vec{u} = \vec{a} \times \vec{b}$)

Again $\vec{V} = (\vec{a} \times \vec{b}) \times \underbrace{(\vec{c} \times \vec{d})}_{\vec{v}} = (\vec{a} \cdot \vec{v})\vec{b} - (\vec{b} \cdot \vec{v})\vec{a} = [\vec{a}\,\vec{c}\,\vec{d}]\vec{b} - [\vec{b}\,\vec{c}\,\vec{d}]\vec{a}$ ...(2)

From (1) and (2), $[\vec{a}\,\vec{b}\,\vec{d}]\vec{c} - [\vec{a}\,\vec{b}\,\vec{c}]\vec{d} = [\vec{a}\,\vec{c}\,\vec{d}]\vec{b} - [\vec{b}\,\vec{c}\,\vec{d}]\vec{a}$ ...(3)

Note that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 0 \Rightarrow$ planes containing the vectors $\vec{a}$ & $\vec{b}$ and $\vec{c}$ & $\vec{d}$ are parallel.

Similarly, $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 0 \Rightarrow$ the two planes are perpendicular.

(i) Equation (3) is suggestive that if $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are four vectors, no three of them are coplanar then each one of them can be expressed as a linear combination of other.

(ii) If $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are p.v.'s of four points then these four points are in the same plane if

$$\left[\vec{a}\,\vec{b}\,\vec{d} \right] - \left[\vec{a}\,\vec{b}\,\vec{c} \right] = \left[\vec{a}\,\vec{c}\,\vec{d} \right] - \left[\vec{b}\,\vec{c}\,\vec{d} \right]$$

Q. **If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} \times \vec{b} = \vec{c}, \vec{b} \times \vec{c} = \vec{a}$, then**

(A) $|\vec{b}| = 1, |\vec{c}| = |\vec{a}|$ **(B)** $|\vec{c}| = 1, |\vec{a}| = |\vec{b}|$ **(C)** $|\vec{b}| = 2, |\vec{c}| = 2|\vec{a}|$ **(D)** $|\vec{a}| = 1, |\vec{b}| = |\vec{c}|$

Sol. **(A)**

Given $\vec{a} \times \vec{b} = \vec{c}$ ...(i)

$\vec{b} \times \vec{c} = \vec{a}$ ...(ii)

From (i), $\vec{c} \perp \vec{a}$ and $\vec{c} \perp \vec{b}$ ...(iii)

From (ii), $\vec{a} \perp \vec{b}$ and $\vec{a} \perp \vec{c}$ ...(iv)

From (iii) and (iv), $\vec{a},\vec{b},\vec{c}$ are mutually perpedicular.

Taking cross product of (i) with (ii), we get,

$(\vec{a}\times\vec{b})\times(\vec{b}\times\vec{c}) = \vec{c}\times\vec{a} \quad \Rightarrow [\vec{a}\,\vec{b}\,\vec{c}]\vec{b} - [\vec{a}\,\vec{b}\,\vec{b}]\vec{c} = \vec{c}\times\vec{a}$

$\Rightarrow [\vec{a}\,\vec{b}\,\vec{c}]|\vec{b}| = |\vec{c}\times\vec{a}|$

$\Rightarrow |[\vec{a}\,\vec{b}\,\vec{c}]||\vec{b}| = |\vec{c}\times\vec{a}| \Rightarrow |[\vec{a}\,\vec{b}\,\vec{c}]||\vec{b}| = |\vec{c}\times\vec{a}|$

$\Rightarrow |\vec{a}||\vec{b}||\vec{c}||\vec{b}| = |\vec{c}||\vec{a}| \Rightarrow |\vec{b}|^2 = 1 \Rightarrow |\vec{b}| = 1$

From (i), $\vec{a}\times\vec{b} = \vec{c}$

$\Rightarrow |\vec{a}\times\vec{b}| = |\vec{c}| \Rightarrow |\vec{a}||\vec{b}| = |\vec{c}|$ [$\because \vec{a}$ and $\vec{b}$ are mutually perpendicular]

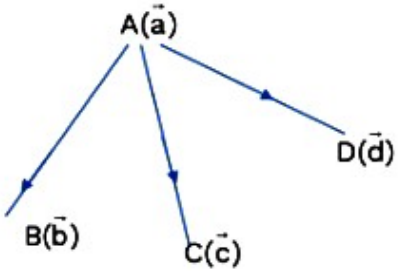
$\Rightarrow |\vec{a}| = |\vec{c}| \quad [\because |\vec{b}| = 1]$

$\because |\vec{b}| = 1$ and $|\vec{a}| = |\vec{c}|$

Condition for coplanarity of four points :

4 points with pv's $\vec{a},\vec{b},\vec{c},\vec{d}$ are coplanar if $\exists$ scalars x, y, z and t not all simultaneously zero and satisfying $x\vec{a} + y\vec{b} + z\vec{c} + t\vec{d} = 0$ where $x + y + z + t = 0$

Case I:



Let the four points A, B, C, D are in the same plane

$\Rightarrow$ the vectors $\vec{b}-\vec{a}, \vec{c}-\vec{a}$ and $\vec{d}-\vec{a}$ are in the same plane.

hence $\vec{d}-\vec{a} = l(\vec{b}-\vec{a}) + m(\vec{c}-\vec{a})$

or

$$\underbrace{(l+m-1)}_{x}\vec{a} - \underset{y}{l}\vec{b} - \underset{z}{m}\vec{c} + \underset{i}{1}\vec{d} = 0 \Rightarrow x\vec{a} + y\vec{b} + z\vec{c} + t\vec{d} = 0$$

where, $x + y + z + t = 0$ and

x, y, z, t not all simultaneous zero.

Case II:

Let $x\vec{a} + y\vec{b} + z\vec{c} + t\vec{d} = 0$ where $x + y + z + t = 0$ and not all simultaneously zero

Let $t \neq 0$

$(-y-z-t)\vec{a}+y\vec{b}+z\vec{c}+t\vec{d}=0$

[putting $x=-y-z-t$]

$(\vec{d}-\vec{a})t+y(\vec{b}-\vec{a})+z(\vec{c}-\vec{a})=0$

$\Rightarrow \vec{d}-\vec{a}, \vec{b}-\vec{a}$ and $\vec{c}-a$ are coplanar points

A, B, C, D are coplanar.

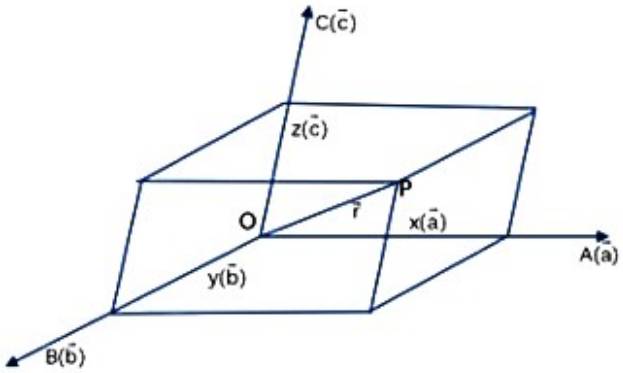
Theorem in space:

If $\vec{a}, \vec{b}, \vec{c}$ are 3 non zero non coplanar vectors then any vector $\vec{r}$ can be expressed as a linear combination :

$\vec{r}=x\vec{a}+y\vec{b}+z\vec{c}$ of $\vec{a}, \vec{b}, \vec{c}$

Example:

Express the non coplanar vectors $\vec{a}, \vec{b}, \vec{c}$ in terms of $\vec{b}\times\vec{c}, \vec{c}\times\vec{a}, \vec{a}\times\vec{b}$



Since $[\vec{a}\,\vec{b}\,\vec{c}]^2=\left[(\vec{a}\times\vec{b})(\vec{b}\times\vec{c})(\vec{c}\times\vec{a})\right]$

$\therefore$ If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar

$\Rightarrow \vec{a}\times\vec{b}, \vec{b}\times\vec{c}, \vec{c}\times\vec{a}$ are also non coplanar.

$\vec{a}=x(\vec{a}\times\vec{b})+y(\vec{b}\times\vec{c})+z(\vec{c}\times\vec{a})$

Taking dot product with $\vec{a}$

$$\vec{a}^2=y\left[\vec{a}\,\vec{b}\,\vec{c}\right] \Rightarrow y=\frac{(\vec{a})^2}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

Taking dot product with $\vec{b}$

$$\vec{a}\cdot\vec{b}=z\left[\vec{b}\,\vec{c}\,\vec{a}\right] \Rightarrow z=\frac{\vec{a}\cdot\vec{b}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

Similarly taking dot product with $\vec{c}$

$$\vec{a}\cdot\vec{c}=x\left[\vec{a}\,\vec{b}\,\vec{c}\right] \Rightarrow x=\frac{\vec{a}\cdot\vec{c}}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

$$\vec{a}=\frac{(\vec{a}\cdot\vec{c})(\vec{a}\times\vec{b})+(\vec{a})^2(\vec{b}\times\vec{c})+(\vec{a}\cdot\vec{b})(\vec{c}\times\vec{a})}{\left[\vec{a}\,\vec{b}\,\vec{c}\right]}$$

Q. **If $\vec{a},\vec{b}$ and $\vec{c}$ are three non-coplanar vectors, then prove that**

$$\vec{d}=\frac{\vec{a}\cdot\vec{d}}{[\vec{a}\vec{b}\vec{c}]}(\vec{b}\times\vec{c})+\frac{\vec{b}\cdot\vec{d}}{[\vec{a}\vec{b}\vec{c}]}(\vec{c}\times\vec{a})+\frac{\vec{c}\cdot\vec{d}}{[\vec{a}\vec{b}\vec{c}]}(\vec{a}\times\vec{b})$$

Sol. Since $\vec{a},\vec{b}$ and $\vec{c}$ are non-coplanar, vectors $\vec{a}\times\vec{b},\vec{b}\times\vec{c}$ and $\vec{c}\times\vec{a}$ are also non-coplanar.

$\vec{d}=l(\vec{b}\times\vec{c})+\vec{m}(\vec{c}\times\vec{a})+\vec{n}(\vec{a}\times\vec{b})$ ...(i)

Now multiplying both sides of (i) scalarly by $\vec{a}$, we have

$\vec{a}\cdot\vec{d}=l\vec{a}\cdot(\vec{b}\times\vec{c})+m\vec{a}\cdot(\vec{c}\times\vec{a})+n\vec{a}\cdot(\vec{a}\times\vec{b})=l[\vec{a}\vec{b}\vec{c}]$ $\because[\vec{a}\vec{c}\vec{a}]=0=[\vec{a}\vec{a}\vec{b}]$

$\Rightarrow l=(\vec{a}\cdot\vec{d})/[\vec{a}\vec{b}\vec{c}]$

Similarly, multiplying (i) scalarly by $\vec{b}$ and $\vec{c}$ successively, we get

$m=(\vec{b}\cdot\vec{d})/[\vec{a}\vec{b}\vec{c}]$ and $n=(\vec{c}\cdot\vec{d})/[\vec{a}\vec{b}\vec{c}]$

Putting these values of *l*, m and n in (i), we get the required relation.

Q. **Show that the p.v. of circumcentre of a tetrahedron OABC is (where 'O' is the origin)**

$$\frac{a^2(\vec{b}\times\vec{c})+b^2(\vec{c}\times\vec{a})+c^2(\vec{a}\times\vec{b})}{2[\vec{a}\,\vec{b}\,\vec{c}]}$$

Sol. If the centre 'P' has position vector, $\vec{r}$ then

$\vec{a}-\vec{r}=\overrightarrow{PA},\vec{b}-\vec{r}=\overrightarrow{PB},\vec{c}-r=\overrightarrow{PC}$

where $|\overrightarrow{PA}|=|\overrightarrow{PB}|=|\overrightarrow{PC}|=|\overrightarrow{OP}|=|\vec{r}|$

Consider $|\vec{a}-\vec{r}|=|\vec{r}|$

$\Rightarrow|\vec{a}-\vec{r}|\cdot|\vec{a}-\vec{r}|=\vec{r}\vec{r}\Rightarrow a^2-2\vec{a}\cdot\vec{r}+r^2=r^2\Rightarrow a^2=2\vec{a}\cdot\vec{r}$

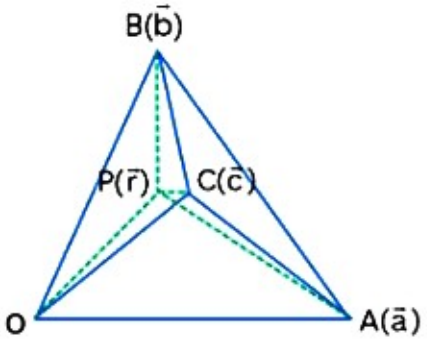
Similarly, $b^2=2\vec{b}\cdot\vec{r}$ and $c^2=2\vec{c}\cdot\vec{r}$

Since $(\vec{b}\times\vec{c}),(\vec{c}\times\vec{a})$ and $(\vec{a}\times\vec{b})$ are non-co-planar, then

$\vec{r}=x(\vec{b}\times\vec{c})+y(\vec{c}\times\vec{a})+z(\vec{a}\times\vec{b})$

$\Rightarrow\vec{a}\cdot\vec{r}=x\vec{a}.(\vec{b}\times\vec{c})+y\cdot0+z\cdot0=x[\vec{a}\vec{b}\vec{c}]$

$\Rightarrow x=\frac{\vec{a}\cdot\vec{r}}{[\vec{a}\vec{b}\vec{c}]}=\frac{a^2}{2[\vec{a}\vec{b}\vec{c}]}$



Similarly, $y = \dfrac{b^2}{2\left[\vec{a}\vec{b}\vec{c}\right]}$ and $z = \dfrac{c^2}{2\left[\vec{a}\vec{b}\vec{c}\right]}$

Hence $\vec{r} = \dfrac{a^2\left(\vec{b}\times\vec{c}\right) + b^2\left(\vec{c}\times\vec{a}\right) + c^2\left(\vec{a}\times\vec{b}\right)}{2\left[\vec{a}\vec{b}\vec{c}\right]}$

Real definition of linearly independence :

If $\vec{V}_1, \vec{V}_2, \ldots\ldots\vec{V}_n$ are vectors and $\lambda_1, \lambda_2, \ldots\ldots\lambda_n$ are scalar and if the linear combination $\lambda_1\vec{V}_1 + \lambda_2\vec{V}_2 + \ldots\ldots + \lambda_n\vec{V}_n = 0$, necessarily implies $\lambda_1 = \lambda_2 = \ldots\ldots\lambda_n = 0$, we say that $\vec{V}_1, \vec{V}_2, \ldots\ldots\vec{V}_n$ are said to constitutes a linearly independent set of vectors.

Point to Remember!!!

(i) 2 non zero, non collinear vectors are linearly independent.

(ii) Three non zero, non coplanar vectors are linearly independent i.e. $\left[\vec{a}\,\vec{b}\,\vec{c}\right] \neq 0 \Leftrightarrow \vec{a}, \vec{b}, \vec{c}$ are linearly independent.

(iii) Four or more vectors in 3D space are always linearly dependent.

Q. **Show that vectors $\vec{i} - 3\vec{j} + 2\vec{k}, 2\vec{i} - 4\vec{j} - \vec{k}$ and $3\vec{i} + 2\vec{j} - \vec{k}$ are linearly independent.**

Sol. Let $\vec{a} = \vec{i} - 3\vec{j} + 2\vec{k}, \vec{b} = 2\vec{i} - 4\vec{j} - \vec{k}$ and $\vec{c} = 3\vec{i} + 2\vec{j} - \vec{k}$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = \begin{vmatrix} 1 & -3 & 2 \\ 2 & -4 & -1 \\ 3 & 2 & -1 \end{vmatrix} \neq 0$$

Q. **If $\vec{a}, \vec{b}$ and $\vec{c}$ are non-coplanar vectors, then prove that $\left|(\vec{a}\cdot\vec{d})(\vec{b}\times\vec{c}) + (\vec{b}\cdot\vec{d})(\vec{c}\times\vec{a}) + (\vec{c}\cdot\vec{d})(\vec{a}\times\vec{b})\right|$ is independent of $\vec{d}$, where $\vec{d}$ is a unit vector.**

Sol. Given $[\vec{a}\vec{b}\vec{c}] \neq 0$ as $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar. Also there does not exist any linear relation between them because if any such relation exists, then they would be coplanar.

Let $\vec{A} = x(\vec{b}\times\vec{c}) + y(\vec{c}\times\vec{a}) + z(\vec{a}\times\vec{b})$

where $x = \vec{a}\cdot\vec{d}, y = \vec{b}\cdot\vec{d}, z = \vec{c}\cdot\vec{d}$

We have to find the value of modulus of $\vec{A}$, i.e., $|\vec{A}|$, which is independent of $\vec{d}$.

Multiplying both sides scalarly by $\vec{a}, \vec{b}$ and $\vec{c}$ and we know that scalar triple product is zero when two vectors are equal.

$\vec{A}\cdot\vec{a} = x[\vec{a}\vec{b}\vec{c}] + 0$

Putting for x, we get

$(\vec{a}\cdot\vec{d})[\vec{a}\vec{b}\vec{c}] = \vec{A}\cdot\vec{a}$

Similarly, we have

$(\vec{b}\cdot\vec{d})[\vec{a}\vec{b}\vec{c}] = \vec{A}\cdot\vec{b}$

$(\vec{c}\cdot\vec{d})[\vec{a}\vec{b}\vec{c}] = \vec{A}\cdot\vec{c}$

Adding the above relations, we get

$[(\vec{a}+\vec{b}+\vec{c})\cdot\vec{d}][\vec{a}\vec{b}\vec{c}] = \vec{A}\cdot(\vec{a}+\vec{b}+\vec{c})$ or $(\vec{a}+\vec{b}+\vec{c})\cdot[\vec{d}[\vec{a}\vec{b}\vec{c}] - \vec{A}] = 0$

Since $\vec{a},\vec{b}$ and $\vec{c}$ are non-coplanar, $\vec{a}+\vec{b}+\vec{c} \neq 0$ because otherwise any one is expressible as a linear combination of other two.

Hence $[\vec{a}\vec{b}\vec{c}]\vec{d} = \vec{A}$

$|\vec{A}| = |[\vec{a}\vec{b}\vec{c}]|$ as $\vec{d}$ is a unit vector.

It is independent of $\vec{d}$.

Reciprocal system of vectors:

1. If $\vec{a},\vec{b},\vec{c}$ and $\vec{a}',\vec{b}',\vec{c}'$ are 2 sets of non coplanar vectors such that $\vec{a}\cdot\vec{a}' = \vec{b}\cdot\vec{b}' = \vec{c}\cdot\vec{c}' = 1$ and $\vec{a}\cdot\vec{b}' = \vec{a}\cdot\vec{c}' = \vec{b}\cdot\vec{a}' = \vec{b}\cdot\vec{c}' = \vec{c}\cdot\vec{a}' = \vec{c}\cdot\vec{b}' = 0$, then $\vec{a},\vec{b},\vec{c}$ and $\vec{a}',\vec{b}',\vec{c}'$ are said to be constitute a reciprocal system of vectors.
2. Reciprocal system of vectors exists only in case of dot product.
3. It is possible to define $\vec{a}',\vec{b}',\vec{c}'$ in terms of $\vec{a},\vec{b},\vec{c}$ as

$$\vec{a}' = \frac{\vec{b}\times\vec{c}}{[\vec{a}\vec{b}\vec{c}]};\ \vec{b}' = \frac{\vec{c}\times\vec{a}}{[\vec{a}\vec{b}\vec{c}]};\ \vec{c}' = \frac{\vec{a}\times\vec{b}}{[\vec{a}\vec{b}\vec{c}]}\ ([\vec{a}\vec{b}\vec{c}] \neq 0)$$

Note: **(i)** $\vec{a}\times\vec{a}' + \vec{b}\times\vec{b}' + \vec{c}\times\vec{c}' = 0$ i. e. $\dfrac{\vec{a}\times(\vec{b}\times\vec{c}) + \vec{b}\times(\vec{c}\times\vec{a}) + \vec{c}\times(\vec{a}\times\vec{b})}{[\vec{a}\vec{b}\vec{c}]}$

(ii) $(\vec{a}+\vec{b}+\vec{c})\cdot(\vec{a}'+\vec{b}'+\vec{c}') = 3$ $(\text{as } \vec{a}\cdot\vec{b}' = \vec{a}\cdot\vec{c}' = 0 \text{ etc})$

(iii) If $[\vec{a}\vec{b}\vec{c}] = V$ then $[\vec{a}'\vec{b}'\vec{c}'] = \dfrac{1}{V} \Rightarrow [\vec{a}\vec{b}\vec{c}][\vec{a}'\vec{b}'\vec{c}'] = 1$

(iv) $\vec{a}'\times\vec{b}' + \vec{b}'\times\vec{c}' + \vec{c}'\times\vec{a}' = \dfrac{\vec{a}+\vec{b}+\vec{c}}{[\vec{a}\vec{b}\vec{c}]},\ [\vec{a}\vec{b}\vec{c}] \neq 0$

Q. **(i) Find the set of vectors reciprocal to the set of vectors**
$2\hat{i}+3\hat{j}-\hat{k}, \hat{i}-\hat{j}-2\hat{k}, -\hat{i}+2\hat{j}+2\hat{k}$

Sol. Let the given vectors be a, b, c

Now, $[abc] = \begin{vmatrix} 2 & 3 & -1 \\ 1 & -1 & -2 \\ -1 & 2 & 2 \end{vmatrix}$

$= 2(-2 + 4) - 3(2 - 2) - 1(2 - 1)$

$= 4 - 1 = 3$

$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -2 \\ -1 & 2 & 2 \end{vmatrix} = 2\hat{i} + \hat{k}$

$\vec{c} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 2 \\ 2 & 3 & -1 \end{vmatrix} = -8\hat{i} + 3\hat{j} - 7\hat{k}$,

$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & -1 & -2 \end{vmatrix} = -7\hat{i} + 3\hat{j} - 5\hat{k}$

Hence, $a' = \dfrac{b \times c}{[abc]} = \dfrac{2\hat{i} + \hat{k}}{3}$

$b' = \dfrac{c \times a}{[abc]} = \dfrac{-8\hat{i} + 3\hat{j} - 7\hat{k}}{3}$

and $c' = \dfrac{a \times b}{[abc]} = \dfrac{-7\hat{i} + 3\hat{j} - 5\hat{k}}{3}$

(ii) Find a set of vector reciprocal to the vectors a, b and a × b.

Sol. Let the given vectors be denoted by a, b and c where c = a × b.

$\therefore \quad [abc] = (\vec{a} \times \vec{b}) \cdot \vec{c} = (\vec{a} \times \vec{b}) = .\left(\vec{a} \times \vec{b}\right) = |\vec{a} \times \vec{b}|^2$ ...(i)

and let the reciprocal system of vector be a', b' and c'.

$\therefore \quad a' = \dfrac{\vec{b} \times \vec{c}}{[\vec{a}\vec{b}\vec{c}]} = \dfrac{\vec{b} \times (\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|^2}$

$b' = \dfrac{\vec{c} \times \vec{a}}{[\vec{a}\vec{b}\vec{c}]} = \dfrac{(\vec{a} \times \vec{b}) \times \vec{a}}{|\vec{a} \times \vec{b}|^2}$

$c' = \dfrac{\vec{a} \times \vec{b}}{[\vec{a}\vec{b}\vec{c}]} = \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|^2}$

$\therefore$ $\vec{a}'$, $\vec{b}'$ and $\vec{c}'$ are required reciprocal system of vectors for $\vec{a}$, $\vec{b}$ and $\vec{a} \times \vec{b}$.

(iii) If $\vec{a}' = \frac{\vec{b}\times\vec{c}}{[\vec{a}\,\vec{b}\,\vec{c}]}, \vec{b}' = \frac{\vec{c}\times\vec{a}}{[\vec{a}\,\vec{b}\,\vec{c}]}, c' = \frac{\vec{a}\times\vec{b}}{[\vec{a}\,\vec{b}\,\vec{c}]}$ **then show that**

$\vec{a}\times\vec{a}'+\vec{b}\times\vec{b}'+\vec{c}\times\vec{c}' = 0$, **where** $\vec{a}, \vec{b}$ **and** $\vec{c}$ **are non-coplanar.**

Sol. Here, $\vec{a}\times\vec{a}' = \frac{\vec{a}\times(\vec{b}\times\vec{c})}{[\vec{a}\,\vec{b}\,\vec{c}]}$ $\vec{a}\times\vec{a}' = \frac{(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}}{[\vec{a}\vec{b}\vec{c}]}$ ...(i)

Similarly, $\vec{b}\times\vec{b}' = \frac{(\vec{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}}{[\vec{a}\vec{b}\vec{c}]}$ ...(ii)

$\vec{c}\times\vec{c}' = \frac{(\vec{c}\cdot\vec{b})\vec{a}-(\vec{c}\cdot\vec{a})\vec{b}}{[\vec{a}\vec{b}\vec{c}]}$ ...(iii)

$\vec{a}\times\vec{a}'+\vec{b}\times\vec{b}'+\vec{c}\times\vec{c}' = \frac{(\vec{a}\cdot\vec{c})\vec{b}-(\vec{a}\cdot\vec{b})\vec{c}+(\vec{b}\cdot\vec{a})\vec{c}-(\vec{b}\cdot\vec{c})\vec{a}+(\vec{c}.\vec{b})\vec{a}-(\vec{c}.\vec{a})\vec{b}}{[\vec{a}\vec{b}\vec{c}]} = 0$

$(\because \vec{a}\cdot\vec{b} = \vec{b}\cdot\vec{a})$

Isolating an known vectors

Satisfying a given relationship with some known vectors:

There is no general method for solving such equations, however dot or cross with known or unknown vectors or dot with $\vec{a}\times\vec{b}$, generally isolates the unknown vector. Use of linear combination also proves to be advantageous.

Q. **Find vector** $\vec{r}$ **if** $\vec{r}\cdot\vec{a} = m$ **and** $\vec{r}\times\vec{b} = \vec{c}$, **where** $\vec{a}\cdot\vec{b} \neq 0$.

Sol. $\vec{r}\cdot\vec{a} = m$ ...(i)

and $\vec{r}\times\vec{b} = \vec{c}$ ...(ii)

From (ii), $\vec{a}\times(\vec{r}\times\vec{b}) = \vec{a}\times\vec{c}$

Or $(\vec{a}\cdot\vec{b})\vec{r}-(\vec{a}\cdot\vec{r})\vec{b} = \vec{a}\times\vec{c}$

Or $(\vec{a}\cdot\vec{b})\vec{r} = \vec{a}\times\vec{c}+(\vec{a}\cdot\vec{r})\vec{b} = \vec{a}\times\vec{c}+m\vec{b}$

$\therefore$ $\vec{r} = \frac{1}{\vec{a}\cdot\vec{b}}(\vec{a}\times\vec{c}+m\vec{b})$

Q. **Find** $\vec{r}$ **such that** $t\vec{r}+\vec{r}\times\vec{a} = \vec{b}$, **where** $\vec{a}$ **&** $\vec{b}$ **are non collinear vectors.**

Sol. Given, $t\vec{r}+\vec{r}\times\vec{a} = \vec{b}$ ...(i)

Since $\vec{a}, \vec{b}, \vec{a}\times\vec{b}$ are non-coplanar vectors therefore $\vec{r}$ can be expressed as linear

combination of $\vec{a}, \vec{b}$ and $\vec{a} \times \vec{b}$

Let $\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$ ...(ii)

Putting the value of $\vec{r}$ in (i), we get

$$t\left[x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})\right] + x(\vec{a} \times \vec{a}) + y(\vec{b} \times \vec{a}) + z(\vec{a} \times \vec{b}) \times \vec{a} = \vec{b}$$

$$t\left[x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})\right] + y(\vec{b} \times \vec{a}) + z\left[(\vec{a} \cdot \vec{a})\vec{b} - (\vec{a} \cdot \vec{b})\vec{a}\right] = \vec{b}$$

$$\Rightarrow \left[tx - z(\vec{a} \cdot \vec{b})\right]\vec{a} + \left[ty + z(\vec{a} \cdot \vec{a}) - 1\right]\vec{b} + (tz - y)(\vec{a} \times \vec{b}) = 0$$

Equating the coefficients of $\vec{a}, \vec{b}$ and $\vec{a} \times \vec{b}$, we get

$tx - z(\vec{a} \cdot \vec{b}) = 0$ ...(iii)

$ty + za^2 - 1 = 0$ ... (iv)

$tz - y = 0$ ...(v)

Solving (iii), (iv) and (v), we get

$$x = \frac{\vec{a} \cdot \vec{b}}{t(t^2 + a^2)},\ y = \frac{t}{t^2 + a^2},\ z = \frac{1}{t^2 + a^2}$$

Putting the values of x, y, z in (ii), we have

$$\vec{r} = \frac{1}{t^2 + a^2}\left[\frac{(\vec{a} \cdot \vec{b})\vec{a}}{t} + t\vec{b} + \vec{a} \times \vec{b}\right]$$

Second method:

Given $t\vec{r} + \vec{r} \times \vec{a} = \vec{b}$ ...(i)

$\therefore \vec{r} \times \vec{a} = \vec{b} - t\vec{r}$ ...(ii)

Taking cross product of both sides of (i) with $\vec{a}$, we get

$t(\vec{r} \times \vec{a}) + (\vec{r} \times \vec{a}) \times \vec{a} = \vec{b} \times \vec{a}$ [Putting the value of $\vec{r} \times \vec{a}$]

$$\Rightarrow\ t(\vec{b} - t\vec{r}) + (\vec{r} \cdot \vec{a})\vec{a} - a^2\vec{r} = \vec{b} \times \vec{a}$$

$$\Rightarrow\ t\vec{b} - (t^2 + a^2)\vec{r} + (\vec{r} \cdot \vec{a})\vec{a} = \vec{b} \times \vec{a}$$ ...(iii)

In (i), taking dot product with $\vec{a}$, we get

$$\vec{r} \cdot (\vec{r} \times \vec{a}) = 0 = \vec{b} \times \vec{a} \quad \therefore \vec{r} \times \vec{a} = \frac{\vec{a} \times \vec{b}}{t}$$

From (ii), $t\vec{b} - (t^2 + a^2)\vec{r} + \frac{\vec{a} \times \vec{b}}{t}\vec{a} = \vec{b} \times \vec{a}$

$$\vec{r} = \frac{1}{t^2 + a^2}\left(\frac{\vec{a} \times \vec{b}}{t}\vec{a} + t\vec{b} + \vec{a} \times \vec{b}\right)$$

Q. **Solve the following simultaneous equations $\vec{x}$ and $\vec{y}$, $\vec{x}+\vec{y}=\vec{a}, \vec{x}\times\vec{y}=\vec{b}$ and $\vec{x}\cdot\vec{a}=1$**

Sol. Given $\vec{x}+\vec{y}=\vec{a}$ ...(i)

$\vec{x}\times\vec{y}=\vec{b}$ ...(ii)

$\vec{x}\cdot\vec{a}=1$ ...(iii)

Putting the value of $\vec{y}$ from (i) in (ii), we get

$\vec{x}\times(\vec{a}-\vec{x})=\vec{b}$

$\Rightarrow \vec{x}\times\vec{a}=\vec{b}$

$\Rightarrow \vec{a}\times(\vec{x}\times\vec{a})=\vec{a}\times\vec{b}$

$\Rightarrow a^2\vec{x}-(\vec{a}\cdot\vec{x})\vec{a}=\vec{a}\times\vec{b} \Rightarrow a^2\vec{x}-\vec{a}=\vec{a}\times\vec{b}$

$\therefore \vec{x}=\frac{1}{a^2}(\vec{a}+\vec{a}\times\vec{b})$ and $\vec{y}=\vec{a}-\vec{x}=\vec{a}-\frac{1}{a^2}(\vec{a}+\vec{a}\times\vec{b})$

Q. **(i) Solve the vector equation $\vec{r}\times\vec{b}=\vec{a}\times\vec{b}, \vec{r}.\vec{c}=0$ provided that $\vec{c}$ is not perpendicular to $\vec{b}$.**

Sol. We are given,

$\vec{r}\times\vec{b}=\vec{a}\times\vec{b}$

$\Rightarrow (\vec{r}-\vec{a})\times\vec{b}=0$

Hence, $(\vec{r}-\vec{a})$ and $\vec{b}$ are parallel.

$\Rightarrow \vec{r}-a=t\vec{b}$ ...(i)

and we know $\vec{r}.\vec{c}=0$

$\therefore$ Taking dot product of Equation (i) by $\vec{c}$, we get

$\Rightarrow \vec{r}.\vec{a}-\vec{a}.\vec{c}=t(\vec{b}.\vec{c})$

$\Rightarrow 0-\vec{a}.\vec{c}=t(\vec{b}.\vec{c})$

$\Rightarrow t=-\left(\frac{\vec{a}\cdot\vec{c}}{\vec{b}\cdot\vec{c}}\right)$ ...(ii)

$\therefore$ From Eqs. (i) and (ii) solution of $\vec{r}$ is

$$\vec{r}=\vec{a}-\left(\frac{\vec{a}\cdot\vec{c}}{\vec{b}\cdot\vec{c}}\right)\vec{b}$$

Q. **(ii) Solve for $\vec{x}$, such that $\vec{A}.\vec{X} = \vec{C}$ and $\vec{A}\times\vec{X} = \vec{B}$ with $\vec{C} \neq 0$.**

Sol. We have, $\vec{A}\times\vec{X} = \vec{B}$

Taking vector product of both sides with $\vec{A}$, we get

$\vec{A}\times\vec{B} = \vec{A}\times(A\times\vec{X})$

$= (\overline{A}\cdot\vec{X})\vec{A} - (\overline{A}\cdot\vec{A})\vec{X}$

$= C\vec{A} - |\vec{A}|^2\vec{X}$ (using $\vec{A}\cdot\vec{X} = C \text{ and } \vec{A}\cdot\vec{A} = |\vec{A}|^2$)

$\Rightarrow |\vec{A}|^2 X = \overline{CA} - \vec{A}\times\vec{B}$

or $\vec{X} = \dfrac{\overline{CA} + \vec{B}\times\vec{A}}{|\vec{A}|^2}$

Q. **(iii) Solve the vecor equation $\vec{r}\times\vec{a} + \overline{kr} = \vec{b}$, where $\vec{a}$ and $\vec{b}$ are two given vector and k is any scalar.**

Sol. Since, $\vec{a}, \vec{b}$ and $\vec{a}\times\vec{b}$ are two non-coplanar vectors.

$\vec{r} = x\vec{a} + y\vec{b} + z(\vec{a}\times\vec{b})$ ...(i)

(where, x, y and z scalars)

On putting r in $\vec{r}\times\vec{a} + k\vec{r}' = \vec{b}$, we get

$\{x\vec{a} + y\vec{b} + z(\vec{a}\times\vec{b})\}\times\vec{a} + k\{x\vec{a} + y\vec{b} + z(\vec{a}\times\vec{b})\} = \vec{b}$

$\Rightarrow \quad y(\vec{b}\times\vec{a}) + z\{(\vec{a}\cdot\vec{a})\vec{b} - (\vec{a}\cdot\vec{b})\vec{a}\} + k\{x\vec{a} + y\vec{b} + z(\vec{a}\times\vec{b})\} = \vec{b}$

$\Rightarrow \quad kx - z(\vec{a}\cdot\vec{b}) = 0, ky + z(\vec{a}\cdot\vec{a}) = 1$

$\Rightarrow \quad -y + zk = 0$

On solving these equations, we get

$z = \dfrac{1}{k^2 + |\vec{a}|^2}$

$x = \dfrac{\vec{a}\cdot\vec{b}}{k(|\vec{a}|^2 + k^2)}$

and $y = \dfrac{k}{k^2 + |\vec{a}|^2}$

On putting theses value in Eq. (i), we get the solution.

$r = \dfrac{(\vec{a}\cdot\vec{b})}{k(k^2 + |\vec{a}|^2)} + \dfrac{k}{k^2 + |\vec{a}|^2}(\vec{b}) + \dfrac{1}{k^2 + |\vec{a}|^2}(\vec{a}\times\vec{b})$

$r = \dfrac{1}{k^2 + |\vec{a}|^2}\left[\dfrac{(\vec{a}\cdot\vec{b})\vec{a}}{k} + (k)\vec{b} + (\vec{a}\times\vec{b})\right]$

Q. **(iv) Solve for vector A and B, where $\vec{A}+\vec{B}=\vec{a}$, $\vec{A}\times\vec{B}=\vec{b}$, $\vec{A}\cdot\vec{a}=1$.**

Sol. We have, $\vec{A}+\vec{B}=\vec{a}$

$\Rightarrow \vec{A}\cdot\vec{a}+\vec{B}\cdot\vec{a}=\vec{a}\cdot\vec{a}$

$\Rightarrow 1+\vec{B}\cdot\vec{a}=|\vec{a}|^2$ (given $\vec{A}.\vec{a}=1$)

$\Rightarrow \vec{B}\cdot\vec{a}=|\vec{a}|^2-1$ ...(i)

Also, $\vec{A}\times\vec{B}=\vec{b}$

$\Rightarrow \vec{a}\times(\vec{A}\times\vec{B})=\vec{a}\times\vec{b}$

$\Rightarrow (\vec{a}\cdot\vec{B})\vec{A}-(\vec{a}\cdot\vec{A})\vec{B}=\vec{a}\times\vec{b}$

$\Rightarrow (|\vec{a}|^2-1)\vec{A}-\vec{B}=\vec{a}\times\vec{b}$ [using Eq. (i) and a. A = 1] ...(ii)

and $\vec{A}+\vec{B}=\vec{a}$

From Eqs. (i) and (ii), we get

$$\vec{A}=\frac{(\vec{a}\times\vec{b})+\vec{a}}{|\vec{a}|^2} \text{ and } \vec{B}=\vec{a}-\left\{\frac{(\vec{a}\times\vec{b})+\vec{a}}{|\vec{a}|^2}\right\}$$

$$\Rightarrow \vec{B}=\frac{(\vec{b}\times\vec{a})+\vec{a}(|\vec{a}|^2-1)}{|\vec{a}|^2}$$

$$\text{Thus } \vec{A}=\frac{(\vec{a}\times\vec{b})+\vec{a}}{|\vec{a}|^2} \text{ and } \vec{B}=\frac{(\vec{b}\times\vec{a})\vec{a}(|\vec{a}|^2-1)}{|\vec{a}|^2}$$

Q. **If $\vec{r}\cdot\vec{a}=0,\vec{r}\cdot\vec{b}=1$ and $[\vec{r}\vec{a}\vec{b}]=1,\vec{a}\cdot\vec{b}\neq 0,(\vec{a}\cdot\vec{b})^2-|\vec{a}|^2|\vec{b}|^2=1$, then find $\vec{r}$ in terms of $\vec{a}$ and .**

Sol. Writing $\vec{r}$ as linear combination of $\vec{a},\vec{b}$ and $\vec{a}\times\vec{b}$, we have $\vec{r}=x\vec{a}+y\vec{b}+z(\vec{a}\times\vec{b})$

For scalars x, y and z

$0=\vec{r}\cdot\vec{a}=x|\vec{a}|^2+y\vec{a}\cdot\vec{b}$ (taking dot product with $\vec{a}$)

$1=\vec{r}\cdot\vec{b}=x\vec{a}\cdot\vec{b}+y|\vec{b}|^2$ (taking dot product with $\vec{b}$)

$$\text{Solving, we get } y=\frac{|\vec{a}|^2}{|\vec{a}|^2|\vec{b}|^2-(\vec{a}\cdot\vec{b})^2}=-|\vec{a}|^2 \text{ and } x=\frac{\vec{a}\cdot\vec{b}}{(\vec{a}\cdot\vec{b})^2-|\vec{a}|^2|\vec{b}|^2}=\vec{a}\cdot\vec{b}$$

Also $1=[\vec{r}\vec{a}\vec{b}]=z|\vec{a}\times\vec{b}|^2$ (taking dot product with $\vec{a}\times\vec{b}$)

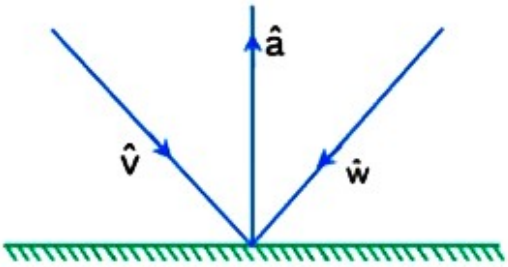
$$\Rightarrow z=\frac{1}{|a\times b|^2}$$

$$\text{thus } \vec{r}=\left((\vec{a}\cdot\vec{b})\vec{a}-|\vec{a}|^2\vec{b}\right)+\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|^2}=\vec{a}\times(\vec{a}\times\vec{b})+\frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|^2}$$

Previous Year's Question

Incident ray is along the unit vector $\hat{v}$ and the reflected ray is along the unit vector $\hat{w}$. The normal is along unit vector $\hat{a}$ outwards. Express $\hat{w}$ in terms of $\hat{a}$ and $\hat{v}$.

[JEE 2005 (Mains), 2 + 4 out of 60]



Q. **If** $p\vec{x}+(\vec{x}\times\vec{a})=\vec{b};(p\neq 0)$ **prove that** $\vec{x}=\dfrac{p^2\vec{b}+(\vec{b}\cdot\vec{a})\vec{a}-p(\vec{b}\times\vec{a})}{p\left(p^2+|\vec{a}|^2\right)}$.

Sol. $p\vec{x}+(\vec{x}\times\vec{a})=\vec{b}$ ...(i)

$p(\vec{x}\cdot\vec{a})+(\vec{x}\times\vec{a})\cdot\vec{a}=\vec{a}\cdot\vec{b}$ (Take dot produt with $\vec{a}$)

$p(\vec{x}\cdot\vec{a})+0=\vec{a}\cdot\vec{b}$

$\vec{x}\cdot\vec{a}=\dfrac{\vec{a}\cdot\vec{b}}{p}$ ...(ii)

$\vec{a}\times(p\vec{x})+\vec{a}\times(\vec{x}\times\vec{a})=\vec{a}\times\vec{b}$ (Take cross product with $\vec{a}$ both side)

$p(\vec{a}\times\vec{x})+\vec{a}\cdot\vec{a}(\vec{x})-(\vec{a}.\vec{x})\vec{a}=\vec{a}\times\vec{b}$

$p(p\vec{x}-\vec{b})+|\vec{a}|^2\,\vec{x}-(\vec{a}\cdot\vec{x})\vec{a}=\vec{a}\times\vec{b}$ [Using (I)]

$p^2\vec{x}-p(\vec{b})+|\vec{a}|^2\,\vec{x}-\left(\dfrac{\vec{a}\cdot\vec{b}}{p}\right)\vec{a}=-(\vec{b}\times\vec{a})$ [Using (II)]

$\left(p^2+|\vec{a}|^2\right)\vec{x}=p(\vec{b})+\left(\dfrac{\vec{a}\cdot\vec{b}}{p}\right)\vec{a}-(\vec{b}\times\vec{a})$

$\vec{x}\left(p^2+|\vec{a}|^2\right)=\dfrac{p^2(\vec{b})+(\vec{b}\cdot\vec{a})\vec{a}-p(\overline{b\times a})}{p}$

$\vec{x}=\dfrac{p^2(\vec{b})+(\vec{b}\cdot\vec{a})\vec{a}-p(\vec{b}\times\vec{a})}{p\left(p^2+|\vec{a}|^2\right)}$